

ORACLE

Oracle Database@Google Cloud

Technical Deep Dive

Version February 2026

Agenda

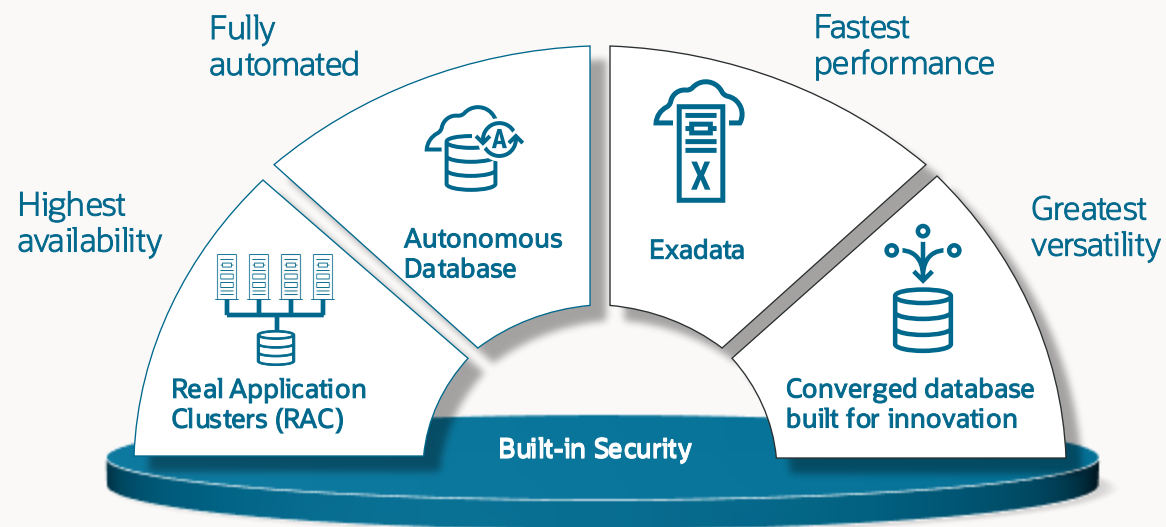
- 1 Service Overview
- 2 Oracle Database Services
 - Autonomous AI Database (ADB-S)
 - Exadata Database Service (ExaDB-D and ExaDB-XS)
 - Base Database Service (BaseDB)
- 3 Architecture, Network Topologies
- 4 Purchase, Enablement, Provisioning
- 5 High Availability and Disaster Recovery
- 6 Migration
- 7 Observability
- 8 Identity and Permissions
- 9 Security
- 10 Automation

Service Overview

Oracle Database@Google Cloud – The best of both worlds

Enterprise grade data management services

Innovative cloud services



 Compute Engine

 Kubernetes Engine

 Cloud Run

 Cloud Storage

 Vertex AI



ORACLE
CLOUD
Infrastructure



Google Cloud



Benefits of Oracle Database@Google Cloud

FULL FUNCTIONALITY

- Run latency-sensitive apps, with μ s latencies
- Highest database performance, availability, security, and scale
- Includes: Oracle RAC, Exadata, Autonomous AI Database, Autonomous Recovery Service
- Embedded AI: Vector search, Select AI

SEAMLESS EXPERIENCE

- Native Google Cloud console and API experience
- One place for monitoring, notifications, operations, billing, etc.
- Leverage both Google Cloud and Oracle skills

MARKETPLACE & PROGRAMS

Pay-as-you-use or custom pricing

Leverage existing investments:

- Google Cloud Commitments
- Oracle Database Licenses (ULA, BYOL)
- Oracle Support Rewards

REGIONS

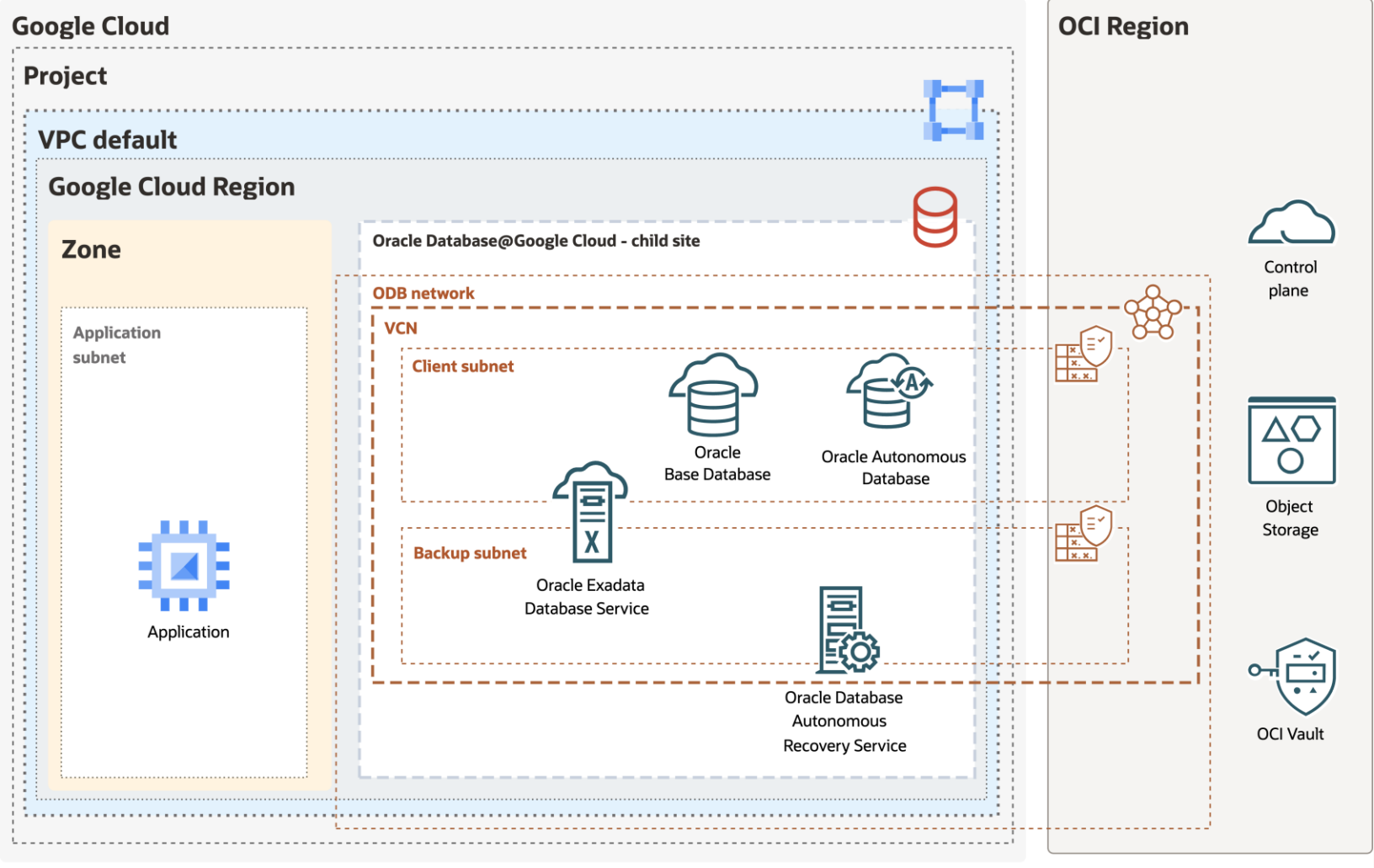
14 regions available
3 regions planned

<https://docs.oracle.com/en-us/iaas/Content/database-at-gcp/get-started-regions.htm>

JOINT SUPPORT

- Collaborative support
- Ability to speak to either Oracle or Google Cloud and seek help

High Level Architecture



The best of Oracle and Google Cloud, all in one place

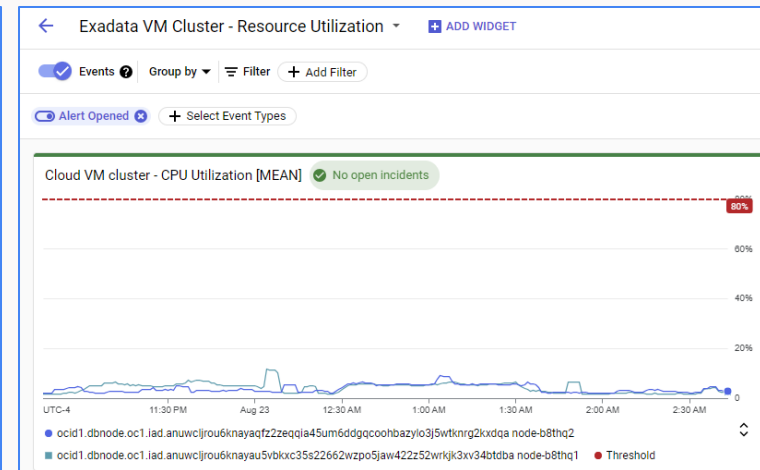
Purchase in Google Cloud Marketplace and parity with OCI

Deploy, manage, and monitor through Google Cloud Console and APIs

Combine with your choice of Google Cloud services

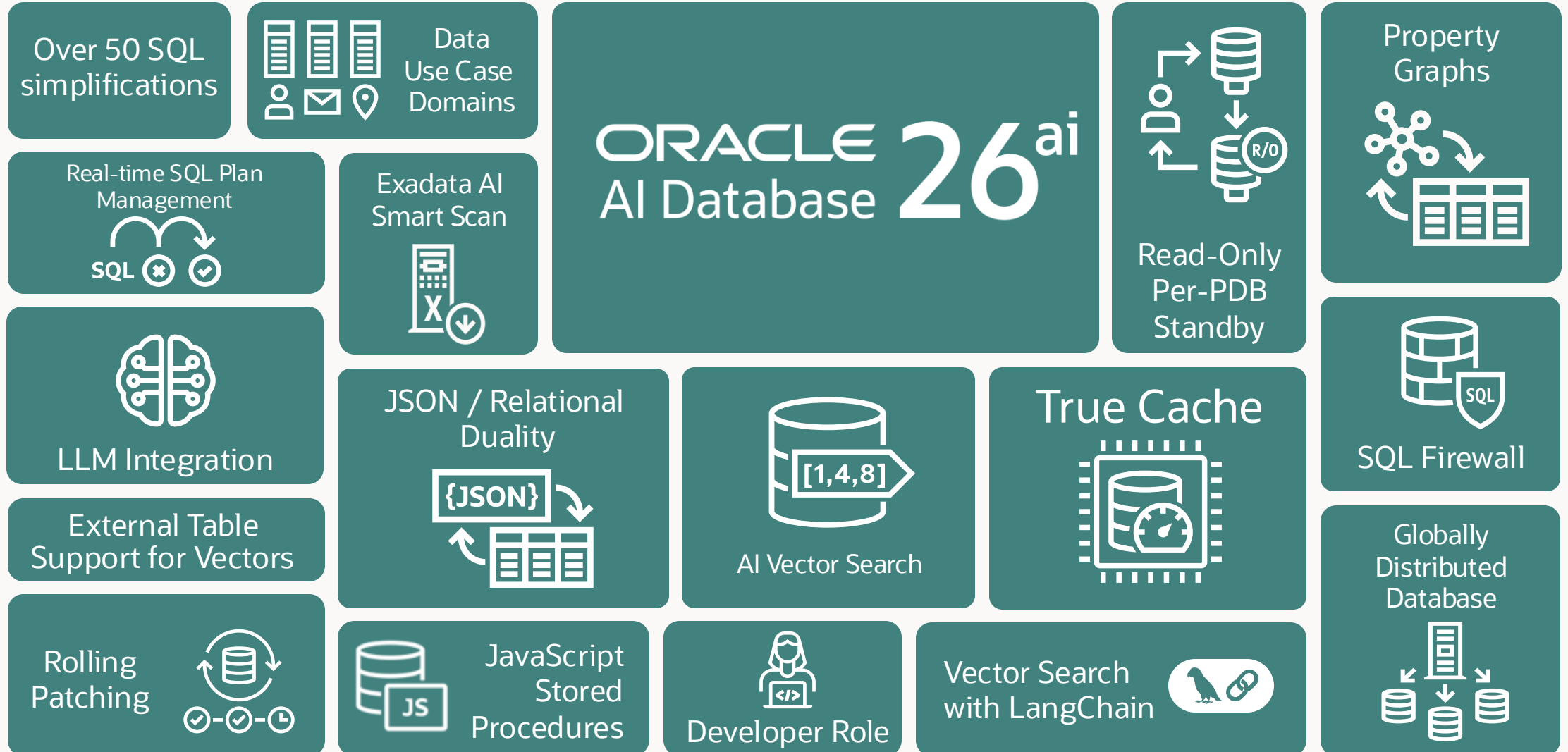
The screenshot shows the Google Cloud Marketplace page for 'Oracle Database@Google Cloud'. The page includes a header with the Google Cloud logo and a user profile 'omcpmout2'. Below the header, there's a 'Product details' section with the Oracle logo and a description: 'Accelerate innovation and supercharge database applications with Oracle Database@Google Cloud'. There are 'SUBSCRIBE' and 'CONTACT SALES' buttons. Below this, there's an 'OVERVIEW' section with a description of the service and its benefits, and an 'Additional details' section with links to 'SaaS & APIs' and 'Databases'.

The screenshot shows the 'Create Exadata Infrastructure' page in the Google Cloud Console. The page has a header with the Google Cloud logo and a user profile 'omcpmout2'. Below the header, there's a 'Create Exadata Infrastructure' section with 'Instance details' and 'Machine configuration' subsections. The 'Instance details' section includes fields for 'Infrastructure display name' (demo-exa-infra), 'Infrastructure ID' (demo-exa-infra), and 'Region' (us-east4). The 'Machine configuration' section includes a dropdown for 'Exadata Infrastructure model' (Exadata.X9M), a field for 'Database servers' (2), 'Storage servers' (3), and 'Storage size' (189 TB).



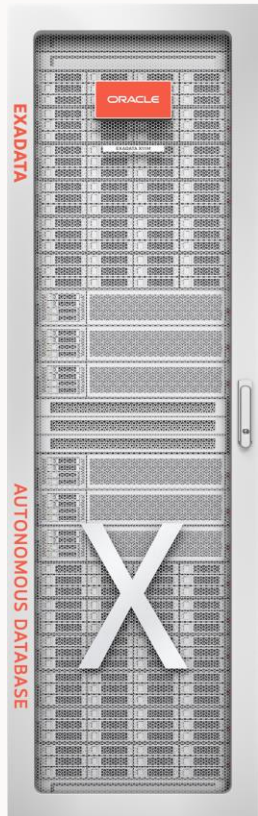
The screenshot shows the 'Cloud overview' page in the Google Cloud Console. The page has a header with the Google Cloud logo and a user profile 'omcpmout2'. Below the header, there's a 'Cloud overview' section with a list of 'PINNED PRODUCTS' including 'APIs & Services', 'Billing', 'IAM & Admin', 'Marketplace', 'Vertex AI', 'Compute Engine', 'Kubernetes Engine', 'Cloud Storage', 'BigQuery', 'VPC Network', 'Cloud Run', 'SQL', 'Security', 'Google Maps Plat...', 'Oracle Database ...', and 'Monitoring'.

Oracle AI Database 26ai : Next Long-term Support Release



Exadata: Premier Platform for Oracle Database Services

Extreme performance, scalability, and availability for data and AI workloads, at a low cost



Data optimized hardware

Scale-out, data optimized compute, networking, and storage

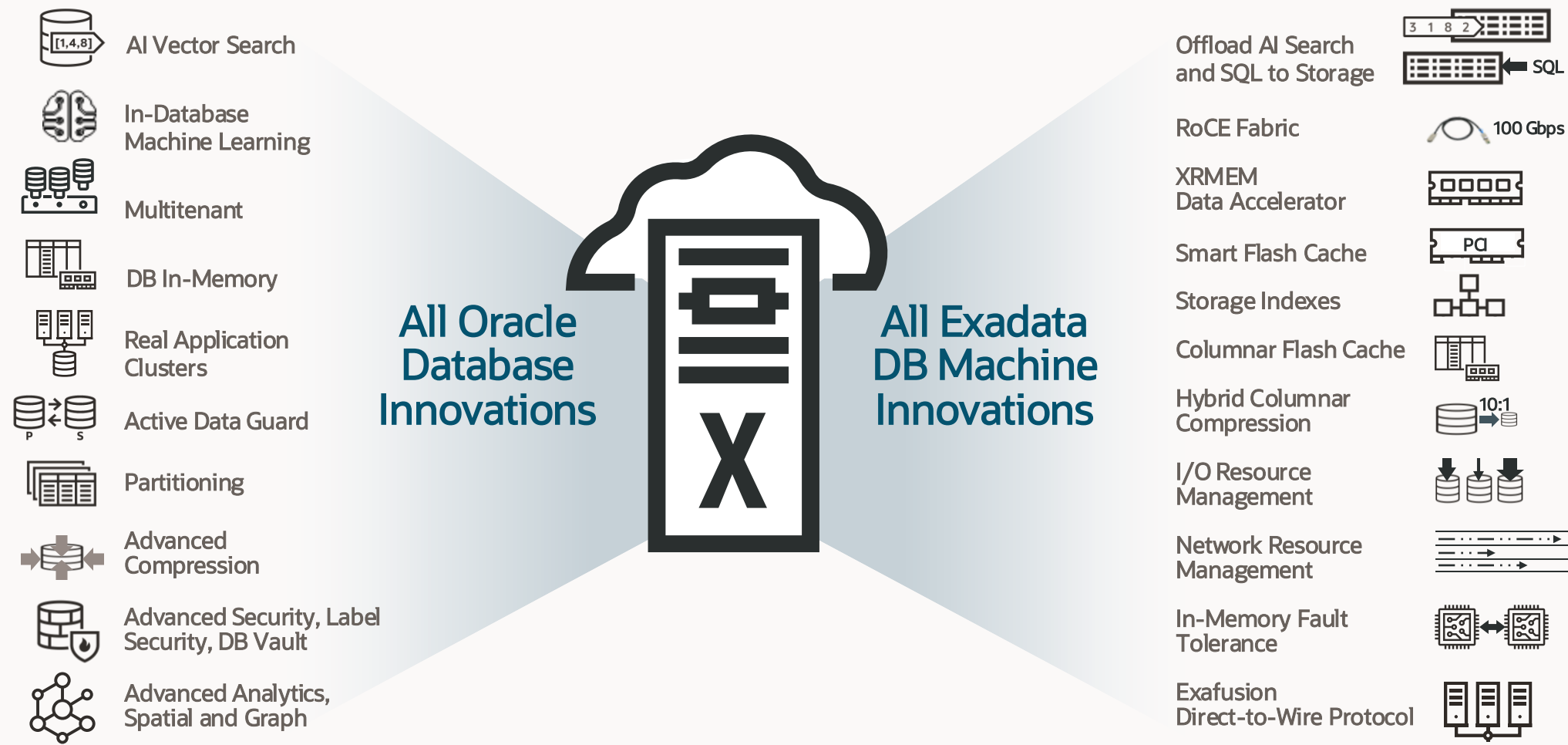
Data intelligent software

Unique algorithms deliver extreme performance and efficiency for AI, analytics, and mission-critical apps, at any scale

Engineered to run everywhere

On-premises, Cloud@Customer, Oracle Cloud, and Multicloud

Complete Oracle Database and Exadata capabilities in the cloud



Oracle Database Services on Oracle Database@Google Cloud

Autonomous AI Database



Fully managed, cloud-native Oracle Database service on **Exadata** with **complete automation** and the **lowest TCO**

Exadata Database Service



Co-managed Oracle Database service on **Exadata** with cloud automation and **full administrative control**

Base Database Service



Co-managed Oracle Database service on **Oracle hardware** with cloud automation and **full administrative control**

Autonomous Recovery Service

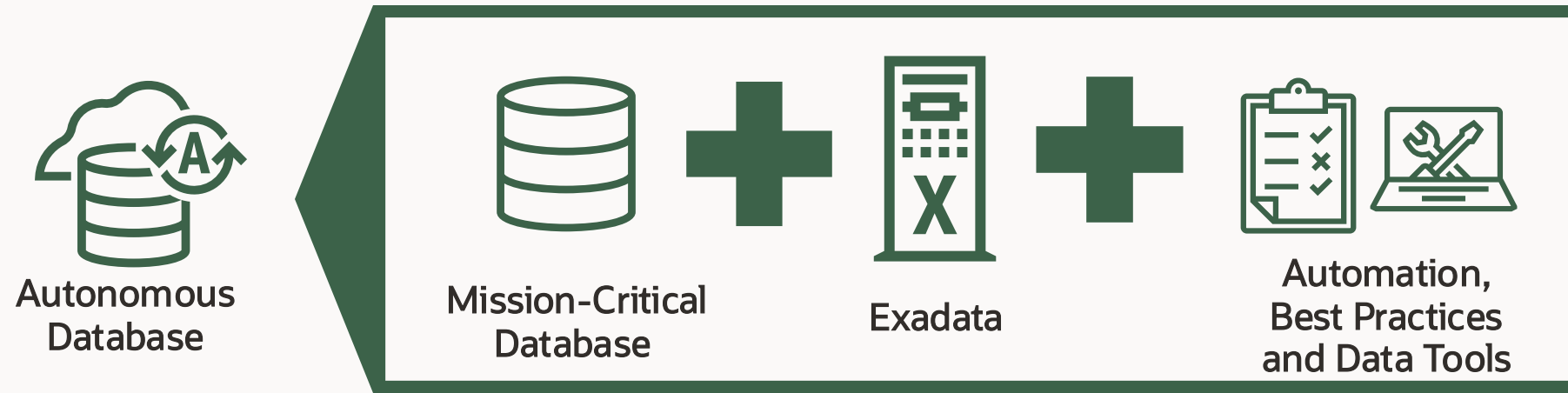


Fully managed zero data loss recovery service with fast **incremental forever backups**, continuous **recovery validation** and data **anomaly detection**

Autonomous AI Database (Serverless)

Autonomous Database Delivers the Best Data Experience

Fully-managed, cloud native, lowest TCO

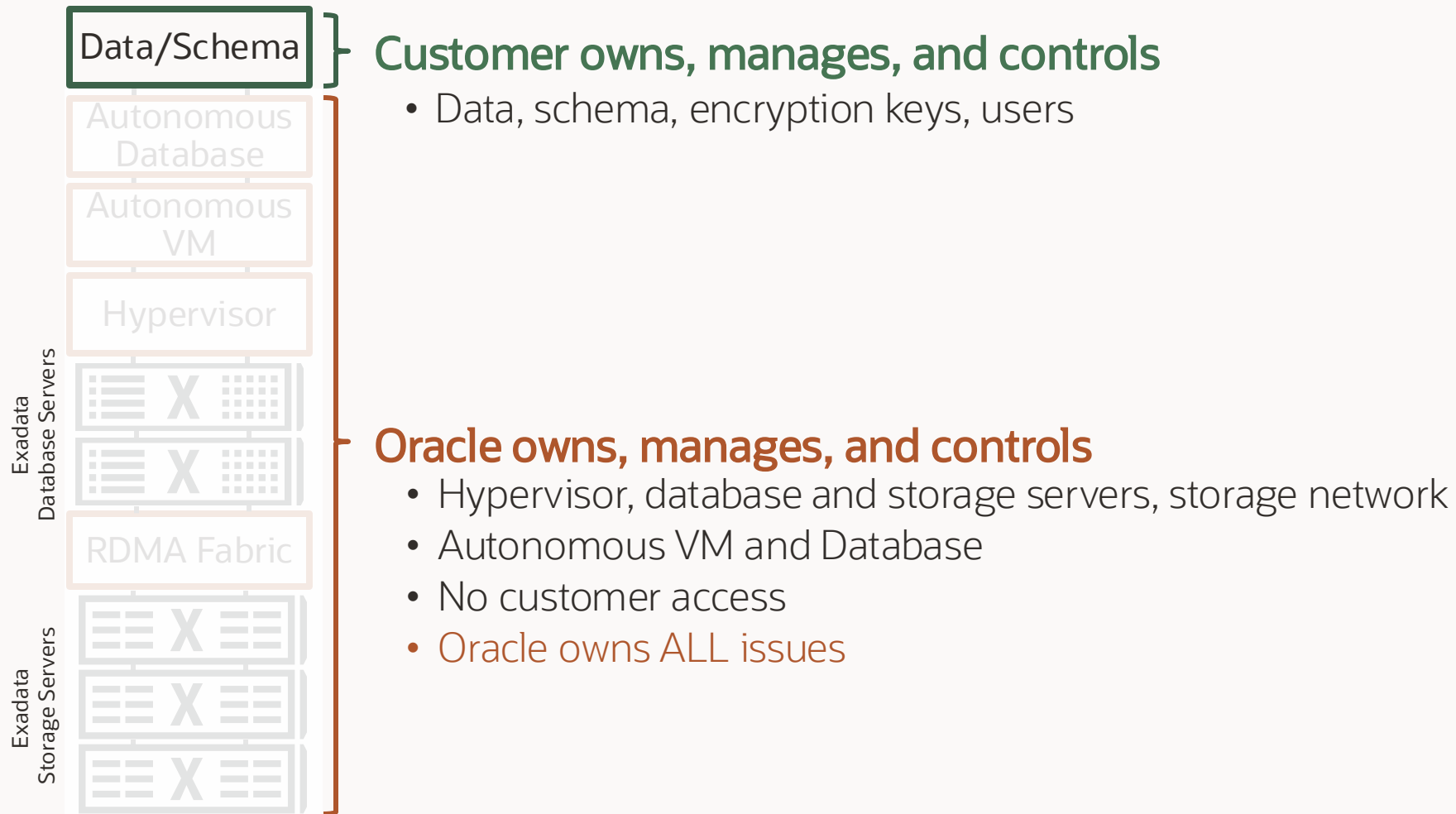


Automatically secures, tunes, and scales

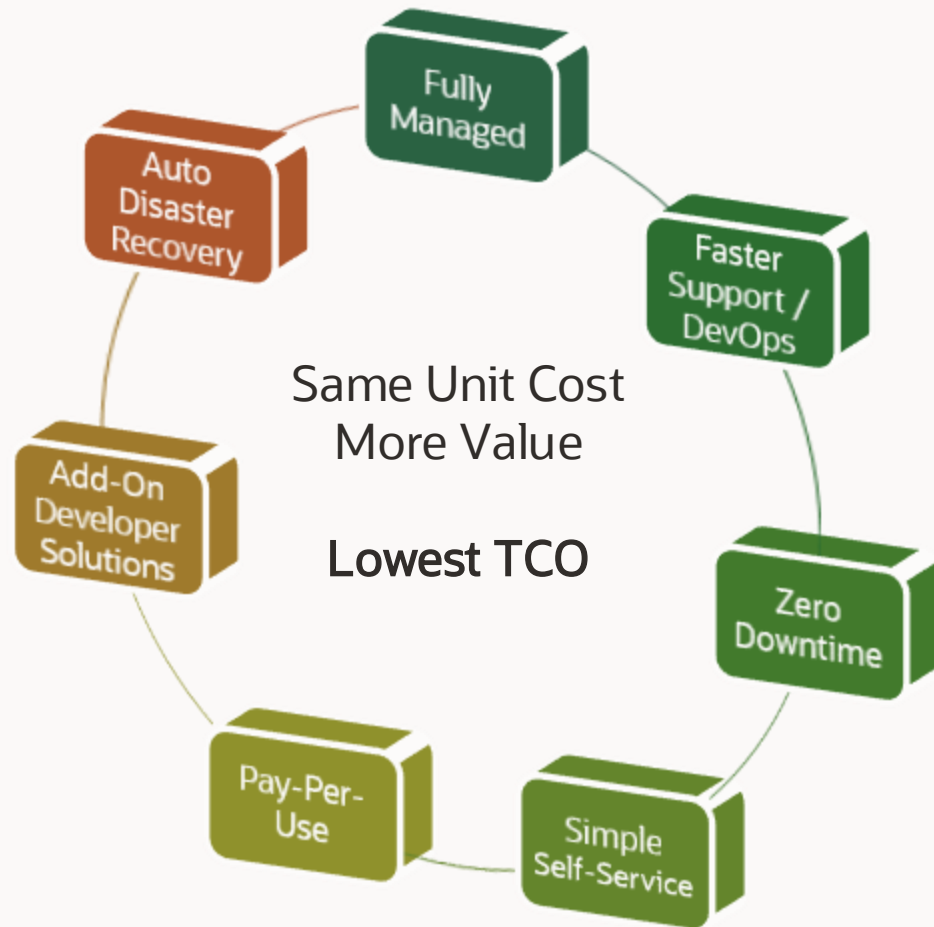
- ✓ Eliminates many manual tasks
- ✓ Reduces human error
- ✓ Simplifies scaling
- ✓ Minimizes downtime
- ✓ Aligns costs with demand

Runs on top of **Real Application Clusters** ♦ Up to **99.995% SLA** ♦ **Zero** regression SLO

Autonomous AI Database management model



What can Autonomous AI Database do for you



Operational Delegation

- **Fully Managed** Database Operations: backup, update, upgrade, OS maintenance, incident management, health monitoring
- **Faster support**, less customer involvement via DevOps incident response and monitoring of health metrics and logs
- Built-in Application Continuity for **zero downtime** via application aware, workload specific and prioritized connection services

Improved Automation

- Zero upfront cost for simple database **self-service** setups, with auto workload configuration and capacity management
- Auto-scaling with **pay-per-use** billing, not on allocation
- Managed **developer add-on's**: APEX, ORDS, DB Actions and more
- MAA disaster recovery setup with FSFO **automated failover**

Autonomous AI Database is workload-optimized



Autonomous AI Lakehouse

Pre-configured for columnar format, partitioning, and large joins to accelerate **analytics, data warehouse, and data lake**



Autonomous AI Transaction Processing

Pre-configured for row format, indexes, and data caching to accelerate **transaction processing and mixed workloads**

Autonomous innovation, database automation for day-to-day operational tasks



Automatic Tuning

Auto-Indexing, Auto-SPM, Auto-Partitioning tunes database performance for the developer



Automatic Provisioning

Automatic deployment of new database architectures without additional DBA work – Support Dev/Ops cycles



Automatic Scaling

Up to 3X automatic, elastic scaling helps you avoid overprovisioning resources.



Automatic Encryption

Native, built-in data protection that meets organizational security requirements without code complexity.



Automatic Updating

Automatic online rolling patching and updating with configurable scheduling.



Automatic Configuration

Built-in best practices for specific workloads and native application connection services for HA, parallelism, request prioritization.

Autonomous AI Database subscription options

Simple price structure: Pay As You Go or Universal Credits

Also available:

- **Bring-your-own-license (BYOL):** Use existing on-premise Oracle Database licenses and save 75%
- **Mix-and-match:** Combine BYOL and license-included to ensure that BYOL limits are not exceeded

Pay per Database

- Starts at 2 ECPUs
- Each database is billed independently
- Available as subscription or BYOL
- BYOL includes Transparent Data Encryption, Data Safe, Partitioning, Advanced Compression, and select management packs at **no additional cost**

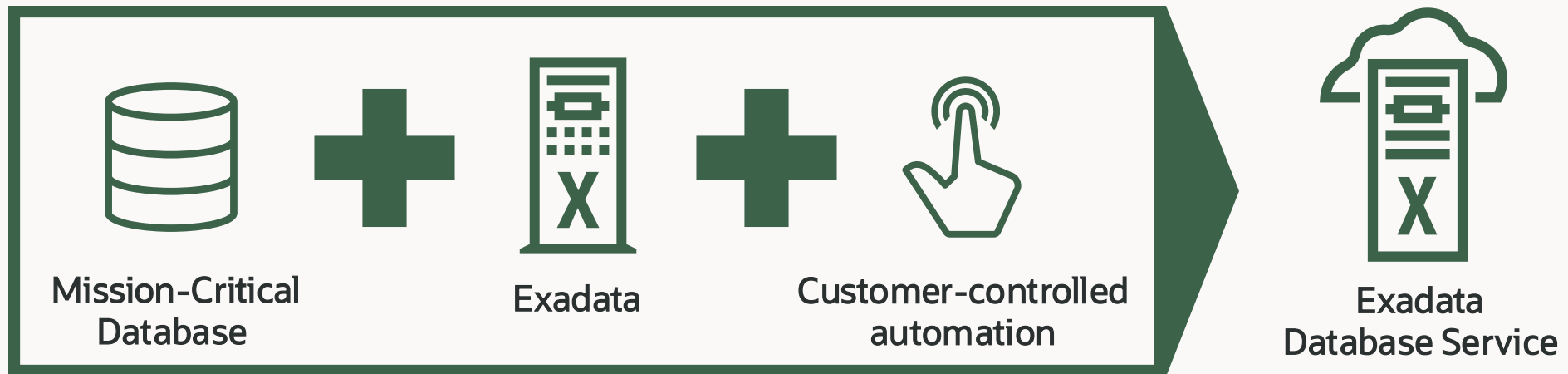
Pay per Elastic Pool

- Starts at 128 ECPUs
- Aggregate dozens of hundreds of databases into a single bill
- **Up to 4-8x cost savings** vs per-database pricing
- No changes to database functionality, scalability or performance
- Available as license-included or BYOL

Exadata Database Service Dedicated Infrastructure (ExaDB-D) and Exascale Infrastructure (ExaDB-XS)

Exadata Database Service

Customer-managed with 100% Oracle Database compatibility and unique Exadata capabilities



- ✓ Intelligent data architecture with unique software optimizations
- ✓ Extreme performance and availability for all workloads
- ✓ Oracle-managed infrastructure
- ✓ All Oracle Database functionality and options

Run mission-critical Oracle Database workloads at any scale with full administrative control

Exadata Cloud Infrastructure

All the power and functionality of Oracle Database and Exadata in the public cloud

Exadata in the public cloud

- Same architecture as on-premises Exadata
- Infrastructure managed by Oracle experts
- Deployed in chosen public cloud (OCI) or multicloud region

Consistent public cloud experience

- Public cloud UI/API-driven system and database management
- Flexible financial model with pay-per-use database cores
- Infrastructure subscription for as low as 48-hour minimum

Efficient cloud adoption strategy

- Meet or even exceed on-premises service levels in the public cloud
- Multicloud extends Exadata benefits to applications in other clouds
- Dedicated Infrastructure and shared Exascale Infrastructure

Exceptional Outcomes

- Performance
- Availability
- Security
- Consolidation



Elastic Scaling

- Add CPU and Storage Online
- Any size workload or database

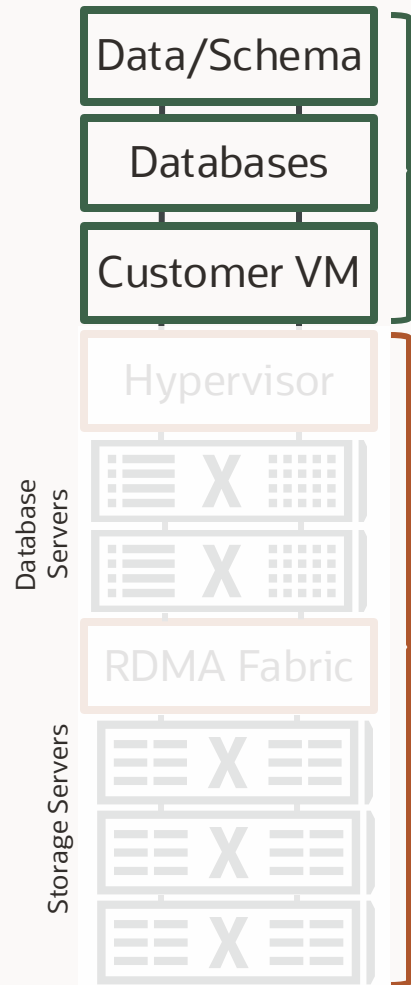


Highest Value

- Best Price/Performance
- Pay-per-use Pricing
- One DB Service vs Many



Simple management model in public cloud



Customer owns everything inside database

- Data, schema, encryption keys

Customer subscribes to database services

- Customer manages VMs and Databases using Cloud Automation (UI / APIs)
- Automation to create, delete, patch, backup, scale up/down, etc.
- Runs all supported Oracle Database versions
- Customer controls access to customer VM
- Customer can install and manage additional software in customer VM
- Oracle staff are not authorized to access customer VM

Oracle owns and manages infrastructure

- Hypervisor, database and storage servers, storage network
- Patching, security scans, security updates
- Monitoring and maintenance
- Customer not authorized to access Oracle infrastructure

Exadata Cloud Infrastructure architecture

Exadata components in the cloud

- Scale-out database compute and storage connected by secure RDMA fabric
- Infrastructure managed by Oracle

Cloud controls and networking

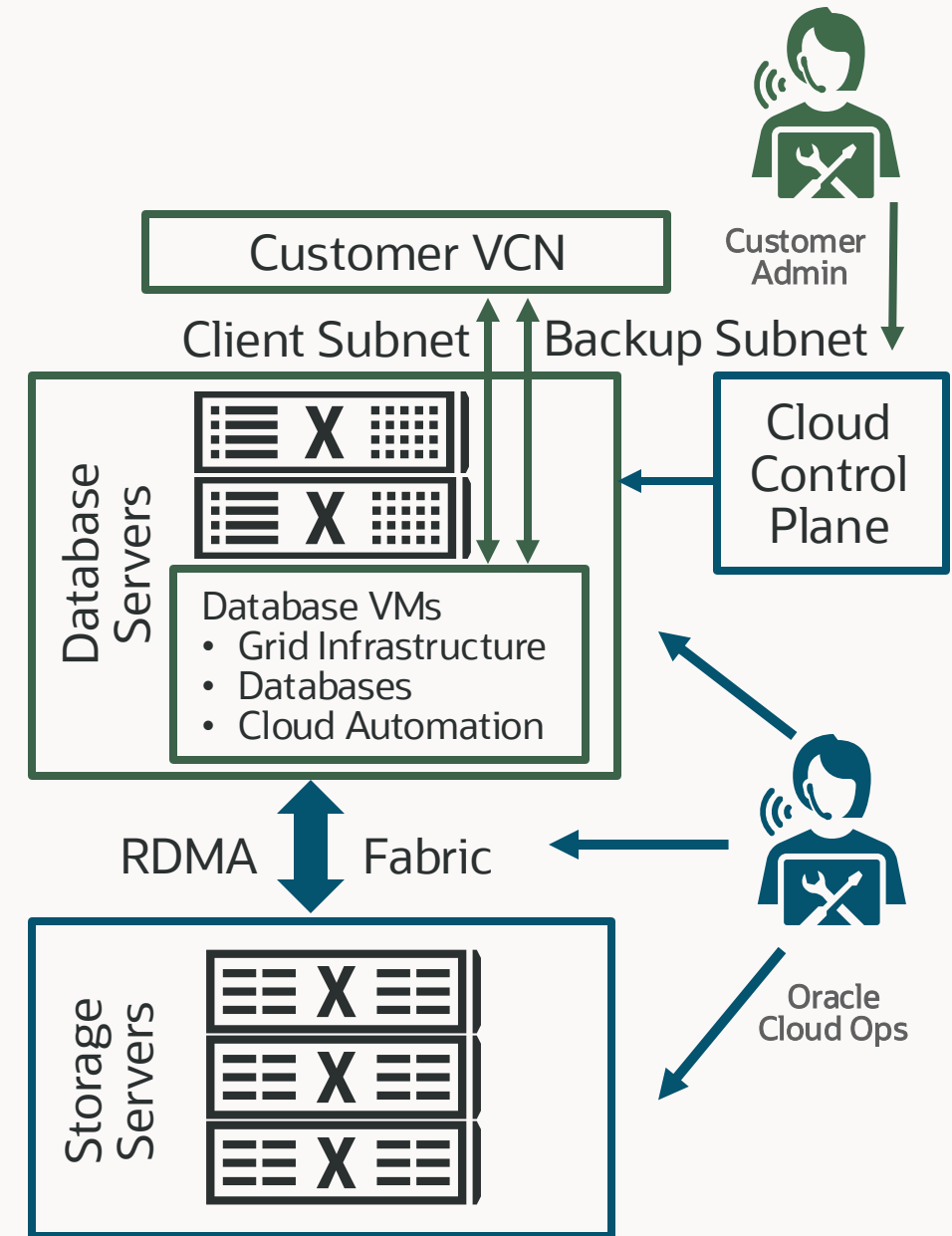
- Identity management and policies to control access
- Virtual Cloud Network (VCN)
- Compartments to organize cloud resources

Secure isolation

- Databases run in secure virtual machines isolated from underlying infrastructure

Control plane

- Controls and manages Exadata infrastructure



Choice of infrastructure for Exadata Database Service in the public cloud

Dedicated Infrastructure



- Complete server and **tenant isolation** with server hardware dedicated to a single customer
- Start with 2 database and 3 storage servers and scale out for the largest workloads, including **extreme-scale** transaction processing and multi-petabyte data warehouses
- Cost-effective **consolidation** by deploying many AI, Analytic, and OLTP databases

Exascale Infrastructure



- **Low entry cost** starting with a single VM cluster with 8 ECPU's and 300 GB of Exadata storage
- Pay only for the compute and storage required, and **granularly scale** without dedicated infrastructure from a pool of shared resources
- All unique **Exadata** performance, reliability, availability and security capabilities built-in

Fully-elastic scale-out dedicated infrastructure configurations

Compute and storage servers in fully-elastic configurations

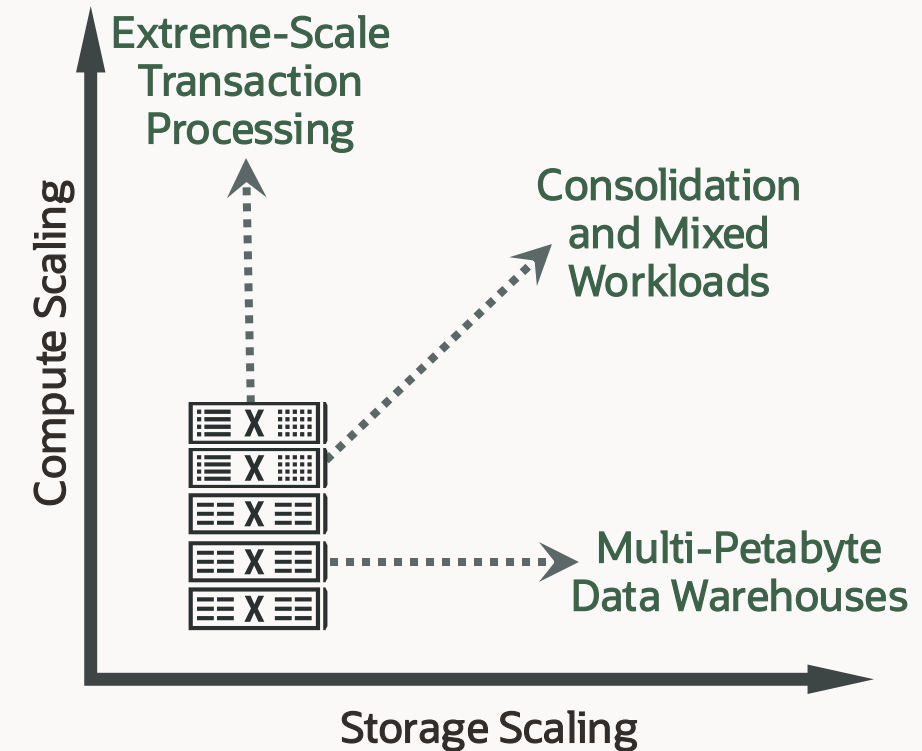
- Start with 2 **compute** and 3 **storage** servers
- Scale-out to 32 **compute** and 64 **storage** servers
- Choose quantity of each type of server based on workloads
- Complete server and tenant isolation with server hardware dedicated to a single customer

Scale-out X11M compute servers for database processing

- 190 AMD EPYC Processor cores (760 ECPUs)
- 1,390 GB Memory

Scale-out X11M intelligent storage servers for database offload

- 64 AMD EPYC Processor cores
- 1.25 TB Exadata RDMA Memory (XRMEM)
- 27.2 TB NVMe Flash
- 80 TB HDD usable storage



Performance scales as more servers are added

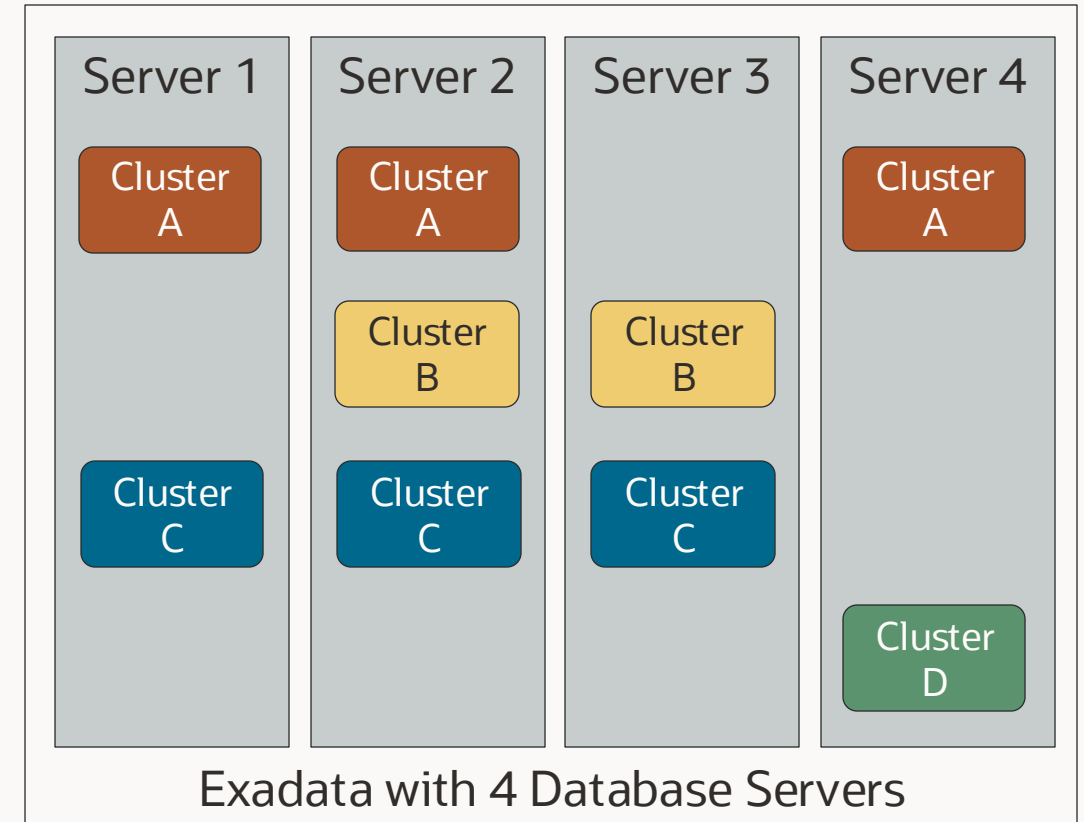
Consolidate databases with Multiple VM Clusters on dedicated infrastructure

Benefits of Multiple VM Clusters

- More efficient utilization of infrastructure
- Isolate databases on the same infrastructure
- Pay only for database licensing on a subset of servers

Multiple VM Clusters Provisioning Options

- Provision multiple VM Clusters per infrastructure
- Provision VM Clusters on a subset of database servers
- Expand and shrink VM Clusters on demand



Choice of infrastructure for Exadata Database Service in the public cloud

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Exascale Infrastructure



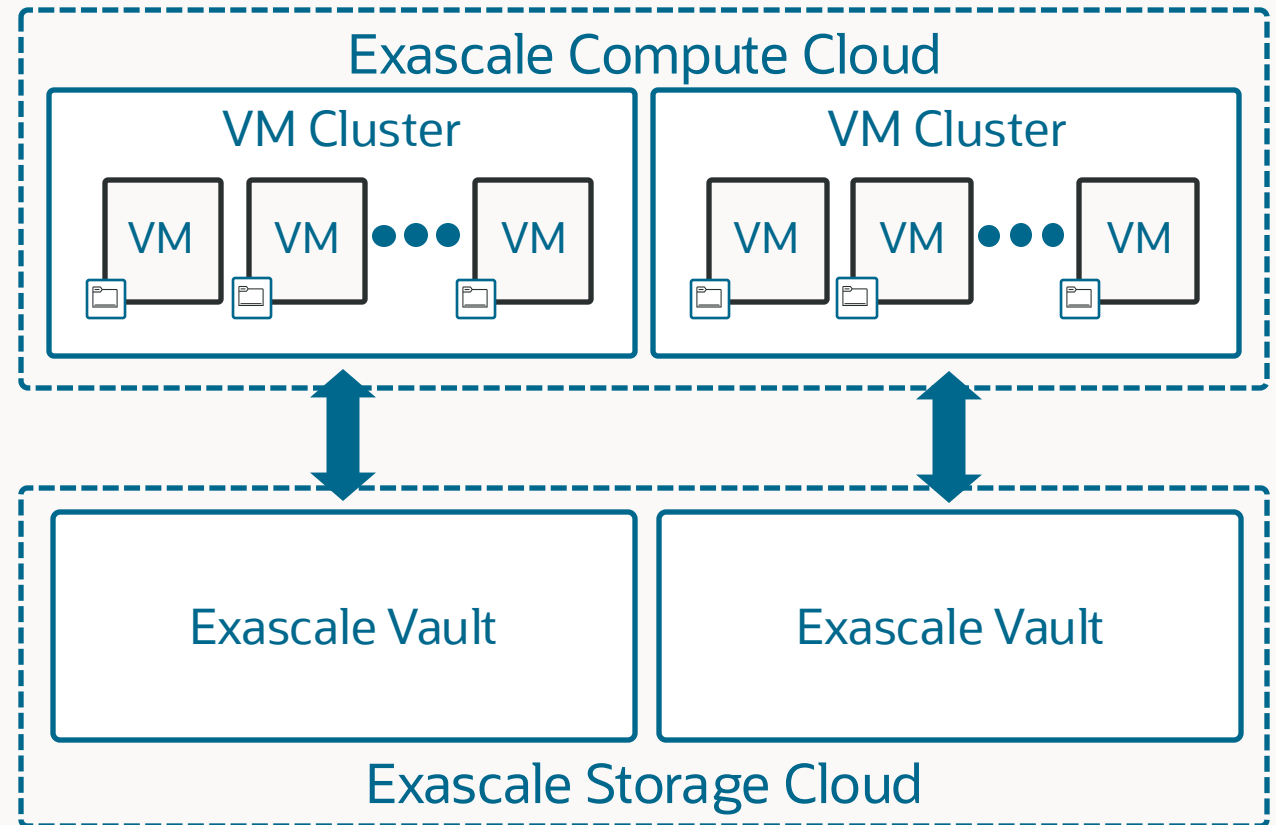
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Building blocks for Exadata Database Service on Exascale Infrastructure

Exascale compute and storage is deployed on powerful, data-optimized shared Exadata infrastructure

Users create VM Clusters across physical database servers with VM Storage attached from Oracle managed shared storage

Database files are stored in logical Storage Vaults and are triple mirrored for redundancy across the Exascale Storage Cloud



Next-generation Exascale technology on shared Exadata Infrastructure

Lower entry cost with no dedicated infrastructure by provisioning only a VM cluster and database storage

Granular elastic and independent scaling of compute and storage resources

Pay only for the resources required from a pool of shared Exascale infrastructure



Minimum Size

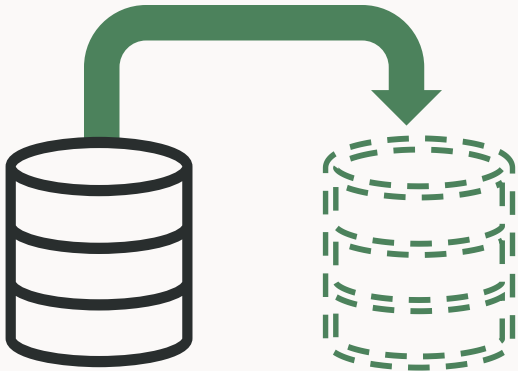
Single VM Cluster
8 Total ECPUs per VM
300 GB Exascale Vault database
storage per VM Cluster



Maximum Size

10-Node VM Cluster
200 Total ECPUs per VM
100 TB Exascale Vault
database storage per VM
Cluster

Database-aware intelligent clones on Exascale



Instantly create PDB clones for Development or Test

- Either a full copy or thin clones
- Clone any read/write or read-only PDB
- Clone across CDBs and VM clusters

Clones leverage Exascale **redirect-on-write** technology

- Clone shares block with parent until they make a change
- Drastically reduces storage capacity needs for cloning

Get native Exadata **performance** on development, test, or recovery copies

Empower developers by creating **thousands** of PDB clones for dev teams

Scale Compute Online with Reserved ECPUs

Enabled ECPUs - in use and active

Reserved ECPUs - guaranteed to be available although not in use

Total ECPUs - sum of enabled and reserved ECPUs

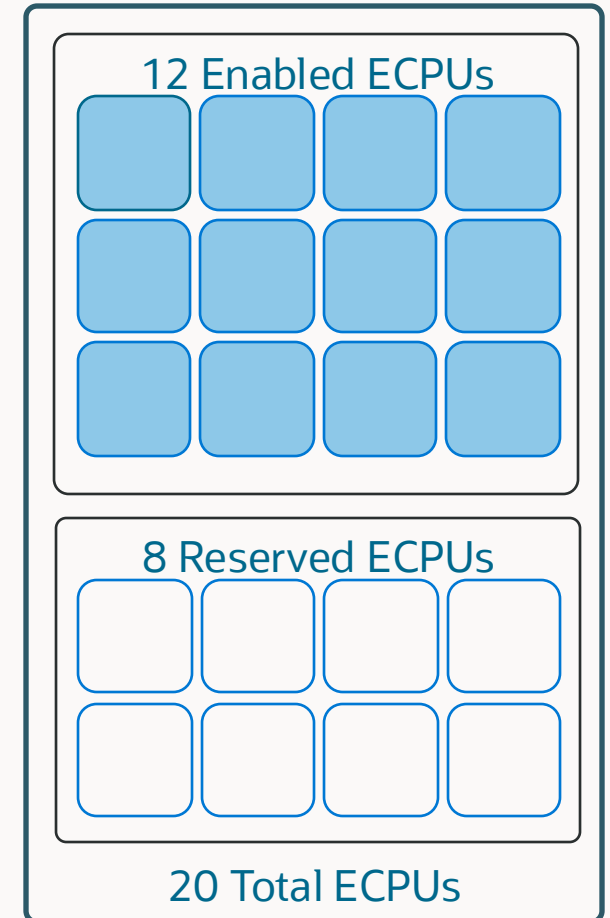
Scale enabled ECPUs online without reboot by enabling reserved ECPUs

Scaling total ECPUs requires a rolling reboot

Total usable memory = total ECPUs * 2.75GB memory per ECPU

Increase total usable memory available to VM by scaling total ECPUs

Licensing costs only apply to enabled ECPUs, Infrastructure costs apply to total ECPUs that include enabled and reserved ECPUs



Exadata Database Service subscription options

1. License Included

Ideal for organizations seeking to build new applications or expand functionality

Includes Exadata System Software, Oracle Database, All Options and Management Packs
at one low price

Pay-per-use pricing for ECPUs, all paid using Universal Credits



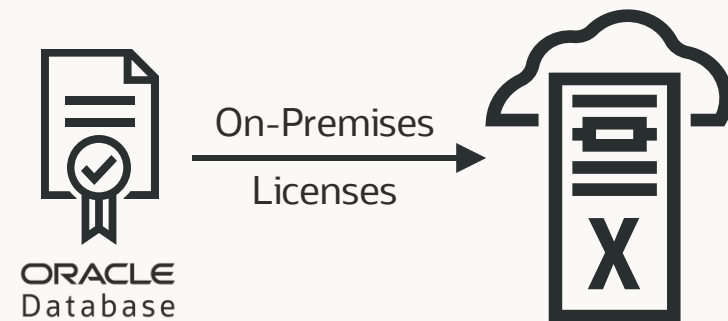
Oracle Database
Enterprise Edition
plus All Options

2. Bring Your Own License

Ideal for organizations moving existing workloads

Utilize **existing on-premises licenses** to save on existing software support

Includes Exadata System Software, Exadata Columnar Cache, TDE, Data Safe, and select database management packs
at no additional cost



Base Database Service

(BaseDB)

Base Database Service

Co-managed with 100% Oracle Database compatibility on Oracle hardware

Full-featured Oracle Database cloud instance

- Flexible single VM on shared hardware generation-agnostic Standard x86 compute with block storage
- Support for Oracle Database 19c and 26ai

Complete customer administrative control

- Customer-controlled automation for create, update, backup, scale, disaster recovery, and more
- Simple deployment in minutes

Cloud economics

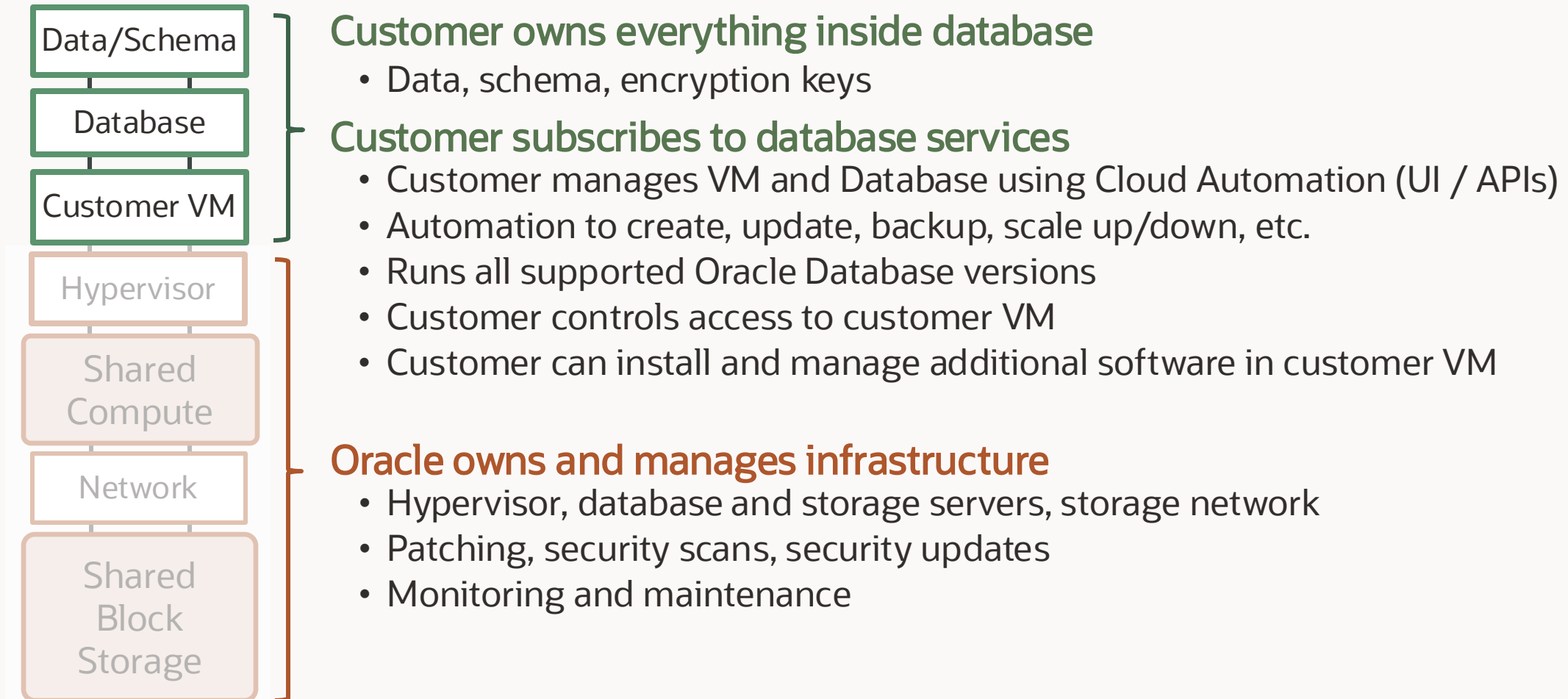
- Low cost with pay-as-you-go pricing and independent compute and storage scaling
- Choice of Oracle Database Standard and Enterprise Edition License-included options or Bring Your Own License (BYOL)

Shape	VM.Standard.x86
Database	Single-instance
ECPU	4 - 256
Memory	8 – 512 GB
Data Storage	256 GB – 40 TB
License Included Editions	Standard, Enterprise, High Performance

* Oracle RAC and Active Data Guard not available on Base Database service on Oracle Database@Google Cloud



Base Database Service management model



Base Database Service subscription options

Tiers of license-included or bring your own license (BYOL) to meet specific application requirements

Standard

Standard Edition 2 Database features plus

- Tablespace Encryption
- Machine Learning
- Spatial and Graph
- Up to 3 PDBs

Enterprise

Enterprise Edition Database features plus

- Data Guard
- Real Application Testing
- Data Masking and Subsetting Pack
- Tuning Pack and Diagnostic Packs
- True Cache

Enterprise High Performance

Enterprise Database Service features plus

- Partitioning
- Advanced Compression
- Advanced and Label Security, DB Vault
- Multitenant
- OLAP
- Lifecycle and Cloud Management Packs
- SQL Firewall

Bring Your Own License (BYOL) includes Transparent Data Encryption (TDE) and Data Safe at no additional cost

Zero Data Loss Autonomous Recovery Service (ZRCV)

Intelligent data protection for Oracle databases

Zero data loss recovery

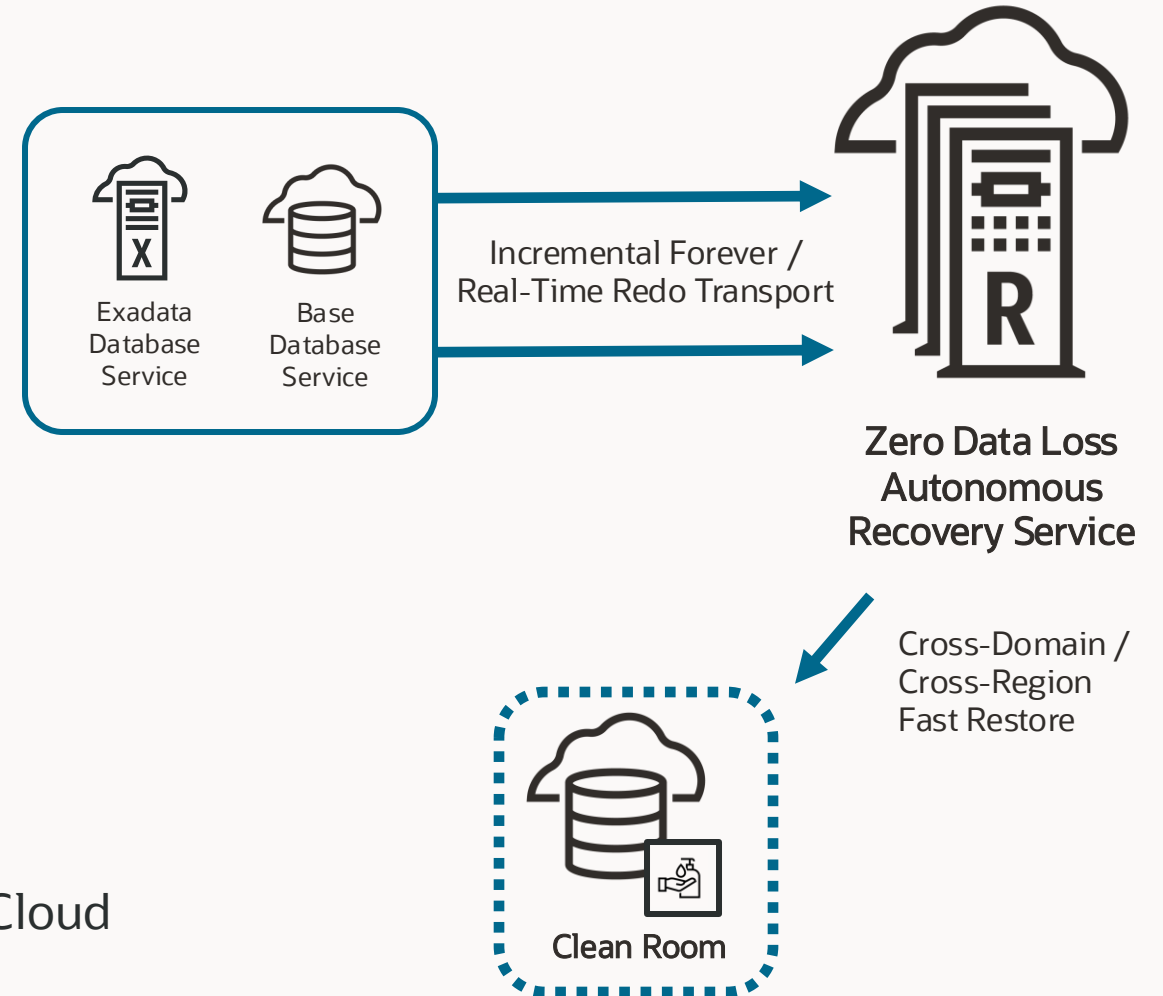
- Real-time transaction protection
- Continuous recovery validation

Fast, low-impact backup and recovery

- Space-efficient incremental forever
- One-step recovery without needing incremental apply

Database-aware ransomware resiliency

- Immutable backups with retention lock
- Separation of duties and granular access control
- Logical airgap between the customer and Oracle Cloud operations tenancies
- Cross-domain and cross-region clean room support

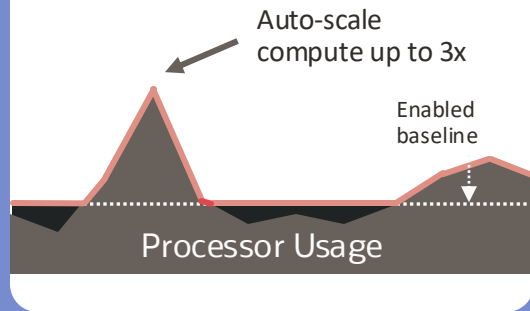


Reduce Costs With Elastic Resource Scaling

Align compute consumption with workload demands

Autonomous AI Database

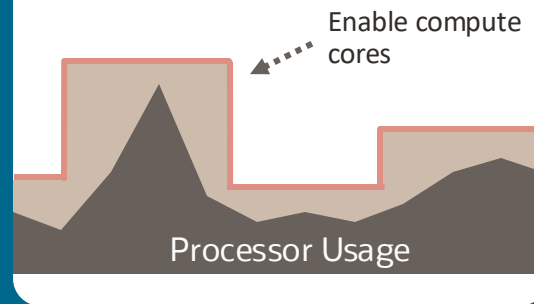
Real-time scaling of consumption and costs based on workload demands



Pay only for CPU resources used with built-in automatic online scaling

Exadata Database Service

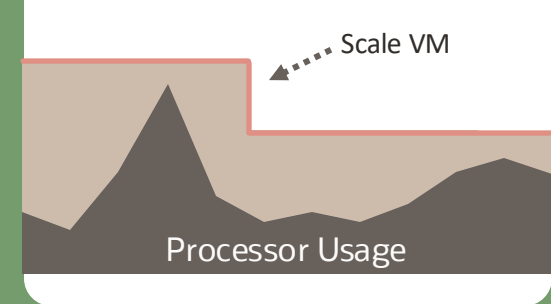
Scale enabled CPU resources based on workload demands



Pay for enabled CPU resources with dynamic online scaling

Base Database Service

Change the VM shape to meet workload demands



Pay for the VM shape and scale with a reboot

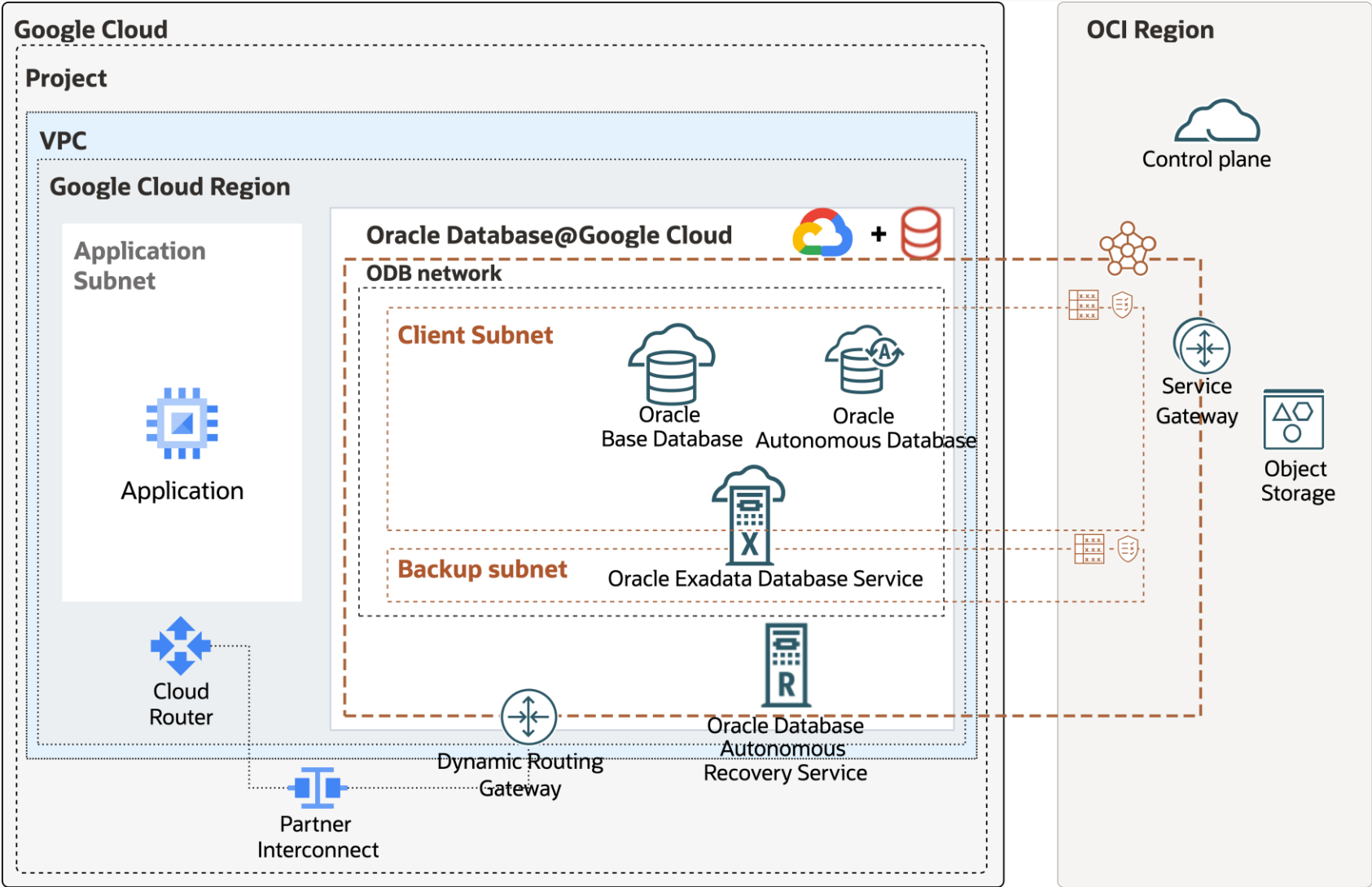
Which Oracle Database Service To Choose

Service	Autonomous AI Database Serverless	Exadata Database Service on Dedicated Infrastructure	Exadata Database Service on Exascale Infrastructure	Base Database Service
Characteristics	- Fully-managed, cloud-native - Exadata performance and availability for mission critical workloads - Elastic auto-scaling - Oracle RAC	- Co-managed - Exadata performance and availability for mission critical workloads - Tenant isolation - Database consolidation - Oracle RAC or single-instance	- Co-managed - Lower entry cost and granular scalability for mission critical workloads requiring Exadata performance and availability - Thin database clones - Oracle RAC or single-instance	- Co-managed - Standard x86 compute with block storage - Single-instance
Oracle DB Version	19c and 26ai	19c and 26ai	19c and 26ai	19c and 26ai
Licensing Options	License Included with all Enterprise Edition database options; BYOL	License Included with all Enterprise Edition database options; BYOL	License Included with all Enterprise Edition database options; BYOL	License Included with Standard and Enterprise Edition with most database options; BYOL

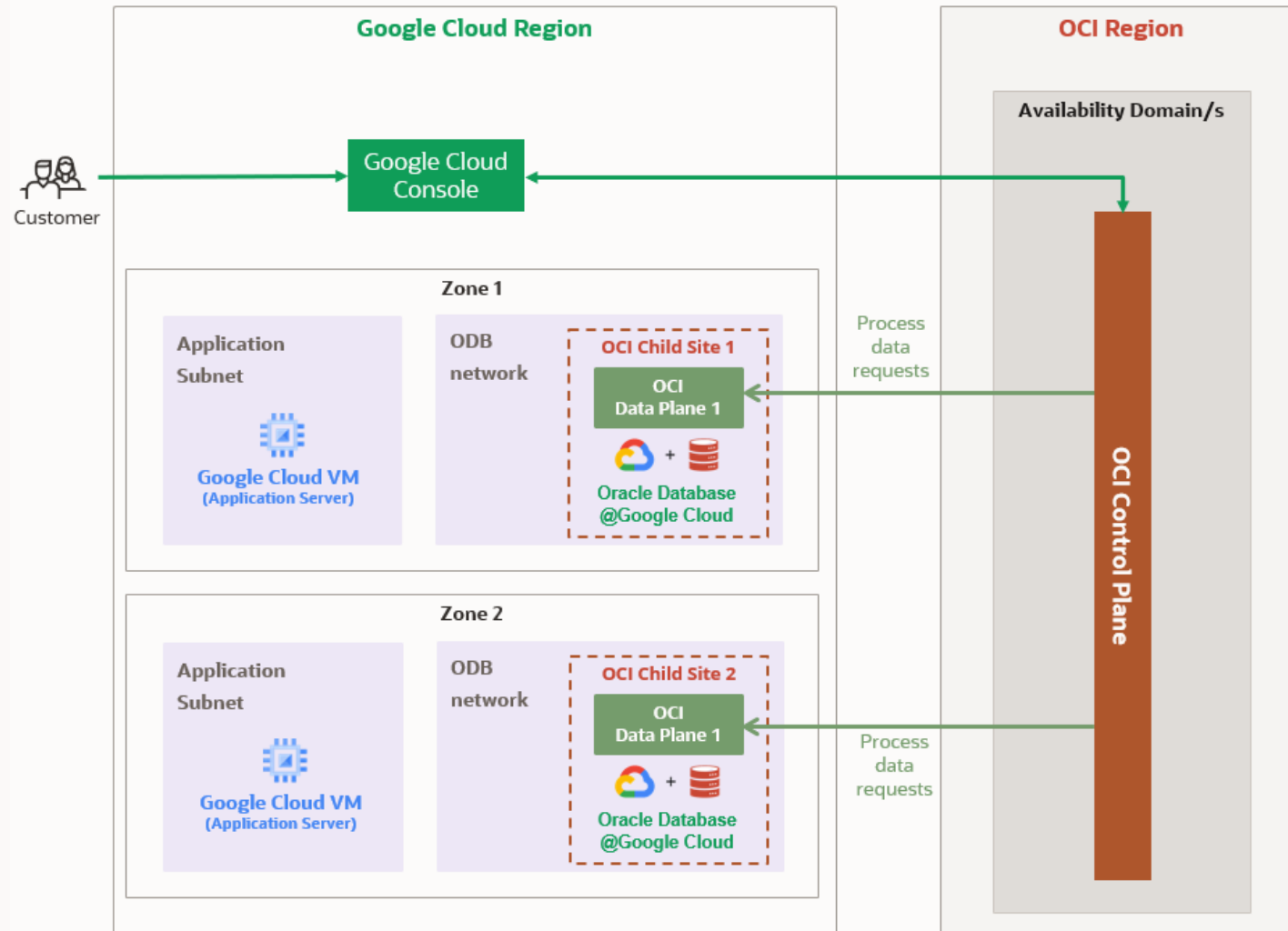


Oracle Database@Google Cloud Architecture

Architecture detailed



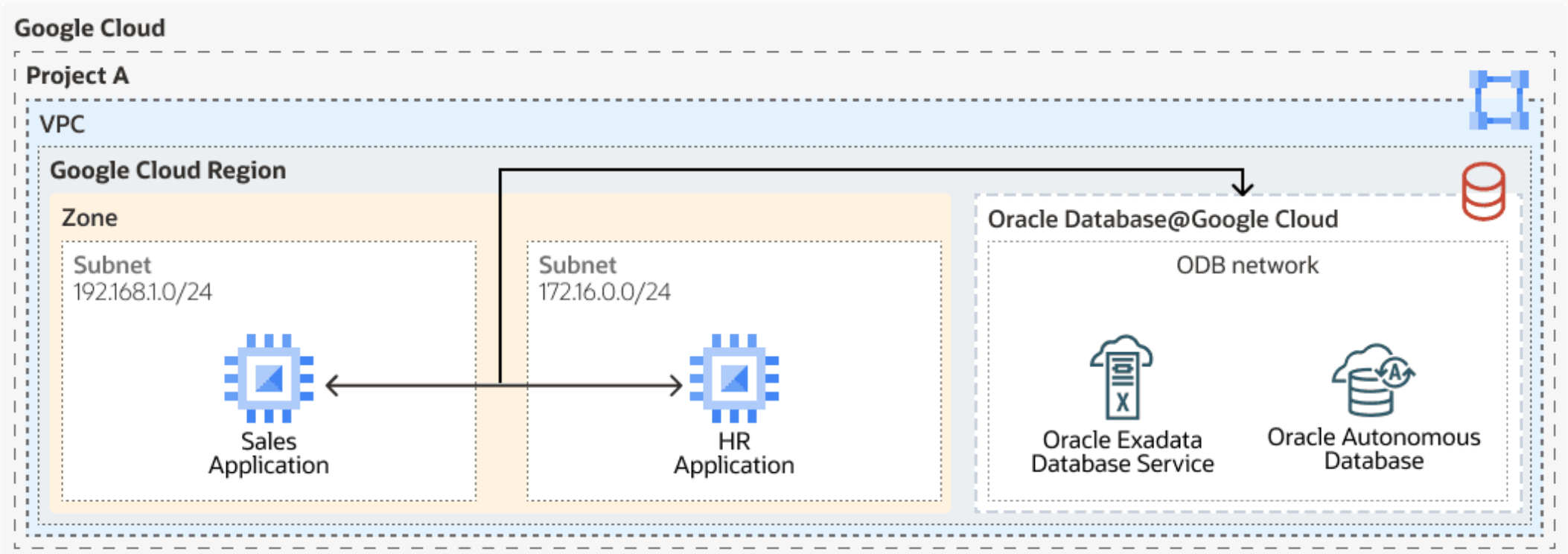
Oracle Database@Google Cloud control plane architecture



Network Topologies

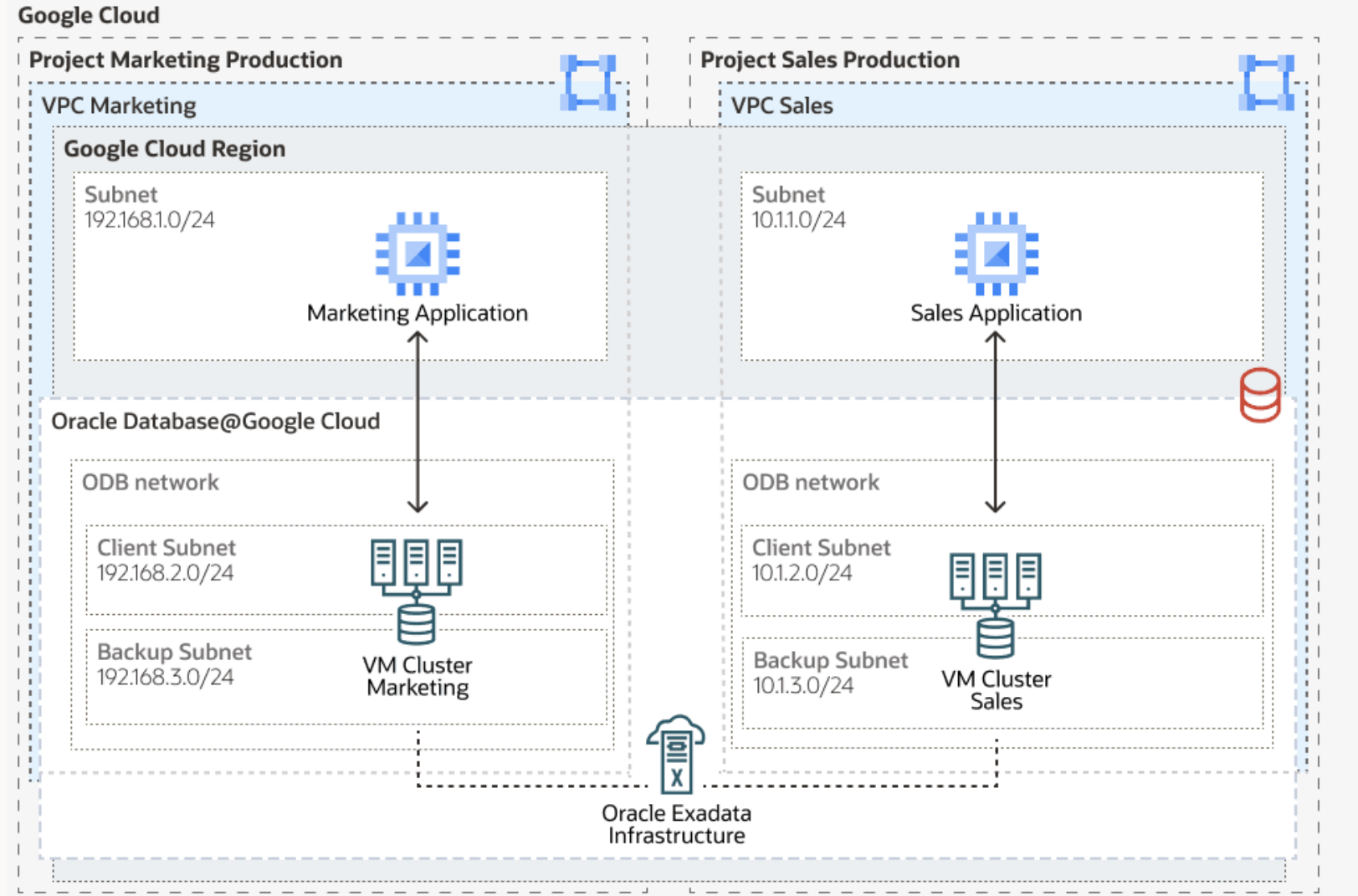
Single VPC – Multiple Subnets

- Oracle Database subnets and application subnets are part of the same Google Cloud Region, VPC, Project
- Subnets in the same VPC have access to the database service
- Firewall rules are defined at the OCI Network Security Group (NSG)



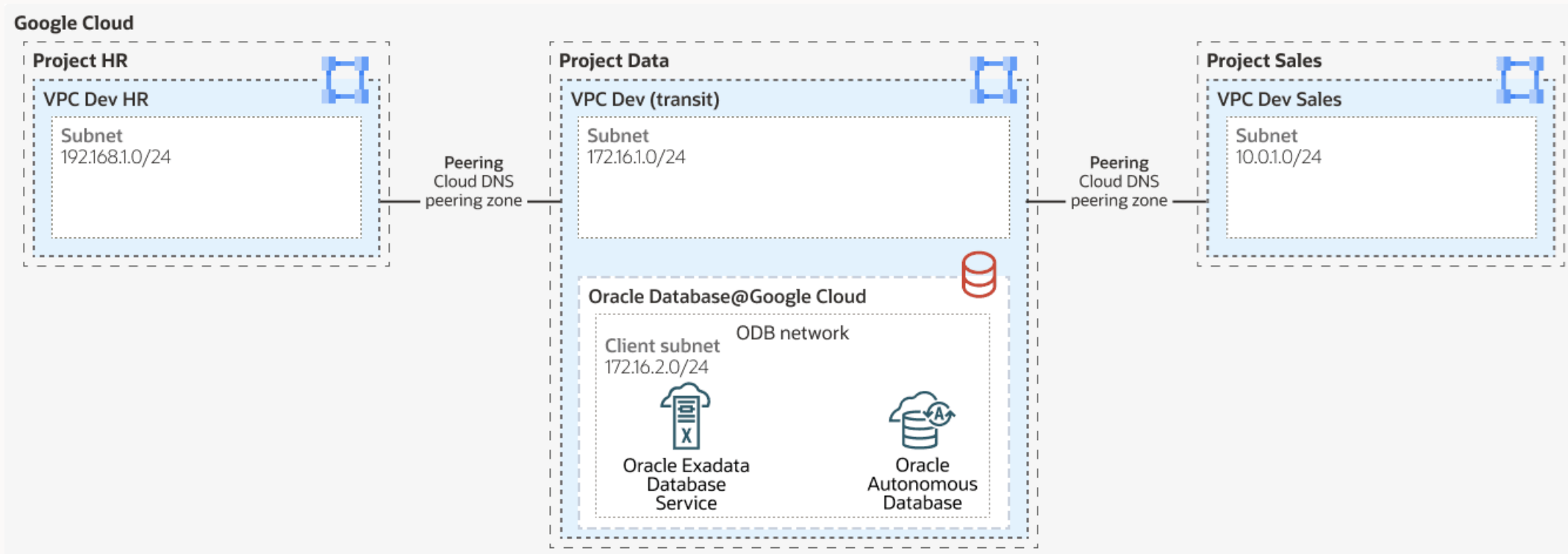
Multiple VPCs

- Multiple VM Clusters share the same Exadata Infrastructure
- Each VM Cluster is connected to a different VPC
- Exadata Infrastructure and VM Cluster are part of the same project



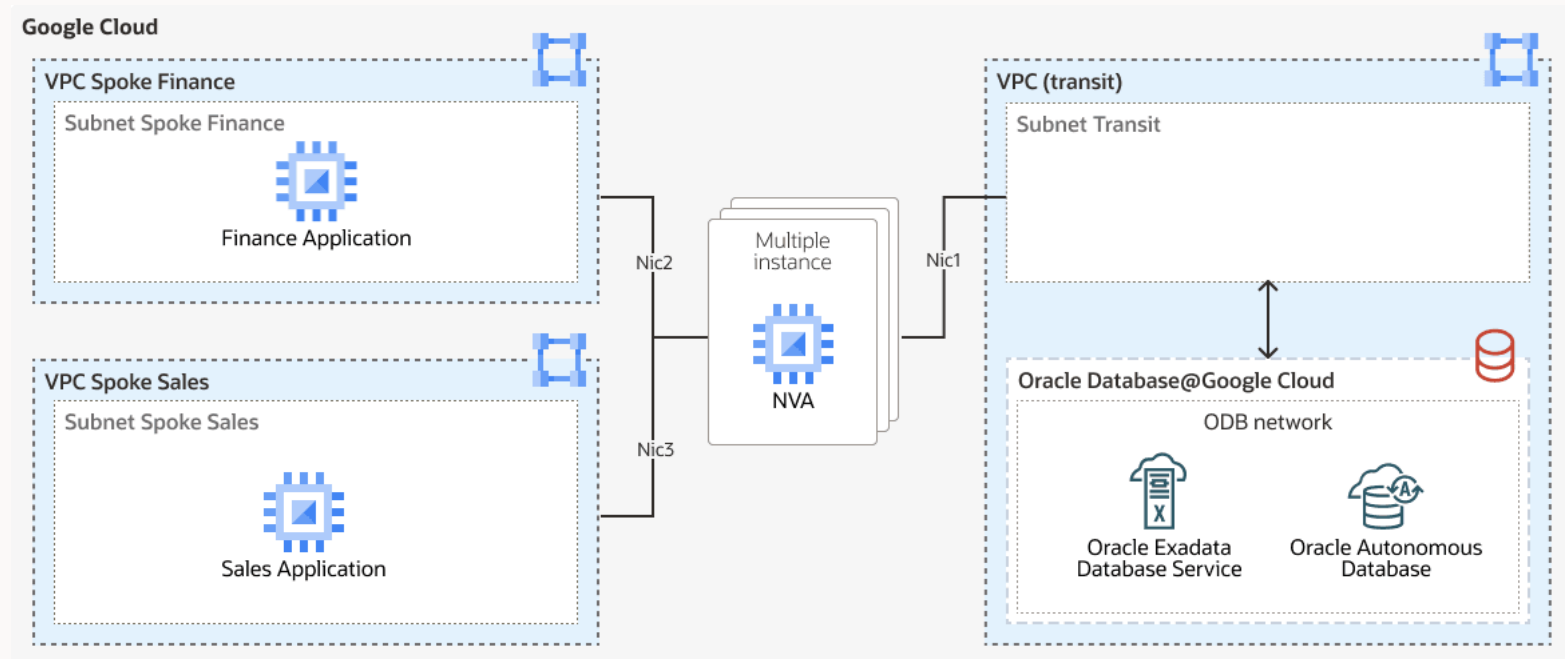
VPC Peering

- Oracle Database@Google Cloud created in the VPC transit.
- The DB is accessible from multiple VPCs by peering.
- VPC transit is peered with other VPCs of the same or different projects and or organizations.



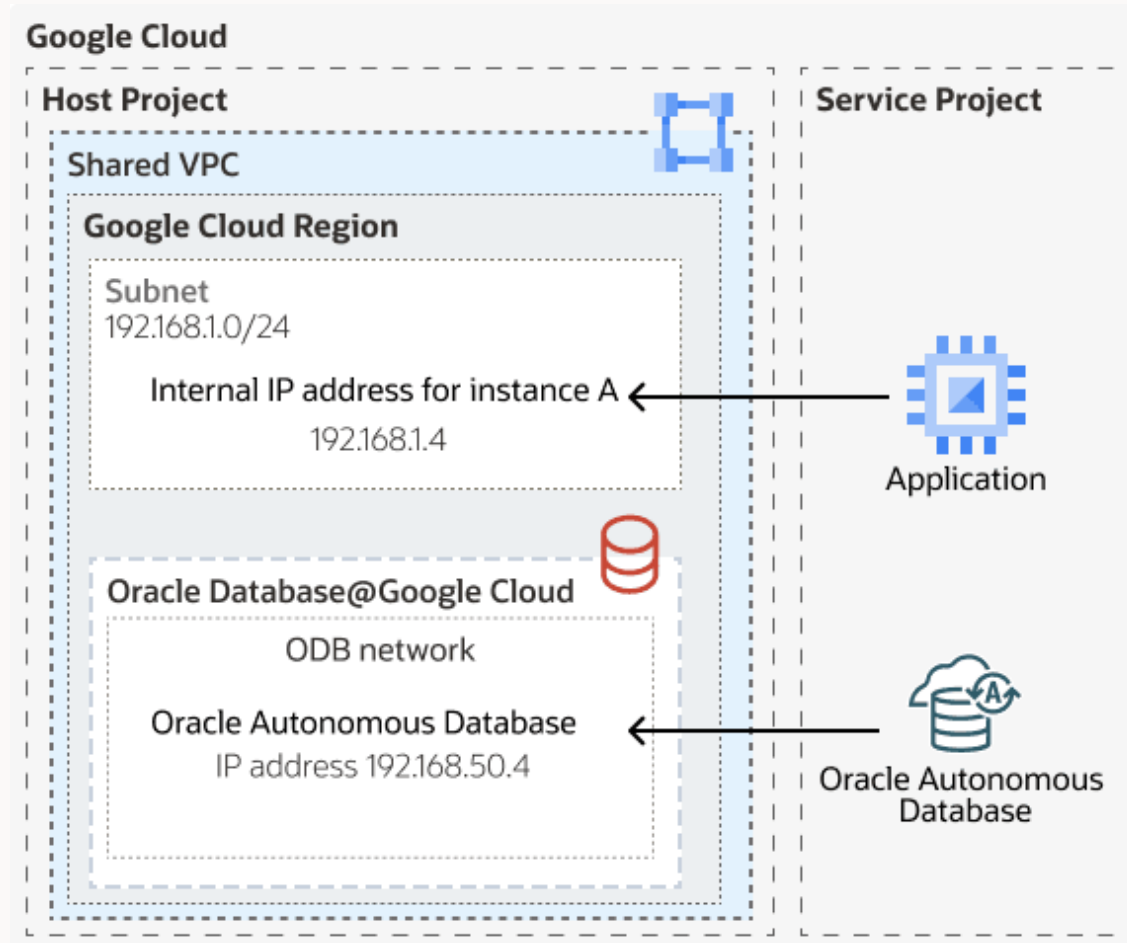
NVA Hub and Spoke

- Hub NVA has multiple VNICs in each spoke subnet.
- Hub NVA supports routing.
- VPC transit is connected to Oracle
- Oracle Database@Google Cloud, as a spoke to the NVA by the Partner Interconnect.

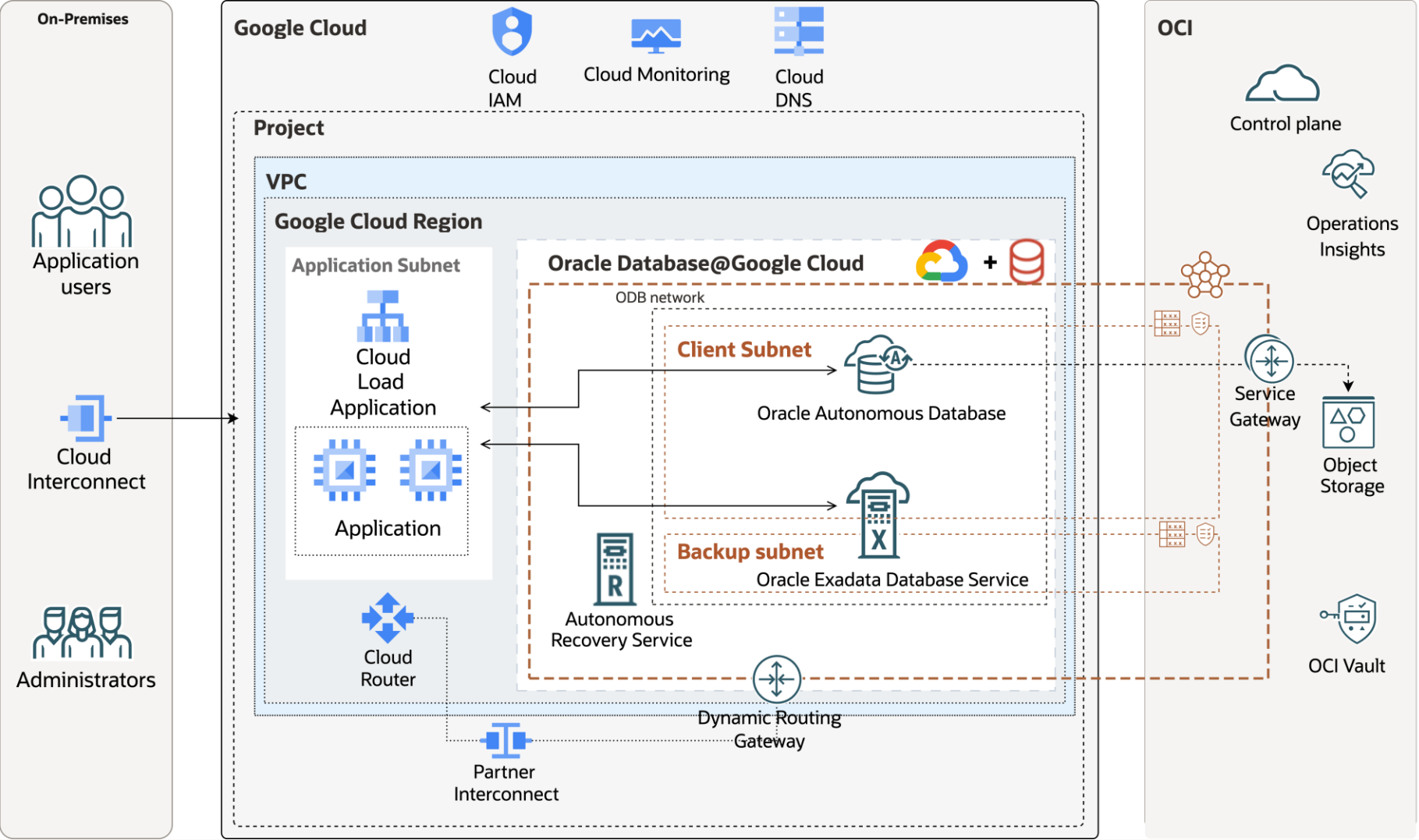


Shared VPC – Oracle Database Network

- VPC is created in the host project shared with services projects by network admins.
- Oracle Database instances are deployed in the services projects.
- Applications instances are connected to Oracle Databases instances in the shared-vpc.



Architecture blueprint



Network Requirements

ODB network

ODB networks provide secure and private connectivity to your Oracle Database resources, giving you control over how they connect and communicate.

- Define the VPC
 - Define client and backup subnet CIDR
 - Subnets must not overlap within your VPC
- Network administrators create the ODB network
- Database administrator uses the defined ODB network

Create ODB Network

- ODB network creates a metadata construct for the Google Cloud VPC: standalone or shared-vpc
- Create up to 5 non-overlapping subnet in one ODB network
- IP address ranges allocated to database services must not overlap with other CIDRs
- CIDR for Client and Backup subnet must be part of the RFC1918 range - 10.0.0.0/8, 172.16.0.0/12 or 192.168.0.0/16.
 - Take DR subnets (primary and standby), on-premises routing into consideration when configuring CIDRs
 - For Exadata X9M, and X11M IP addresses 100.64.0.0/10 are reserved for the cluster interconnect by OCI automation and can't be allocated to client or backup networks.
- IAM roles can be applied to dissociate network administrator and database administrators' roles

Client Subnet Requirements

The minimum CIDR size for the client subnet is /27, and the maximum size is /22.

Number of IP addresses	Consumed by	Summary
3	Oracle Database@Google Cloud Network service	These IP addresses are reserved regardless of how many VM clusters you provision in the Client Subnet. Oracle Database@Google Cloud consumes the following: • 2 IP addresses at the beginning of the CIDR range • 1 IP address at the end of the CIDR range
3	Each VM cluster	Each VM cluster requires 3 IP addresses for Single Client Access Names (<u>SCANs</u>), regardless of how many virtual machines are present in the VM cluster.
4	Each VM	4 IP addresses are needed for each virtual machine.
1	Oracle Autonomous Database	Each Oracle Autonomous Database instance consumes 1 IP address
1	Each VM for Base Database	Each VM created consumes 1 IP address for database



Backup Subnet Requirements

The minimum CIDR size for the backup subnet is /27, and the maximum size is /22.

Number of IP addresses	Consumed by	Summary
3	Oracle Database@Google Cloud Network service	These IP addresses are reserved regardless of how many VM clusters you provision in the Backup Subnet. Oracle Database@Google Cloud consumes the following: • 2 IP addresses at the beginning of the CIDR range • 1 IP address at the end of the CIDR range
3	Exadata database servicer VM cluster	VM clusters have a minimum of 2 virtual machines. Therefore, a VM cluster with 2 virtual machines requires 6 IP addresses in the backup subnet. Each additional virtual machine added to a VM cluster increases the number of IP addresses needed in the Backup Subnet by 3 IPs.
N/A	Base Database and Autonomous Database	Base Database and Autonomous Database Doesn't require backup subnet

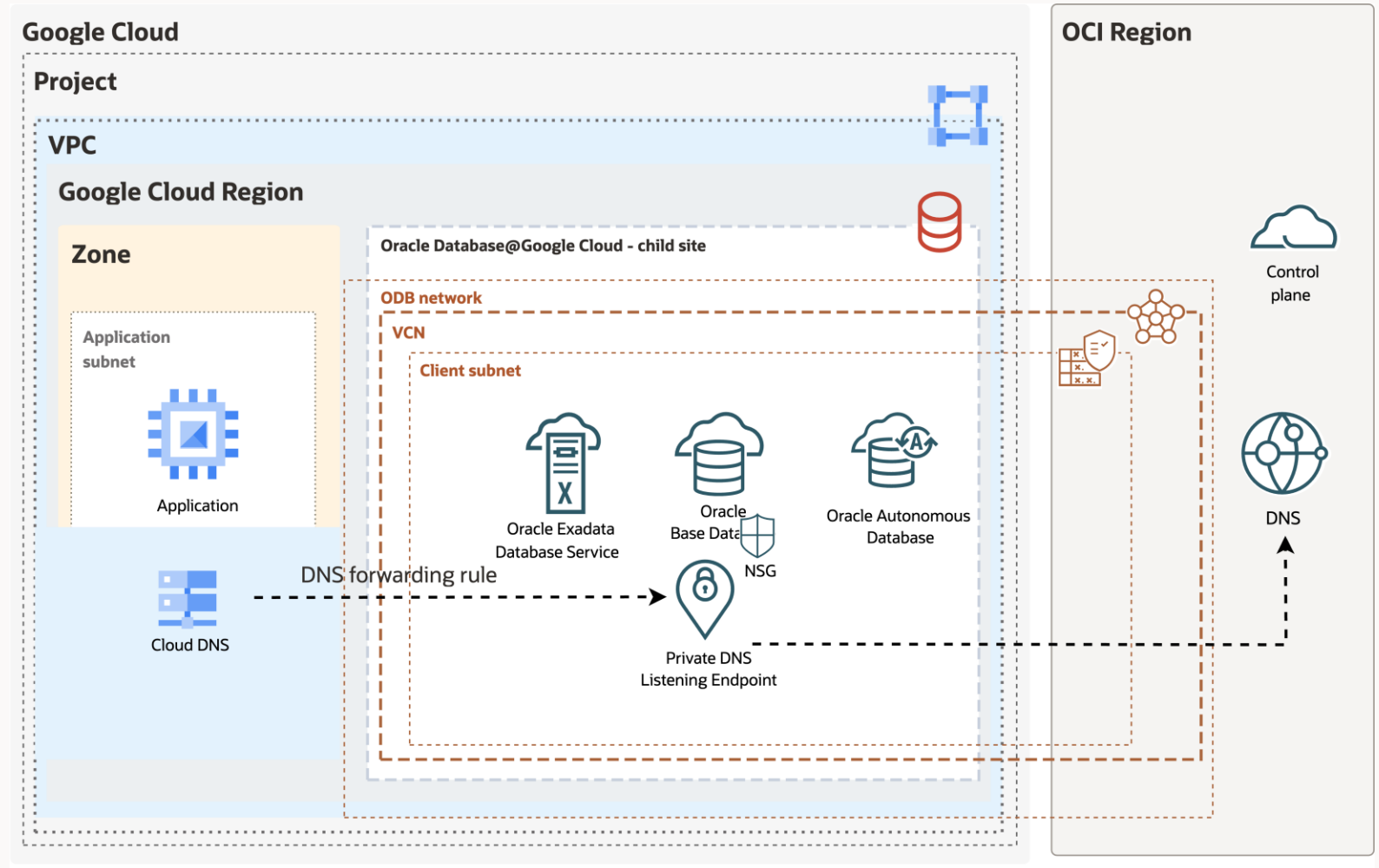


DNS

The private forwarder is deployed in the Project and associated to the VPC.

DNS resolution is forwarded to OCI DNS Private zone

- To resolve from different networks configure DNS Zone peering



DNS Zone peering

From Network Services, Cloud DNS, create a DNS Zone,

- Select Private zone type
- DNS name (VM Cluster Hostname domain name)
- DNS Peering
- Peer project
- Peer network of the project

Create a DNS zone

A DNS zone is a container of DNS records for the same DNS name suffix. In Cloud DNS, all records in a managed zone are hosted on the same set of Google-operated authoritative name servers. [Learn more](#)

If you don't have a domain yet, purchase one through [Cloud Domains](#).

Zone type

- ☒ Private
☐ Public

Zone name *
wyhrrbjebh-jwxfzygreo-oraclevcn-com-app-peer-vpc
Lowercase, no spaces.

DNS name *
wyhrrbjebh-jwxfzygreo-oraclevcn.com.
Example: myzone.example.com

Description
DNS peering

Options *
DNS Peering

Networks

Your managed zone will be visible to the selected networks.

[ADD NETWORKS](#)

Peer project *
omcpmpoc2 [SELECT](#)

Peer network *
demo-vpc

After creating your zone, you can add resource record sets and modify the networks your zone is visible on.

[CREATE](#) [CANCEL](#)



Network latency consideration

Select the closest zone between application VMs and Oracle database

Benchmark with latency tools: sockperf, netperf, latte or adbping

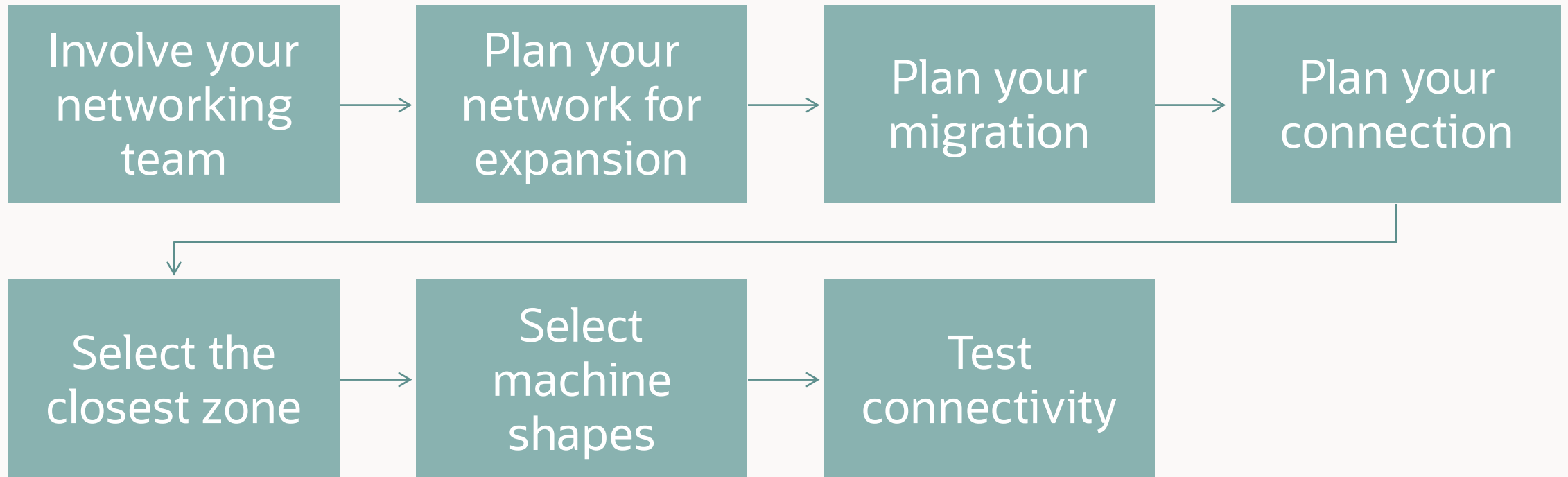
2. Google Cloud offers different machine series, with specific properties like High-bandwidth network, vCPU, Network interfaces, Network performance.

Based on your workload, choose the Machine series accordingly.

➤ In internal benchmarking: C2-standard-60 improves by 30% the latency versus e2-medium.

https://cloud.google.com/compute/docs/machine-resource#machine_type_comparison

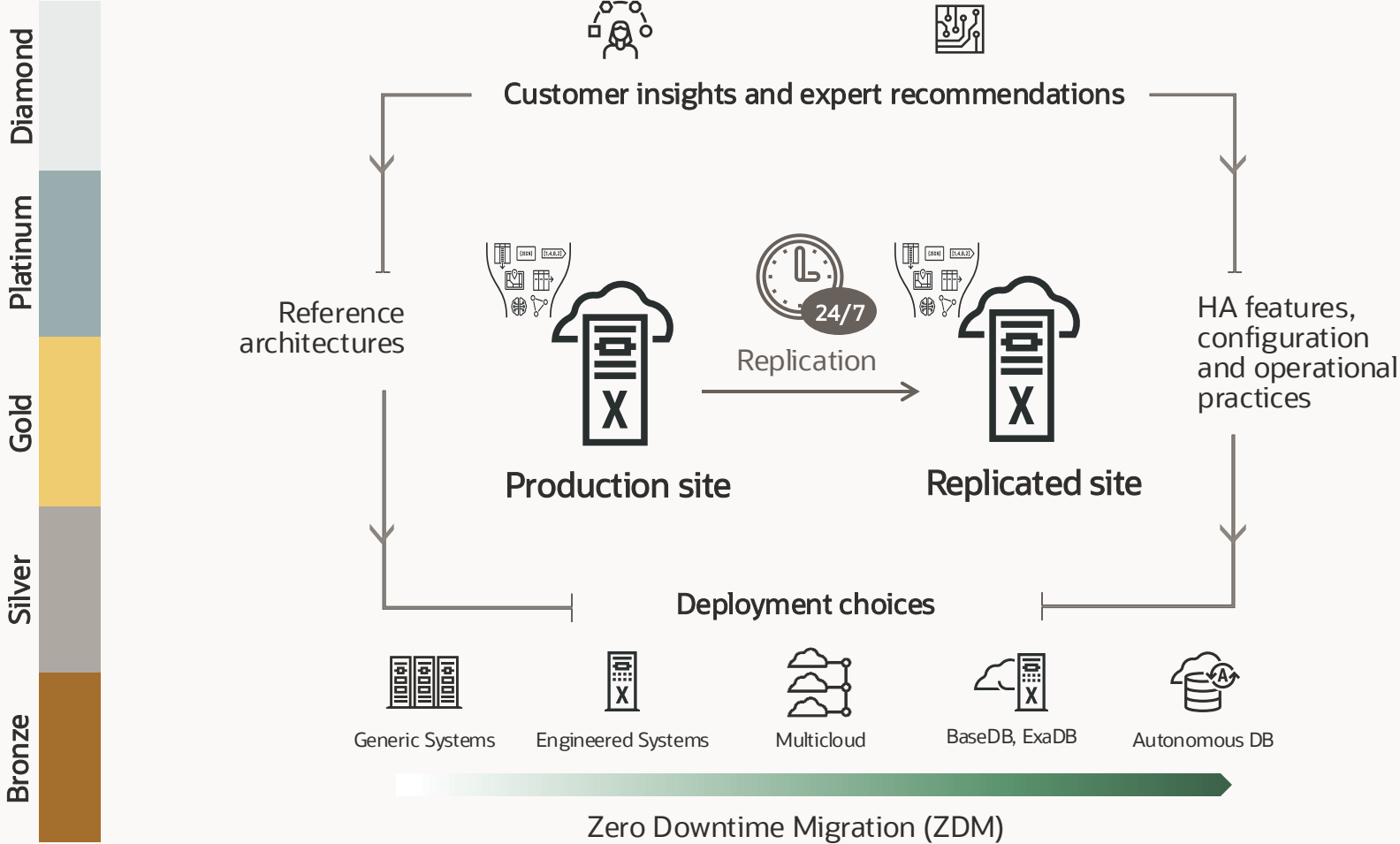
Recommendations



Maximum Availability Architecture

Oracle Maximum Availability Architecture (MAA)

Standard Reference Architectures for Never-Down Deployments



High performance

Resource Management

Database In-Memory

True Cache

Continuous availability

Application Continuity

Online Redefinition

Edition-based Redefinition

Data protection

Flashback

RMAN

ZDLRA+ ZRCV

Active replication

Active Data Guard

Full Stack DR

GoldenGate

Scale out & Lifecycle

RAC

Globally Distributed Database

FPP

Real Application Testing



Oracle MAA Reference Architectures

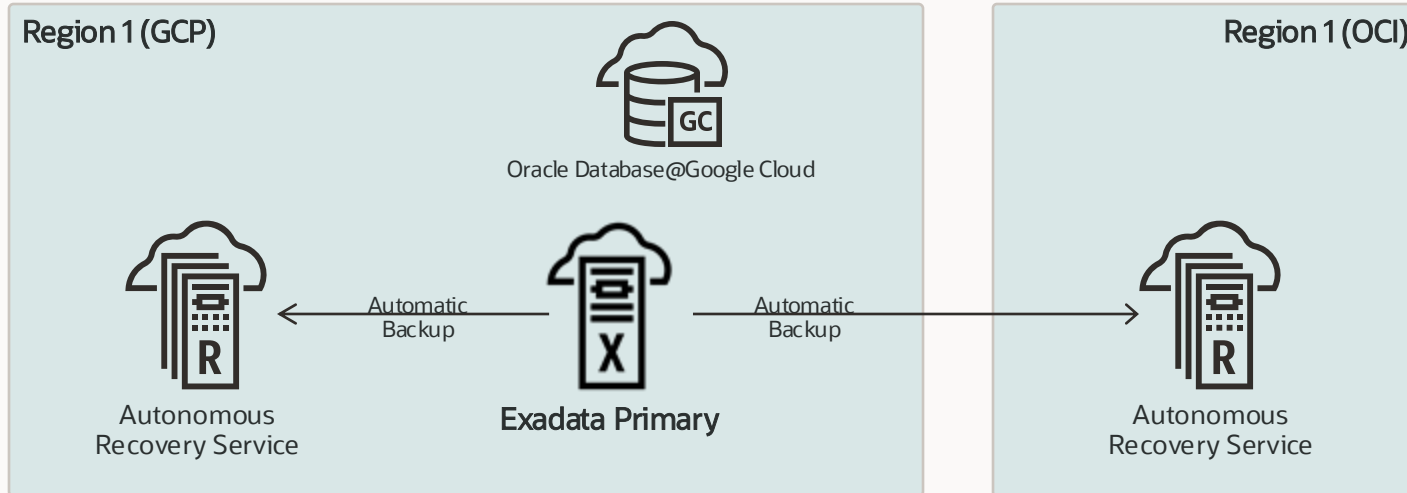
Availability service levels for Oracle AI Database

Bronze	Silver	Gold	Platinum	Diamond (NEW)
Dev, test, prod Single instance DB Restartable Backup/restore 	Prod/departmental Bronze + Database HA with RAC or Local Data Guard Client failover HA best practices Application Continuity (optional) 	Business critical Silver with RAC + DB replication with (Active) Data Guard with automatic failover Client failover DR best practices 	Mission critical Gold with Exadata and either option: Option 1: GoldenGate with Oracle Database 19c OR Option 2: (Active) Data Guard with Oracle AI Database 26ai 	Extreme availability Configuration GoldenGate 26ai replicas, each running. Oracle AI Database 26ai + RAC on Exadata + (Active) Data Guard 
Recoverable local failure: Minutes to hour Disasters: Hours to days RPO < 15 min	Recoverable local failure: seconds to minutes Disasters: Hours to days RPO < 15 min	Recoverable local failure: Less than 60 seconds Disasters: < 5 min RPO = zero or near zero	Recoverable local failure: Less than 20 seconds Disasters: < 30 secs RPO = zero or near zero	Recoverable local failures: Less than 10 seconds Disasters zero to 10 secs RPO = zero or near zero



Oracle **Exadata** Database Service on **Dedicated** Infrastructure

Oracle MAA Silver level architecture



High availability built-in by default

Oracle Exadata and Oracle RAC

- ✓ Agility to scale storage, compute, and memory without downtime
- ✓ Node failure protection
- ✓ Zero downtime maintenance

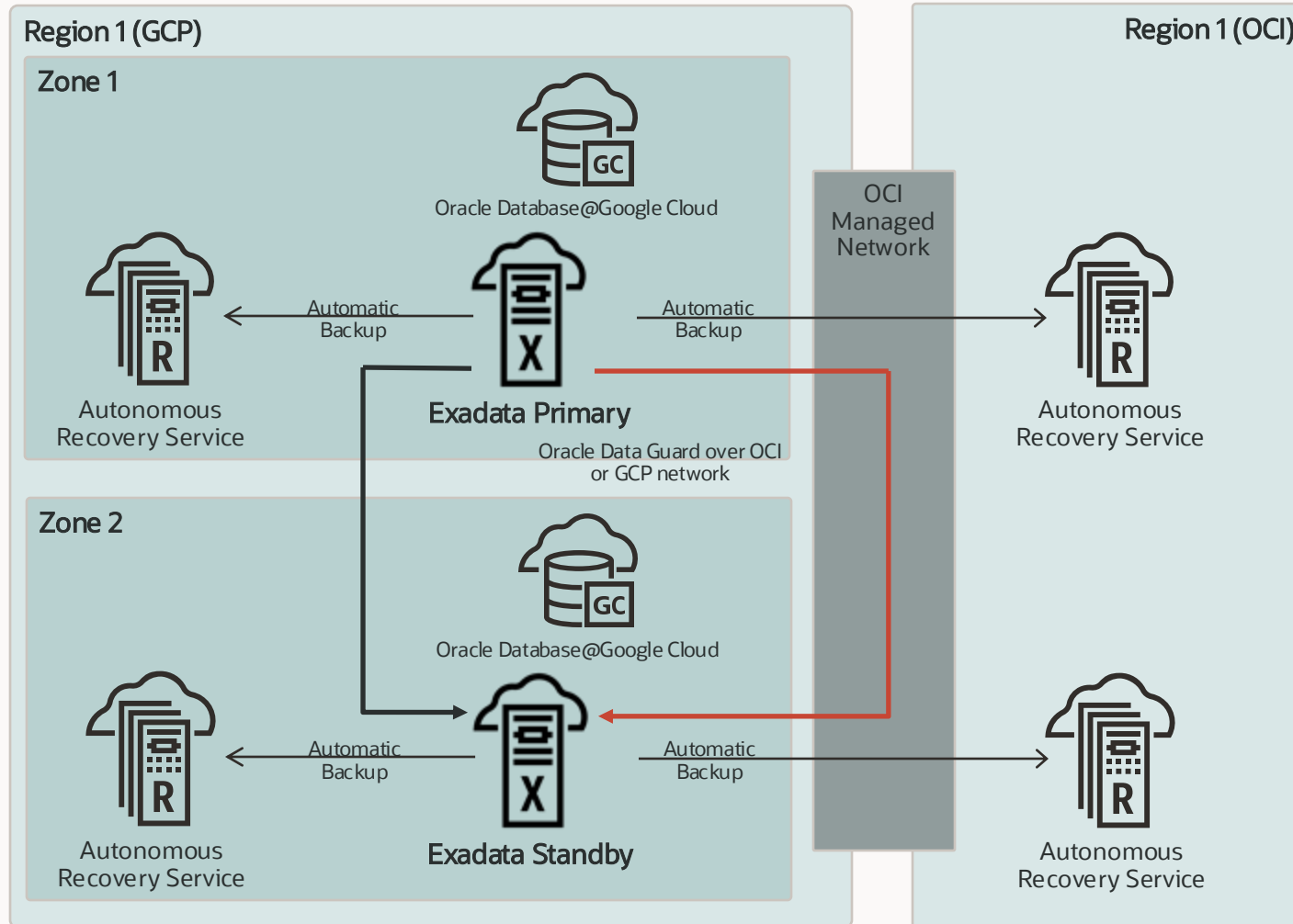
+ Zero Data Loss Autonomous Recovery Service

- ✓ Zero data loss recovery
- ✓ Fast, low impact backup & recovery
- ✓ Database-aware ransomware resiliency



Oracle Exadata Database Service on Dedicated Infrastructure

Oracle MAA Gold level cross-zones architecture



Enhancing availability and ensuring zero data loss

MAA Silver Level +

- ✓ Site failure protection
- ✓ Oracle Data Guard's In-Memory replication with zero data loss

+ Active Data Guard

- ✓ Comprehensive data corruption prevention
- ✓ Offload read-mostly workloads to standby
- ✓ Online updates

+ Multiple Standbys

- ✓ Read-mostly scale-out

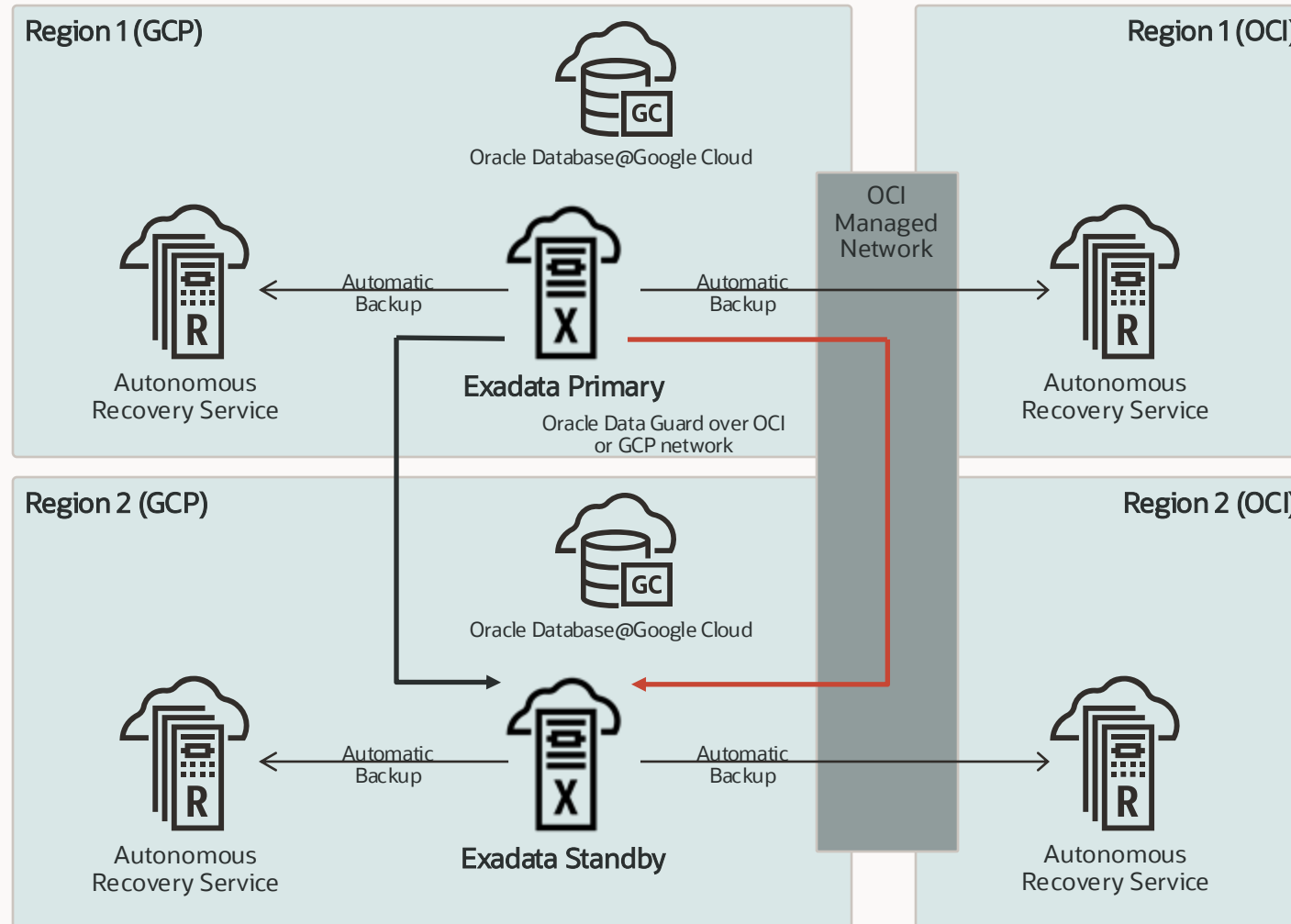
*MAA cross-zones certification for GCP in progress

Oracle Exadata Database Service on Dedicated Infrastructure

Oracle MAA Gold level cross-region architecture



GOLD



Safeguarding continuous availability and real-time data protection

MAA Silver Level +

- ✓ Regional disaster recovery protection
- ✓ Oracle Data Guard's In-Memory replication with (near) zero data loss

+ Active Data Guard

- ✓ Comprehensive data corruption prevention
- ✓ Offload read-mostly workloads to standby
- ✓ Online updates

+ Multiple Standbys

- ✓ Zero data loss across regions with minimal impact on primary performance
- ✓ Read-mostly scale-out

Oracle Data Guard Network traffic configuration options

Via OCI or Google Cloud

Network Traffic through OCI (recommended)

- ✓ Automated Data Guard one-click setup
 - ✓ MAA best practices by default
 - ✓ VCN peering required
- ✓ Oracle controls the network and ensures reliability
- ✓ No charge back for network traffic across availability domains in the same region
- ✓ First 10TB/month cross-region traffic for free
- ✓ Multiple standby databases via cloud tooling
- ✓ Optional: Fast-Start Failover (FSFO) manual setup

Network Traffic through Google Cloud

- ✓ Automated Data Guard one-click setup
 - ✓ MAA best practices by default
 - ✓ VPC peering required
- ✓ Google controls the network and ensures reliability
- ✓ Charge back for cross-region traffic
- ✓ Multiple standby databases via cloud tooling
- ✓ Optional: Fast-Start Failover (FSFO) manual setup

Fully automated Oracle Active Data Guard setup

Using OCI Console, OCI CLI, SDKs, or REST APIs

Database information Backups **Data Guard Group** Connections Pluggable

Add standby operation will also do a precheck. [Learn more](#)

Q Search and Filter

Add standby

1. Click
"Add standby"

Member	VM Cluster	Role
DB0714	vmc	Primary
DB0714	maa	Standby

Add standby

Peer VM Cluster

2. Choose your
standby zone
or region

Peer region
US East (Ashburn)

Canada Southeast (Toronto)

UK South (London)

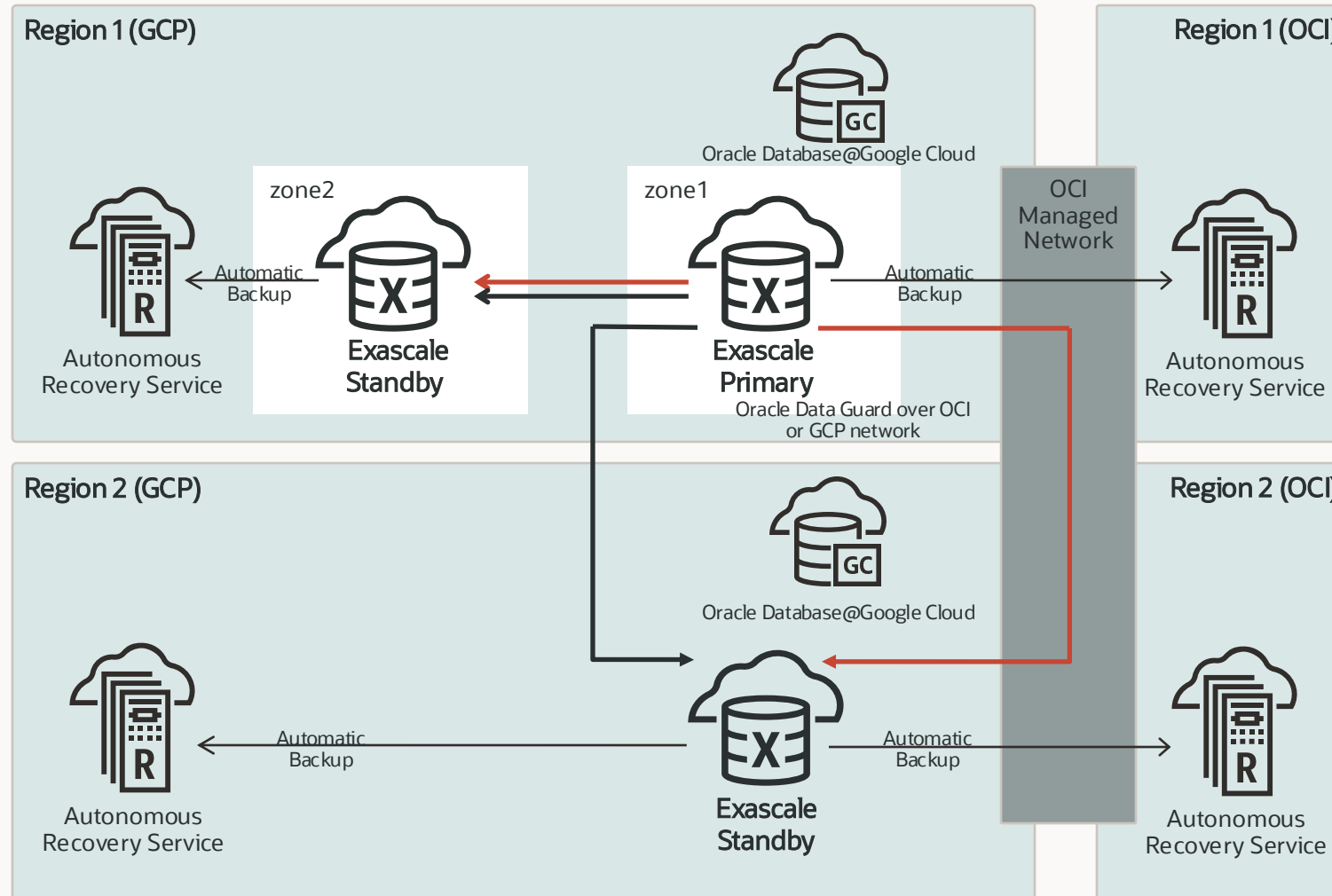
US East (Ashburn)

Primary database is in region UK South (London)



Oracle Exadata Database Service on Exascale Infrastructure

Lower entry point and efficient database clones



Oracle Exadata and Oracle RAC

- ✓ Agility to scale storage, compute, and memory without downtime
- ✓ Node failure protection
- ✓ Zero downtime maintenance

+ Data Guard

- ✓ Site failure protection
- ✓ Regional disaster recovery protection
- ✓ (near) zero data loss

+ Active Data Guard

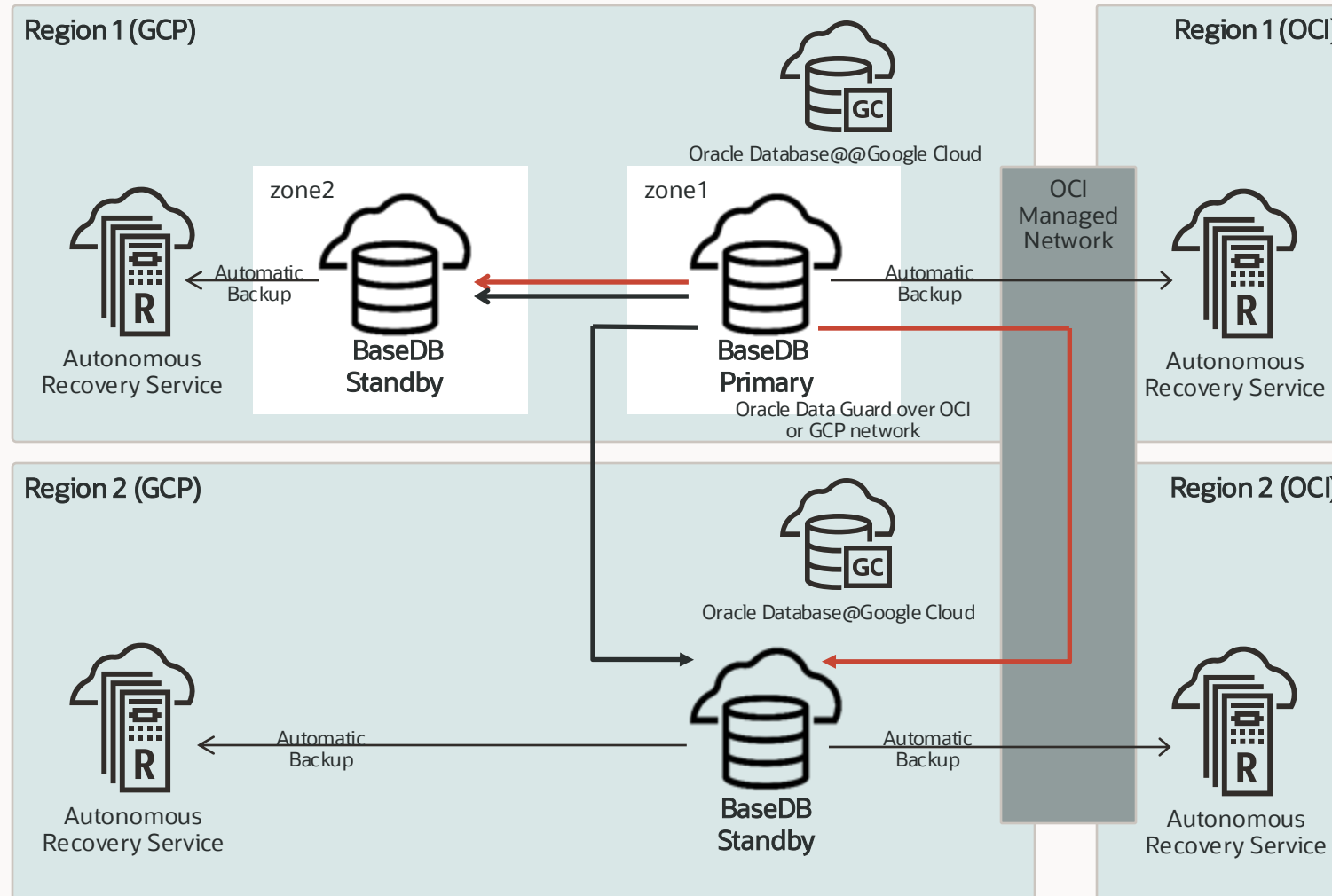
- ✓ Comprehensive data corruption prevention
- ✓ Offload read-mostly workloads to standby
- ✓ Online updates

+ Zero Data Loss Autonomous Recovery Service

- ✓ Database-aware ransomware resiliency

Oracle Base Database Service

Flexible VM-based infrastructure with scalable block storage



Data Guard

- ✓ Site failure protection
- ✓ Regional disaster recovery protection
- ✓ (near) zero data loss

+ Multiple Standbys

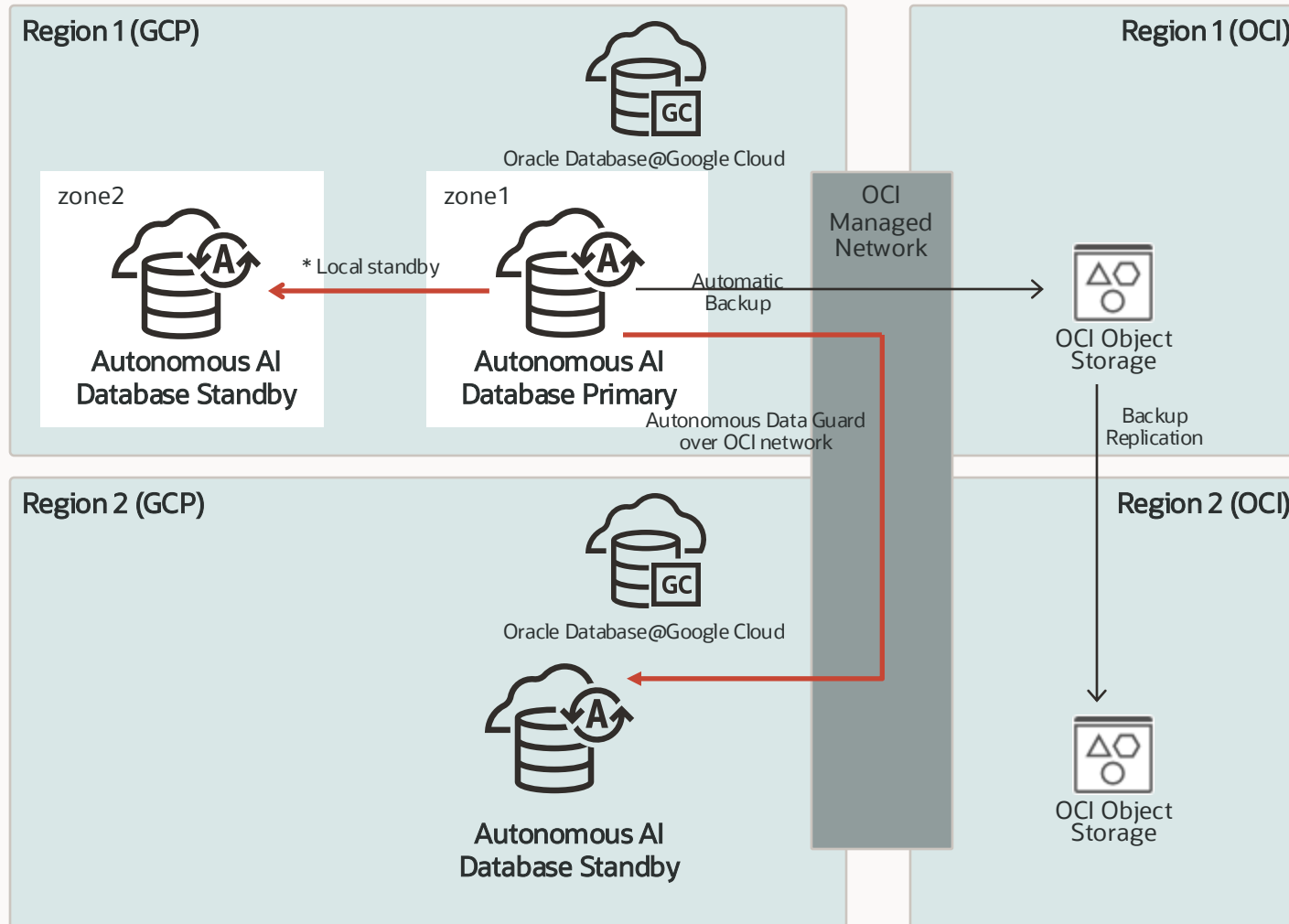
- ✓ Zero data loss across regions with minimal impact on primary performance

+ Zero Data Loss Autonomous Recovery Service

- ✓ Database-aware ransomware resiliency

Oracle Autonomous AI Database Serverless

High availability and data protection built-in by default



Oracle Exadata and Oracle RAC

- ✓ Full elasticity with online and auto-scaling of compute and storage
- ✓ Node failure protection
- ✓ Zero downtime maintenance

Backup-based disaster recovery

- ✓ Automatic backup by default

+ Autonomous Data Guard

- ✓ Local and regional standbys one-click setup
- ✓ Automatic failover for lower RTO
- ✓ Real-time data replication for lower RPO

* Local standby in GCP coming soon

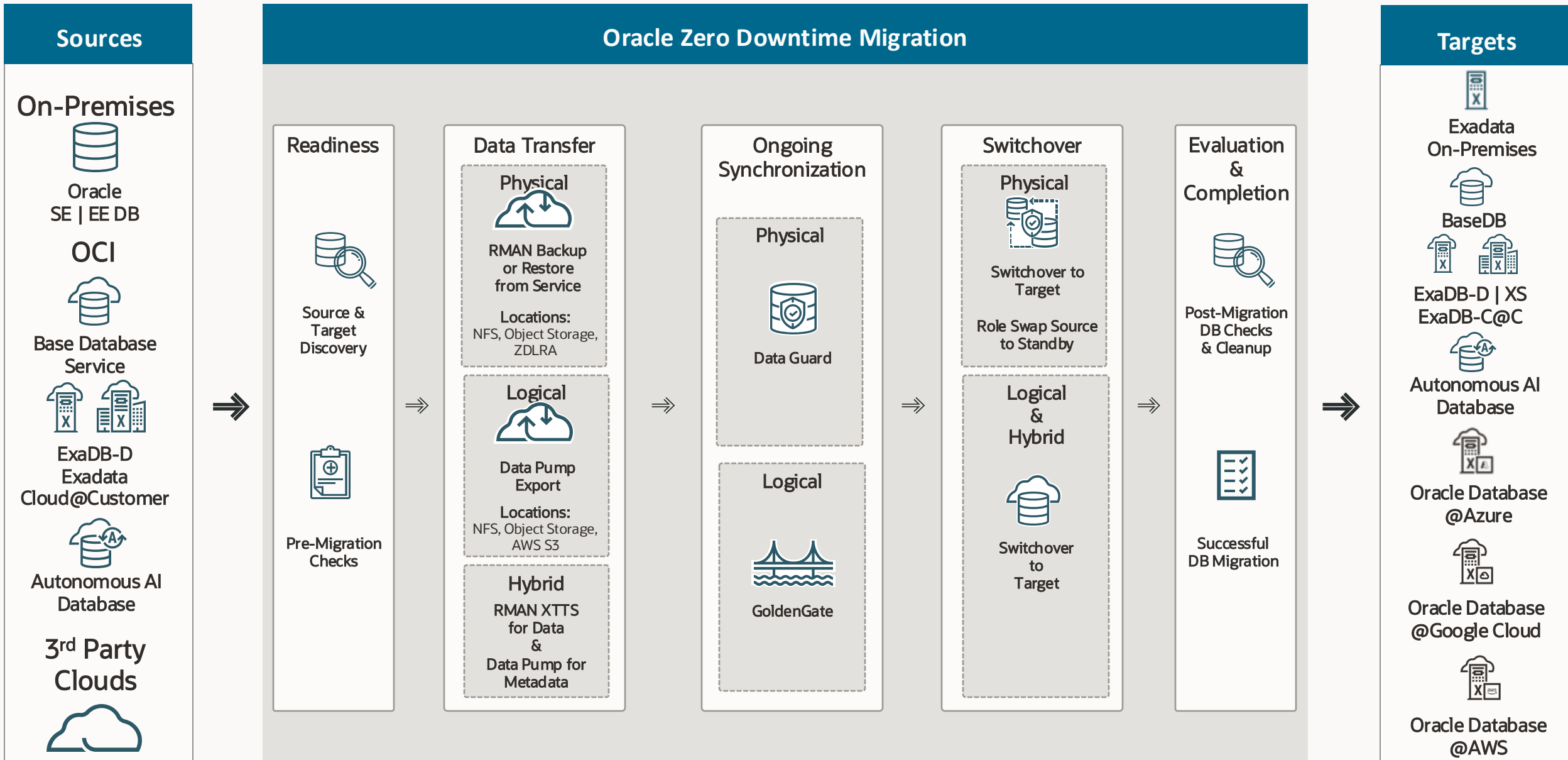
Migration

Migration Methods

Oracle Zero Downtime Migration – Solves all the aforementioned challenges!

	Migration	Online	Downtime *
ZDM: Recommended (Oracle Zero Downtime Migration)	Physical/Logical	✓	Minimal
Oracle Data Guard	Physical	✓	Minimal
Oracle GoldenGate	Logical	✓	Minimal
RMAN	Physical	✗	Required
Oracle Data Pump	Logical	✗	Required
Oracle Database Multitenant Movement	Physical	✗	Required

* Downtime – The duration depends on the volume and use case.



Oracle Database@Google Cloud

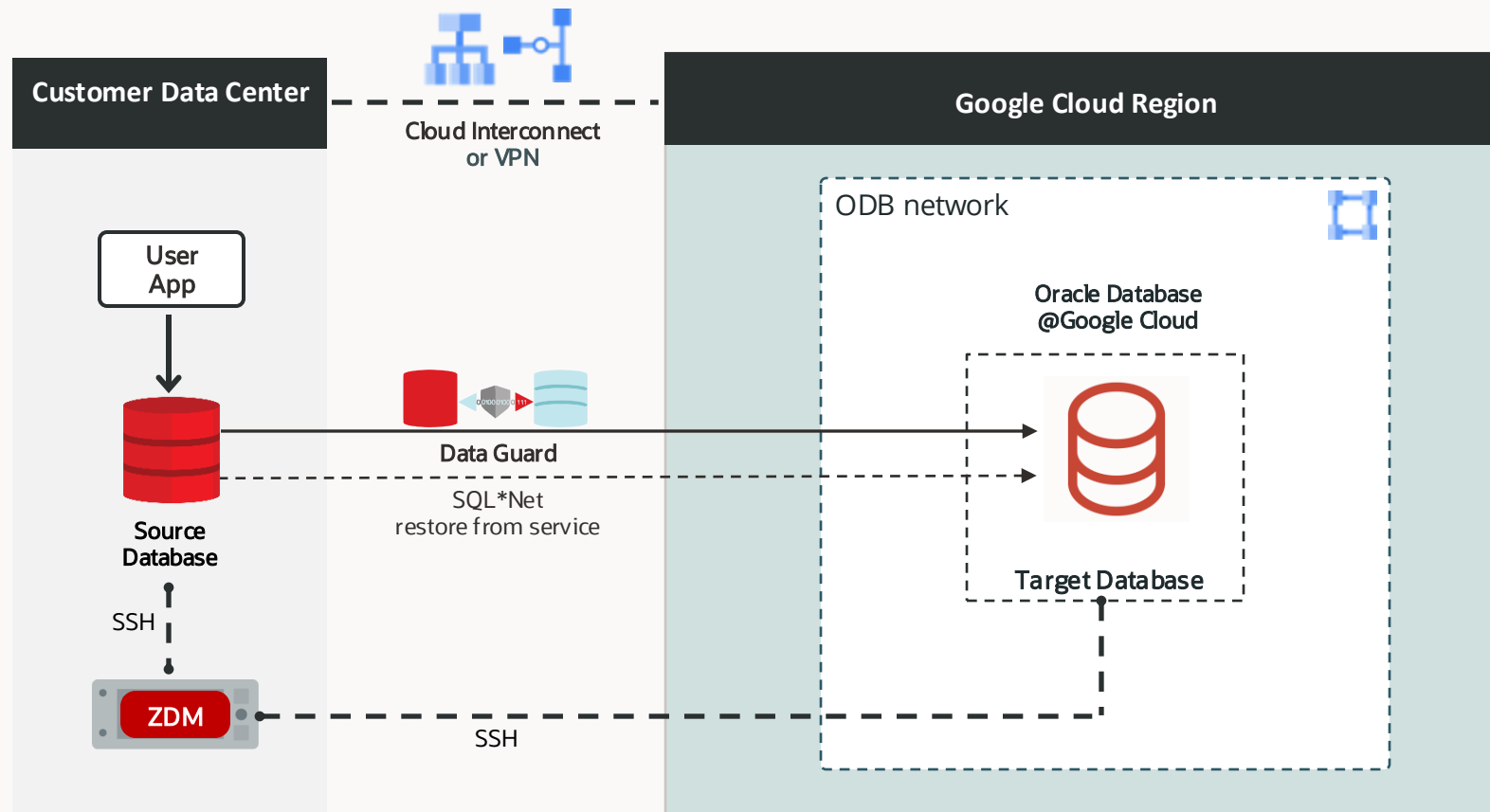
Oracle Zero Downtime Migration – Migration Methodologies

Method	Exadata Database Service Base Database Service	Autonomous AI Database
Physical Online		
Physical Offline		
Logical Online		
Logical Offline		

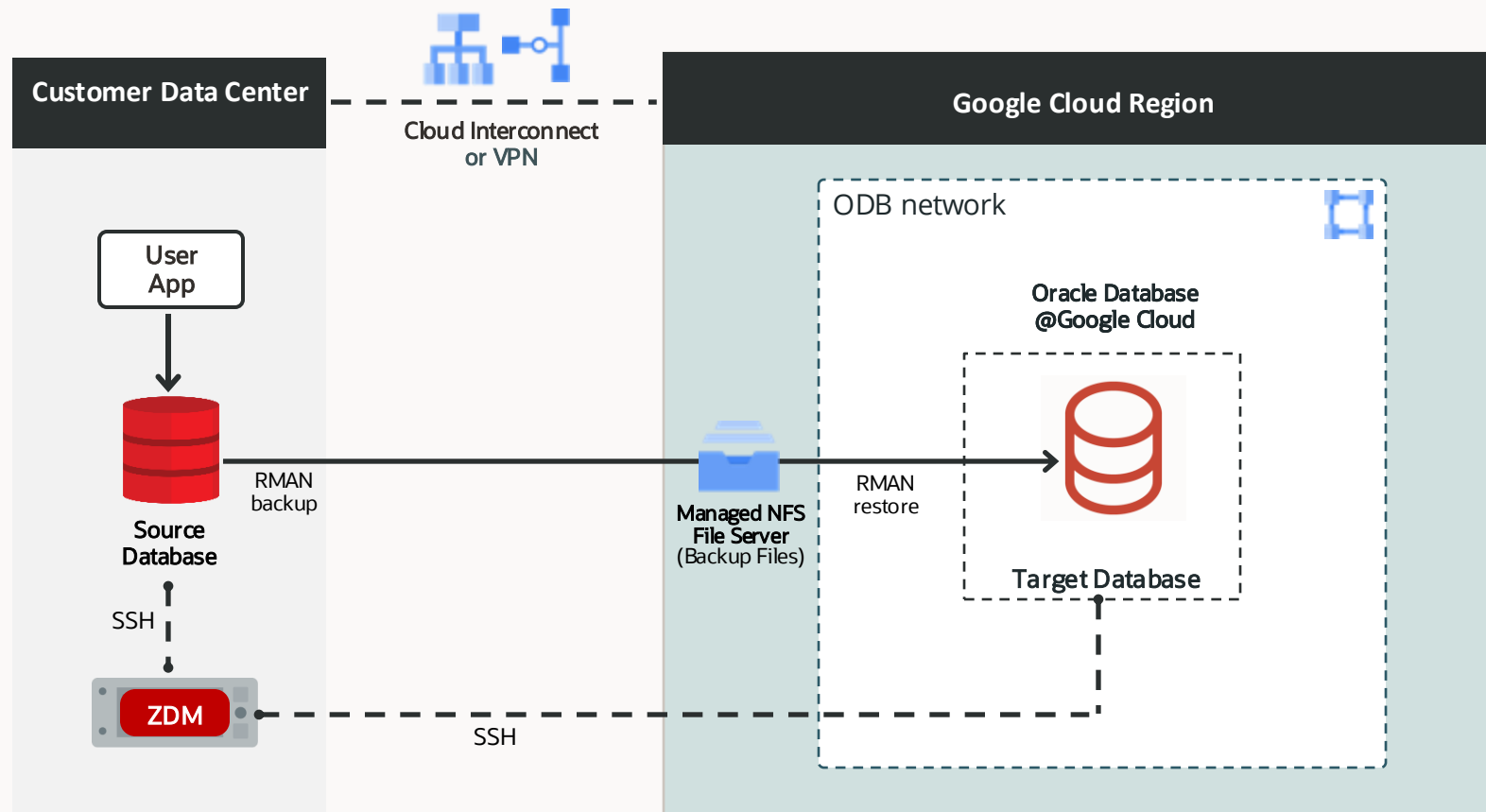


ZDM Physical Online Migration to Oracle Database@Google Cloud

Direct Data Transfer

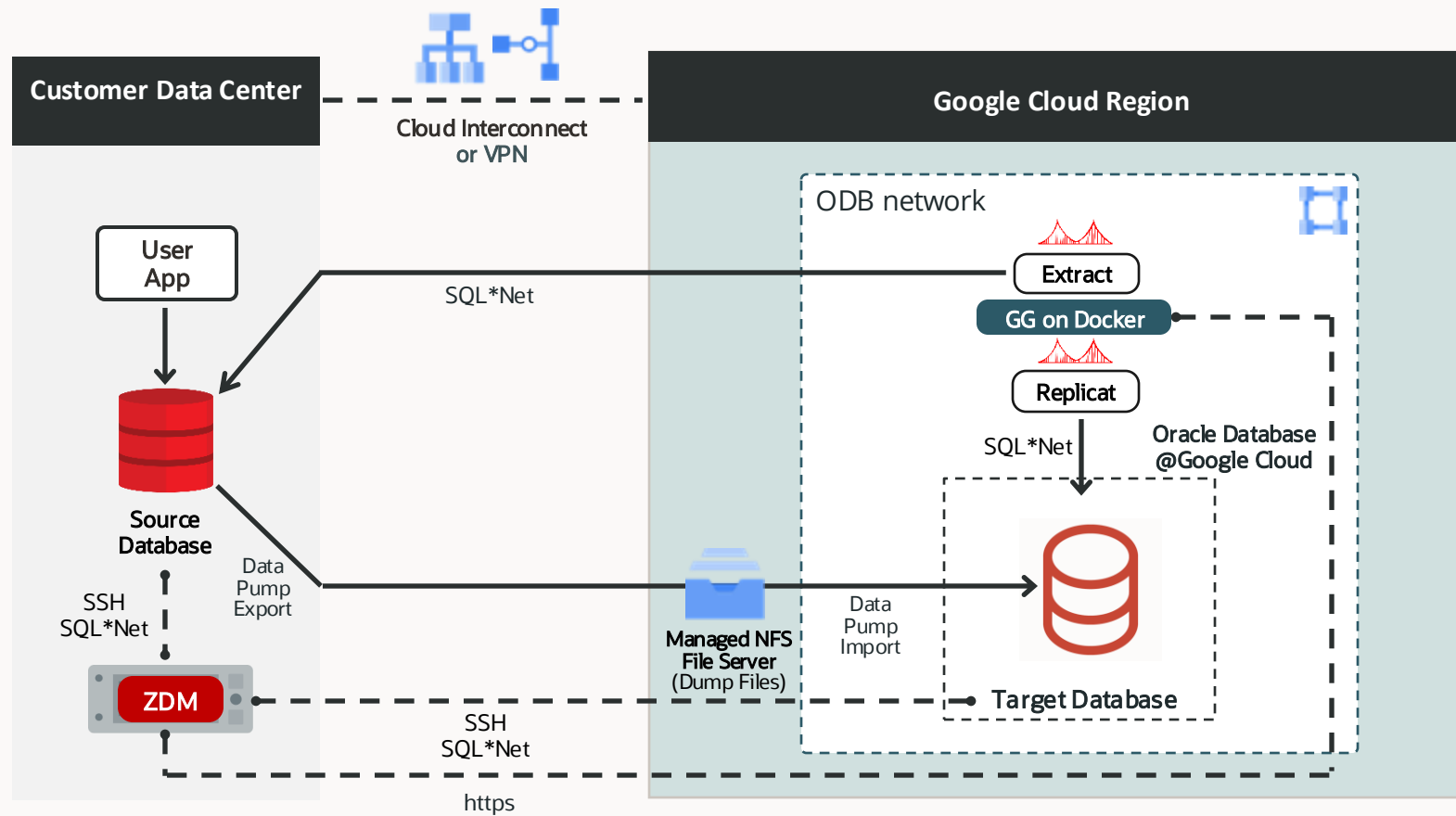


ZDM Physical Offline Migration to Oracle Database@Google Cloud Managed NFS File Server



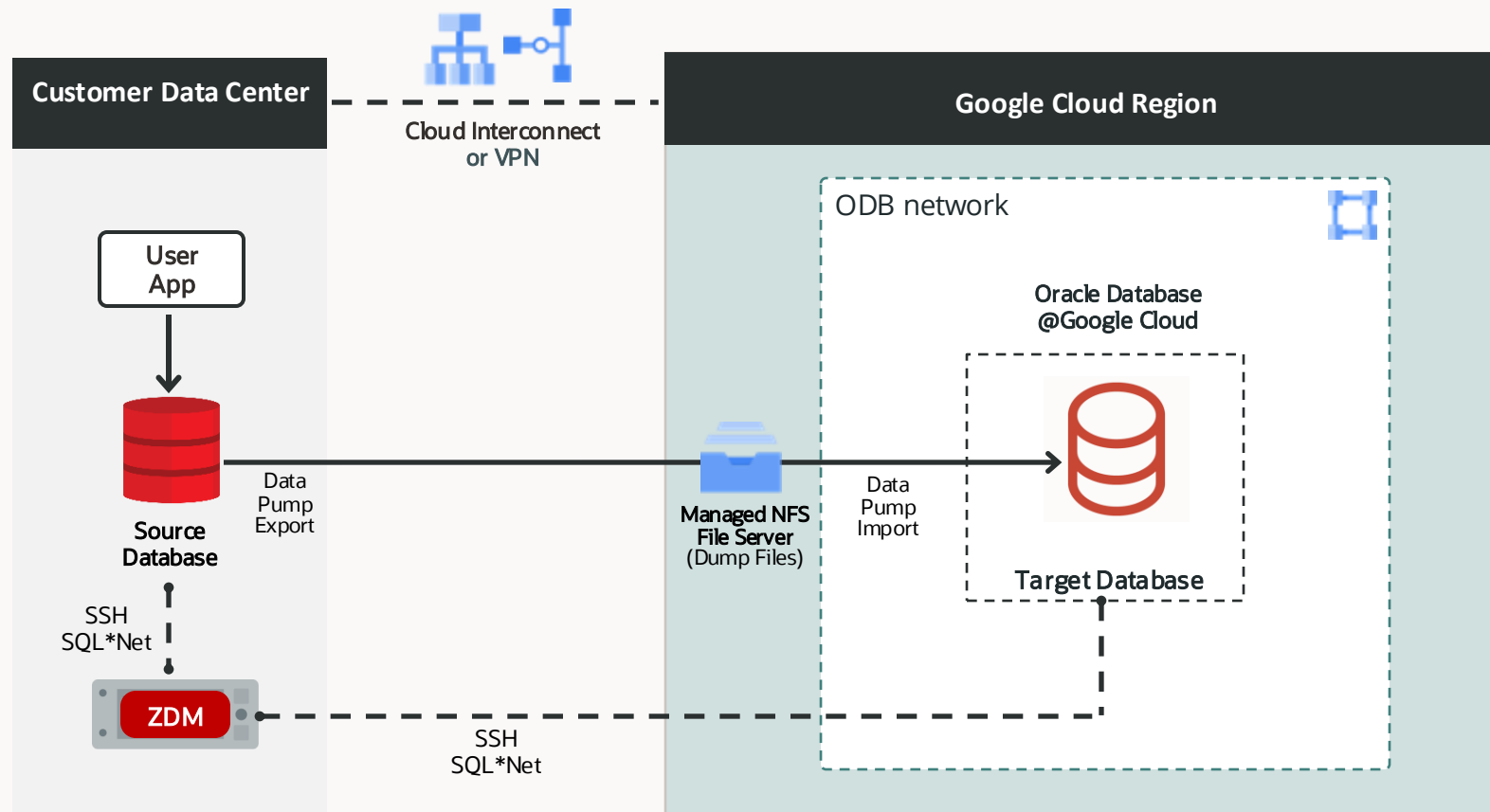
ZDM Logical Online Migration to ExaDB-D@Google Cloud

Managed NFS File Server with GoldenGate



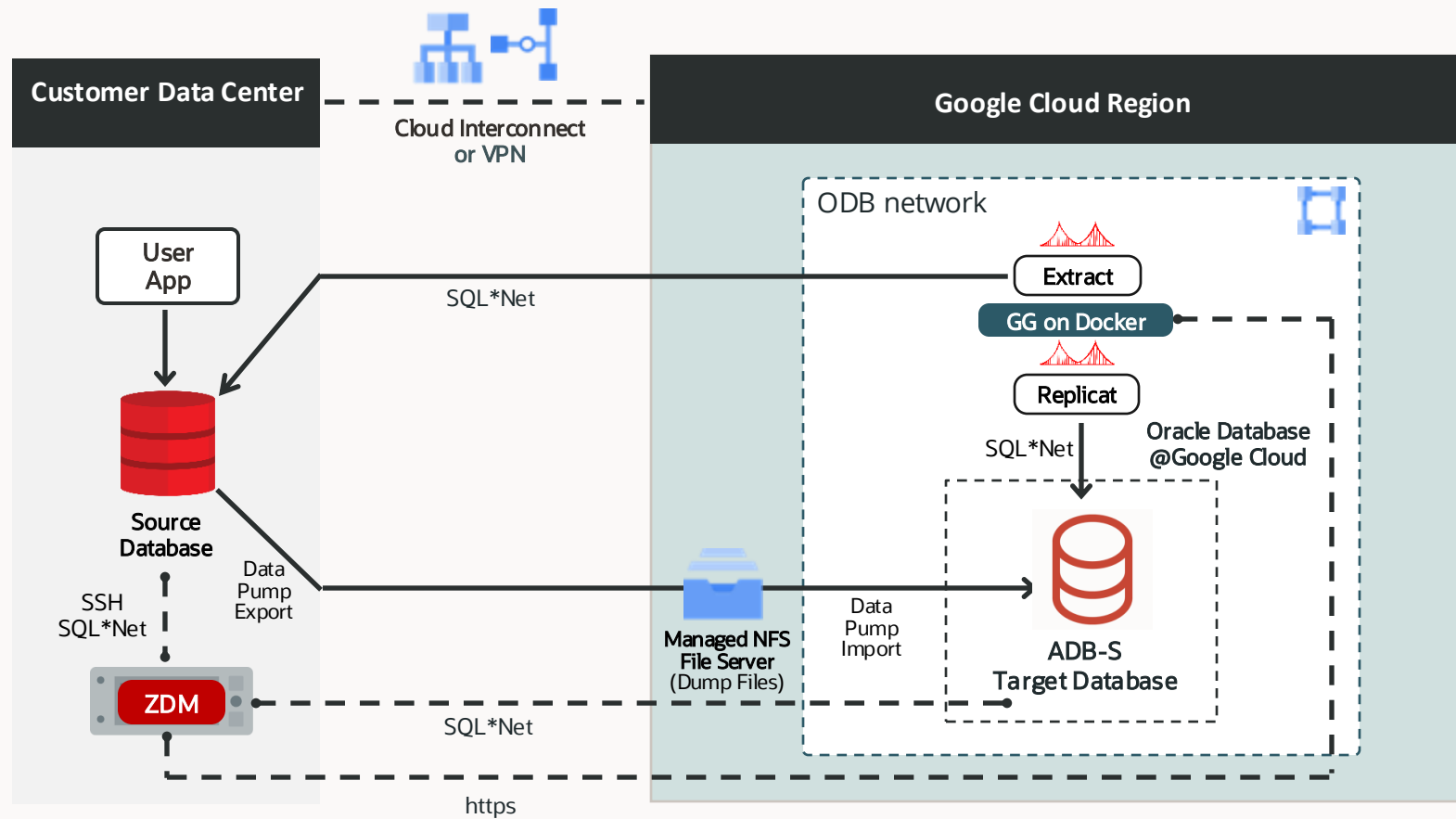
ZDM Logical Offline Migration to ExaDB-D@Google Cloud

Managed NFS File Server



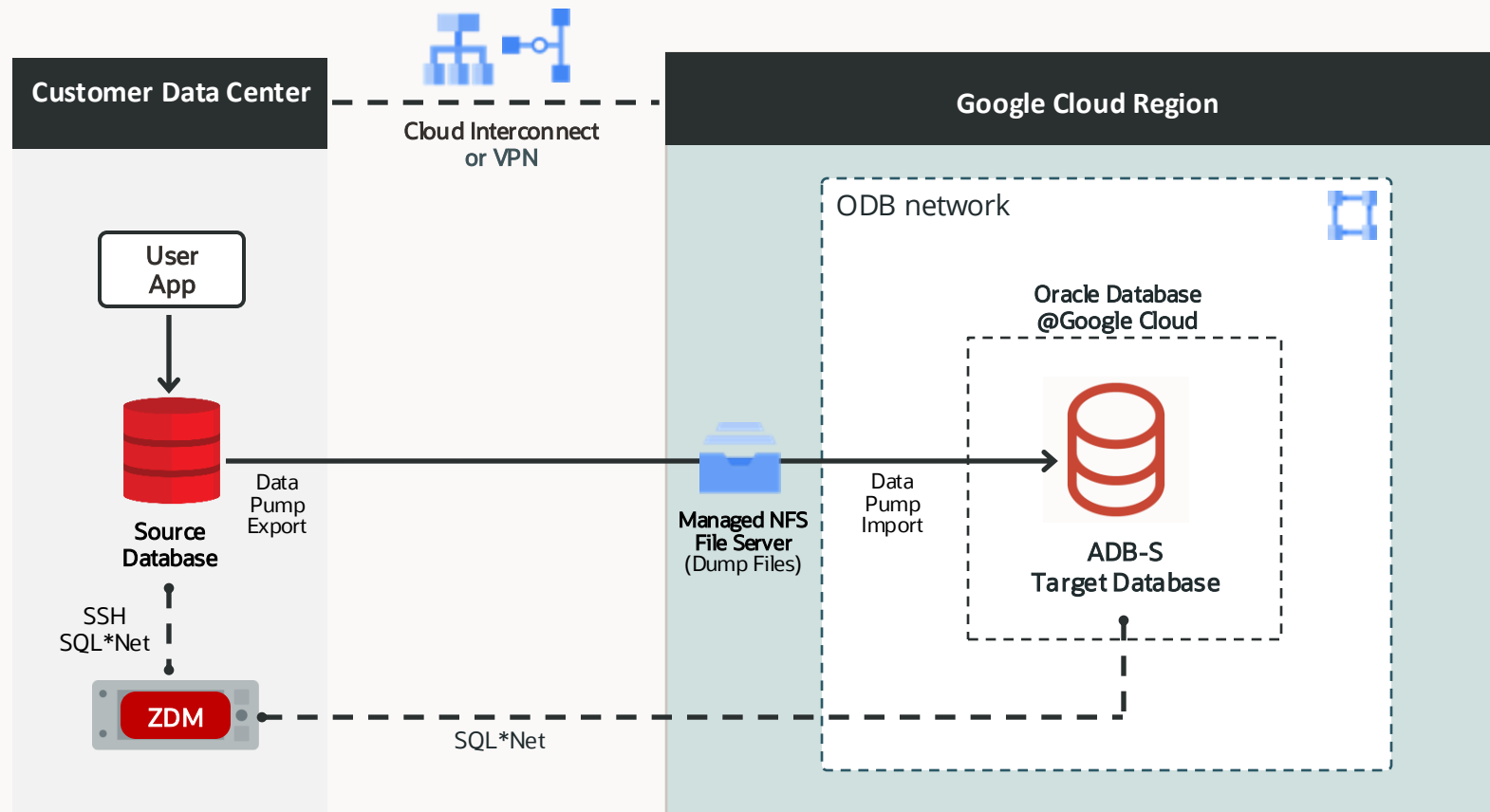
ZDM Logical Online Migration to ADB-S@Google Cloud

Managed NFS File Server with GoldenGate



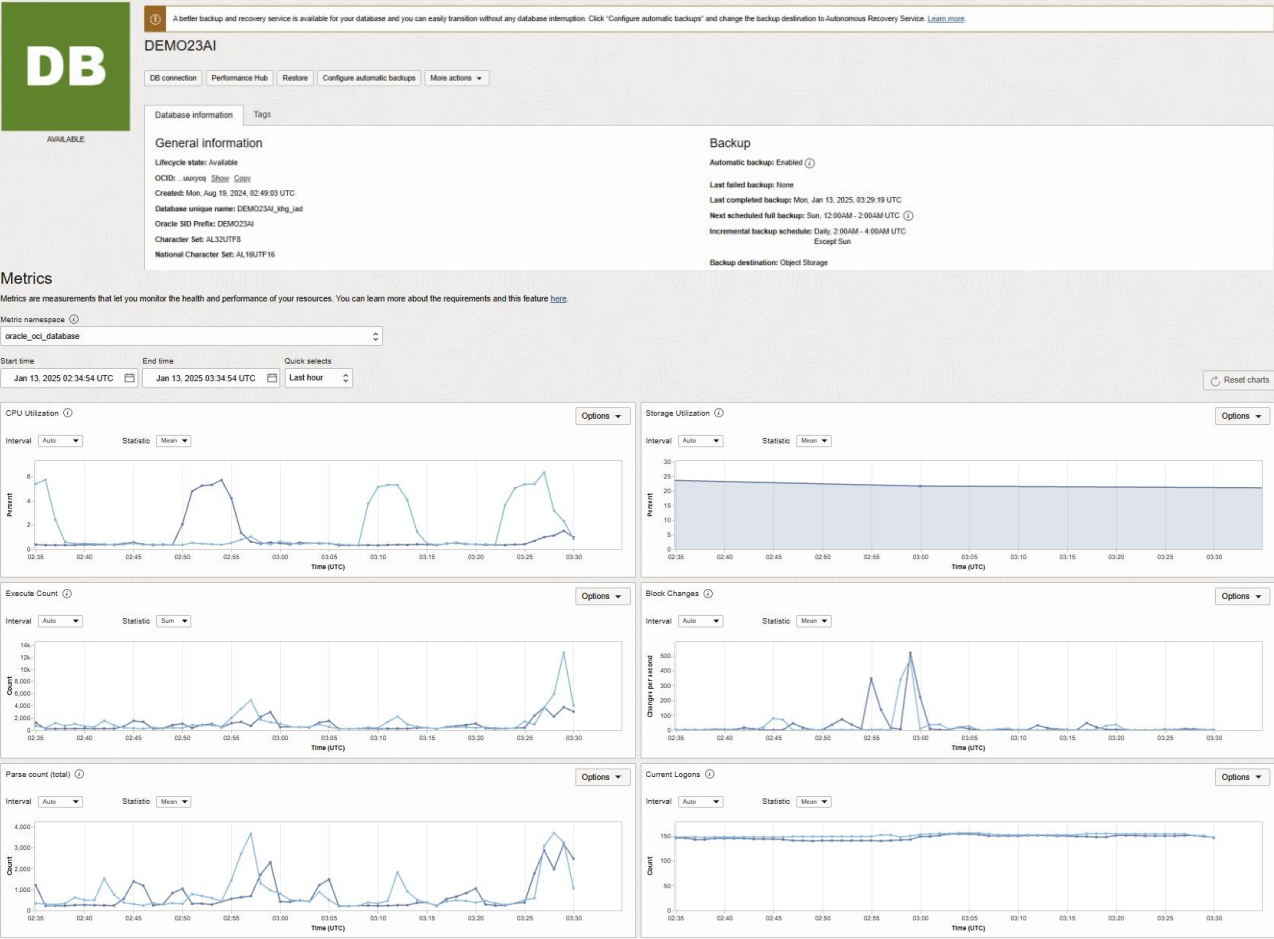
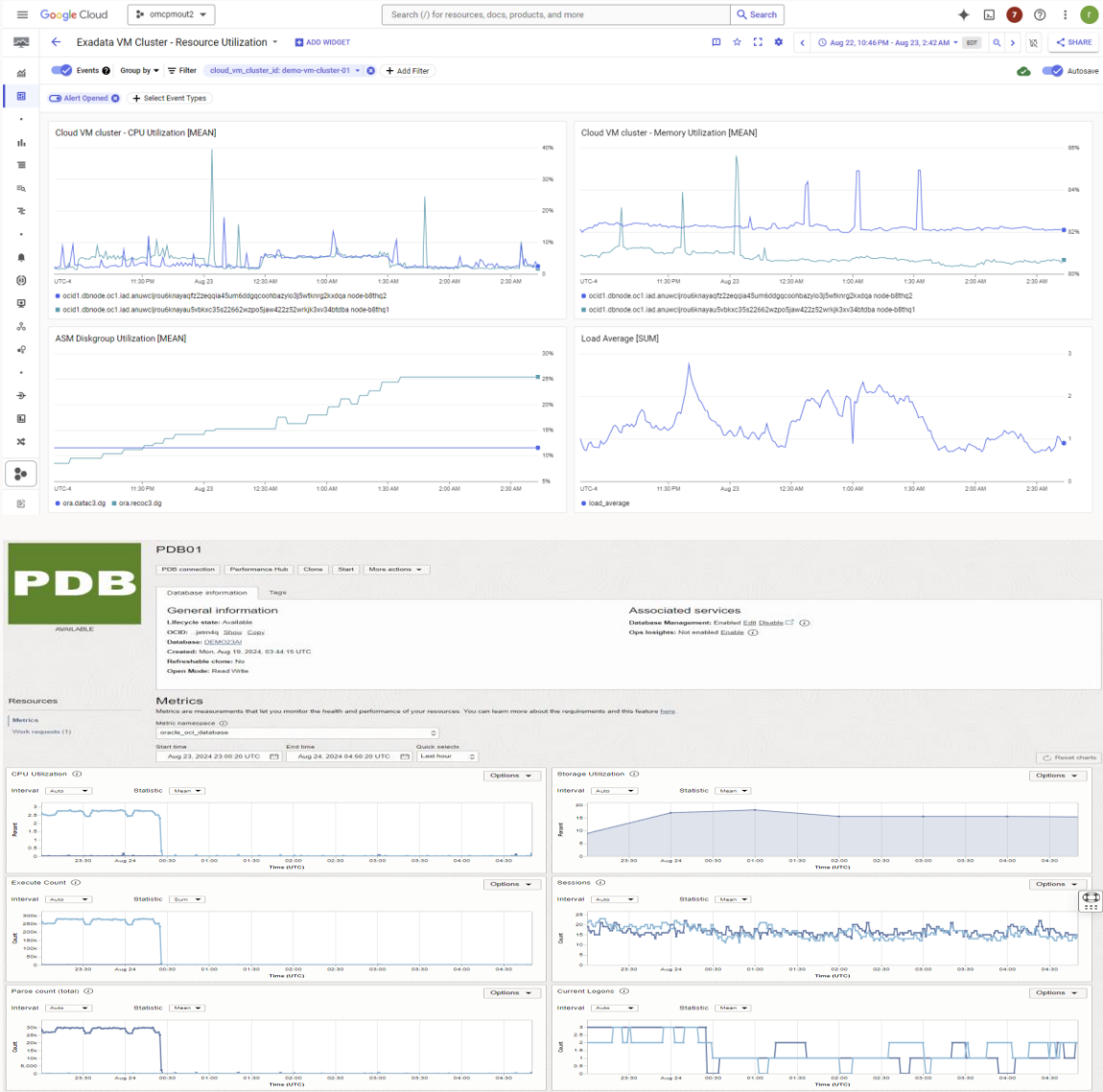
ZDM Logical Offline Migration to ADB-S@Google Cloud

Managed NFS File Server

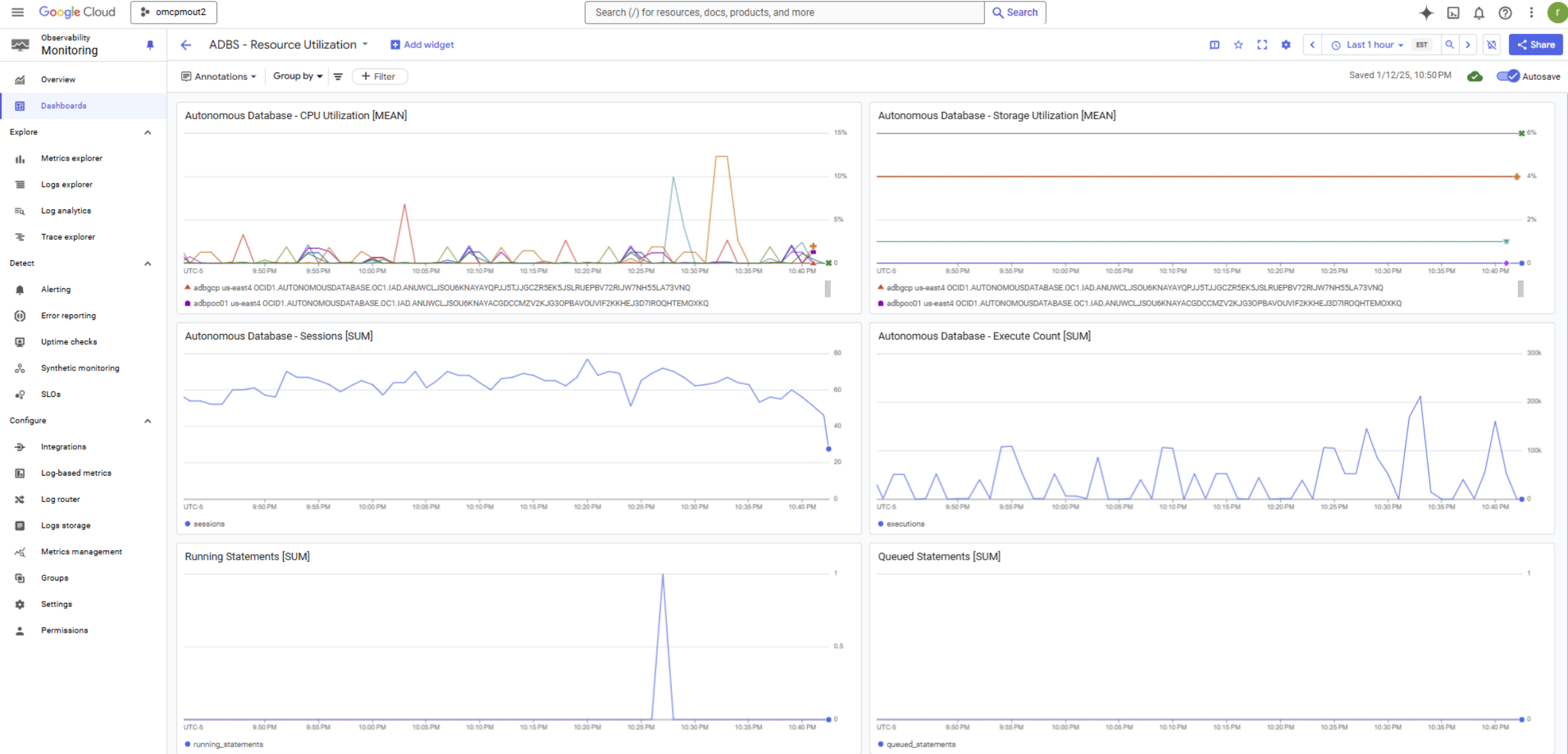


Observability

Oracle Database@Google Cloud – ExaDB Dedicated Observability Dashboard



Oracle Database@Google Cloud – ADB Serverless Observability Dashboard



Observability – Alerting

Google Cloud

omcpmout2

Search (/) for resources, docs, products, and more

Search

◆ 📄 🔔 ⓘ ⋮ 👤

Observability Monitoring

Alerting

+ CREATE POLICY

✎ EDIT NOTIFICATION CHANNELS

🎓 LEARN

Overview

Dashboards

Explore

Metrics explorer

Logs explorer

Log analytics

Trace explorer

Detect

Alerting

Error reporting

Uptime checks

Synthetic monitoring

SLOs

Configure

Integrations

Log-based metrics

Log router

Observability Scopes

omcpmout2

Release Notes

<|

Monitoring now supports both user-scoped and device-scoped Cloud Console Mobile notification channels

MANAGE CHANNELS

LEARN MORE

DISMISS

Summary

Incidents firing

1

Incidents acknowledged

0

Alert policies

1

View all

Incidents

HIDE CLOSED INCIDENTS

State	Severity	Policy name	Incident summary	Opened	Closed	
🚨	Critical	Cloud VM c...	CPU Utilization for omcpmout2 Cloud VM cluster labels (project_id=omcpmout2, resource_container=358321687794, location=us-east4, cloud_ex...	Aug 26, 2024, 5:28:10 PM	-	▼
✅	Critical	Cloud VM c...	CPU Utilization for omcpmout2 Cloud VM cluster labels (project_id=omcpmout2, resource_container=358321687794, location=us-east4, cloud_ex...	Aug 26, 2024, 5:17:07 PM	Aug 26, 2024, 5:22:09 PM	▼

→ See all incidents

Snoozes

CREATE SNOOZE

Show past snoozes

State	Name	Start time	End time
No rows to display			

→ See all snoozes

Policies

CREATE POLICY

Display Name	Type	Last Modified By	Created On
Cloud VM cluster - CPU Utilization [MEAN]	Metrics	rajib.sadhu@oracle.com	August 26, 2024

→ See all policies



Observability – Alerting (Emails)

Google Cloud

VIEW INCIDENT

Alert firing

Critical

Cloud VM cluster - CPU Utilization is above threshold of 4 with a value of 5.202

Start time

Aug 26, 2024 at 9:17PM UTC (less than 1 sec ago)

Policy

Cloud VM cluster - CPU Utilization [MEAN]

Project

omcpmout2

Condition

Cloud VM cluster - CPU Utilization [MEAN]

metric : oracledatabase.googleapis.com/cpu_utilization

cloud_exadata_infrastructure_id : exa-infra-1-us-east4

cloud_vm_cluster_id : demo-vm-cluster-01

dbnode_resource_id : ocid1.dbnode.oc1.iad.anuwcljrou6knayaqfz2zeqqia45um6ddgqcoohbazylo3j5w

deployment_type : exadata hostname : node-b8thq2

location : us-east4 project_id : omcpmout2

resource_container : 358321687794

resource_id : ocid1.cloudvmcluster.oc1.iad.anuwcljrou6knayazy2uyvzwopw4vuxfyujcfbtk32r

VIEW INCIDENT

Google Cloud

VIEW INCIDENT

Alert recovered

Critical

Cloud VM cluster - CPU Utilization is below threshold of 4 with a value of 3.036

Start time

Aug 26, 2024 at 9:17PM UTC (~5 minutes ago)

Policy

Cloud VM cluster - CPU Utilization [MEAN]

Project

omcpmout2

Condition

Cloud VM cluster - CPU Utilization [MEAN]

duration : 5 mins 2 secs

metric : oracledatabase.googleapis.com/cpu_utilization

cloud_exadata_infrastructure_id : exa-infra-1-us-east4

cloud_vm_cluster_id : demo-vm-cluster-01

dbnode_resource_id : ocid1.dbnode.oc1.iad.anuwcljrou6knayaqfz2zeqqia45um6ddgqcoohbazylo3j5w

deployment_type : exadata hostname : node-b8thq2

location : us-east4 project_id : omcpmout2

resource_container : 358321687794

resource_id : ocid1.cloudvmcluster.oc1.iad.anuwcljrou6knayazy2uyvzwopw4vuxfyujcfbtk32r

VIEW INCIDENT

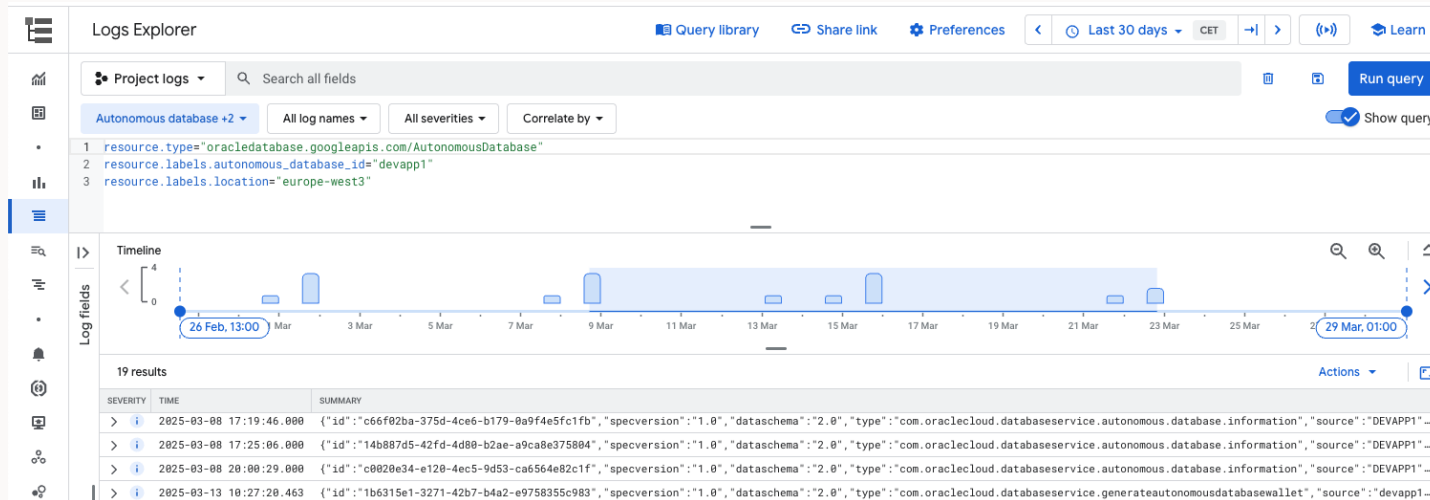


Accessing Oracle Database@Google Cloud Resource Logs

Accessing resource logs for Oracle Database@Google Cloud with Google Cloud Logging
The following logs and events are captured by default

Google Logs Explorer

- `oracledatabase.googleapis.com/CloudExadataInfrastructure`
- `oracledatabase.googleapis.com/CloudVmCluster`
- `oracledatabase.googleapis.com/ContainerDatabase`
- `oracledatabase.googleapis.com/PluggableDatabase`
- `oracledatabase.googleapis.com/AutonomousDatabase`



Identity and Permissions

Identity – Google Cloud roles

Google Cloud permissions allowing to accept "Free + Paid" offers in Google Cloud Marketplace

Agreement type	Permissions to accept offer	Permissions to purchase or subscribe
Private or Public Offer	roles/billing.admin OR (roles/billing.user AND roles/consumerprocurement.orderAdmin)	Billing Admin

To view and accept a private offer, complete the following steps based on the type of product that you're purchasing:

1. Click and open the link that the vendor sent you.

★ **Note:** If you receive an error when opening the offer, you don't have the correct access to the associated Cloud Billing account. To view and purchase the offer, you must have the Billing Administrator (**roles/billing.admin**) role OR both the Billing User (**roles/billing.user**) role AND the Consumer Procurement Order Administrator (**roles/consumerprocurement.orderAdmin**) role.

- 2. Verify the details of the offer, such as the pricing details and contract duration. If this offer replaces an active offer, or if you're using this offer after purchasing the product at list price, review the new offer and previous purchase details. Click **Next**.
- 3. In the **Settings** section of the offer, confirm the Cloud Billing account that you want to charge for the offer.
- 4. Optionally, select the checkbox if you want the offer to automatically renew at the end of the contract term. Auto-renew is only available if the vendor configures it. You can also update your auto-renew preferences at any time from the [Your orders page](#).
- 5. Click **Next**.
- 6. In the **Review and subscribe** section, review the final details, including the plan policy. To acknowledge the additional requirements, select the checkboxes.
- 7. Click **Subscribe**. A window appears to confirm your subscription.
- 8. In the confirmation window, click **View orders** to see the order status for all of your private offers, or click **Go to Marketplace** to browse Cloud Marketplace.
- 9. Depending on the product, you might need to activate your subscription directly with the vendor.

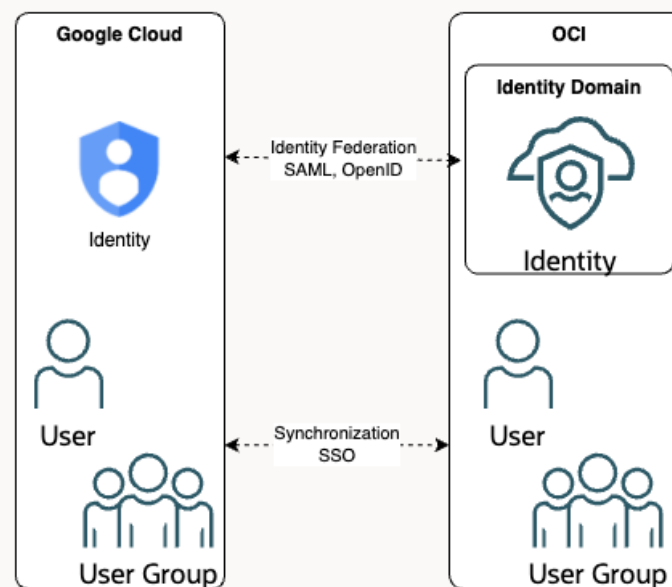
After you subscribe, the vendor is prompted to accept the offer. The offer pricing takes effect on the contract start date, as outlined in the offer.



Identity Roles and Groups

OCI Group name	Description	Google Cloud Role
odbg-exa-infra-administrators	User of this group will be able to administrate ExaDB-D resources in Google Cloud	Oracle Database@Google Cloud Exadata Infrastructure Admin
odbg-vm-cluster-administrators	VM Cluster administrators (including db-homes, databases, backups management) + Infrastructure reader. Manage objects and read buckets for managing backups in Object Storage.	Oracle Database@Google Cloud VM Cluster Admin
odbg-db-family-administrators	This role provides full access to any Exadata resource available in OCI.	Oracle Database@Google Cloud admin
odbg-db-family-readers	Exadata readers.	oracledatabase.googleapis.com/viewer
odbg-exa-cdb-administrators	CDB administrators (includes db homes and backups management). Manage objects and read buckets for managing backups in Object Storage.	
odbg-exa-pdb-administrators	PDB administrators.	
odbg-adbs-db-administrators	ADB-S administrators (includes ADB-S backups management)	Oracle Database@Google Cloud Autonomous Database Admin
odbg-adbs-db-readers	ADB-S readers (includes ADB-S backups read)	Oracle Database@Google Cloud Autonomous Database Viewer
odbg-network-administrators / readers	Network administrators / readers	
odbg-dbmgmt-administrators	Database Management Administrators (Allows to enable full set of metrics in CDB and PDB)	
odbg-metrics-readers	Metrics readers	
odbg-costmgmt-administrators	Cost management in OCI	
odbg-exa-infra-readers	This group is for viewers who need to view all Oracle Exadata Database Service resources in Google Cloud	Oracle Database@Google Cloud Exadata Infrastructure Viewer
odbg-vm-cluster-readers	This group is for viewers who need to view VM Clusters resources in Google Cloud	Oracle Database@Google Cloud VM Cluster Viewer

Identity Federation configuration



Example group combinations:

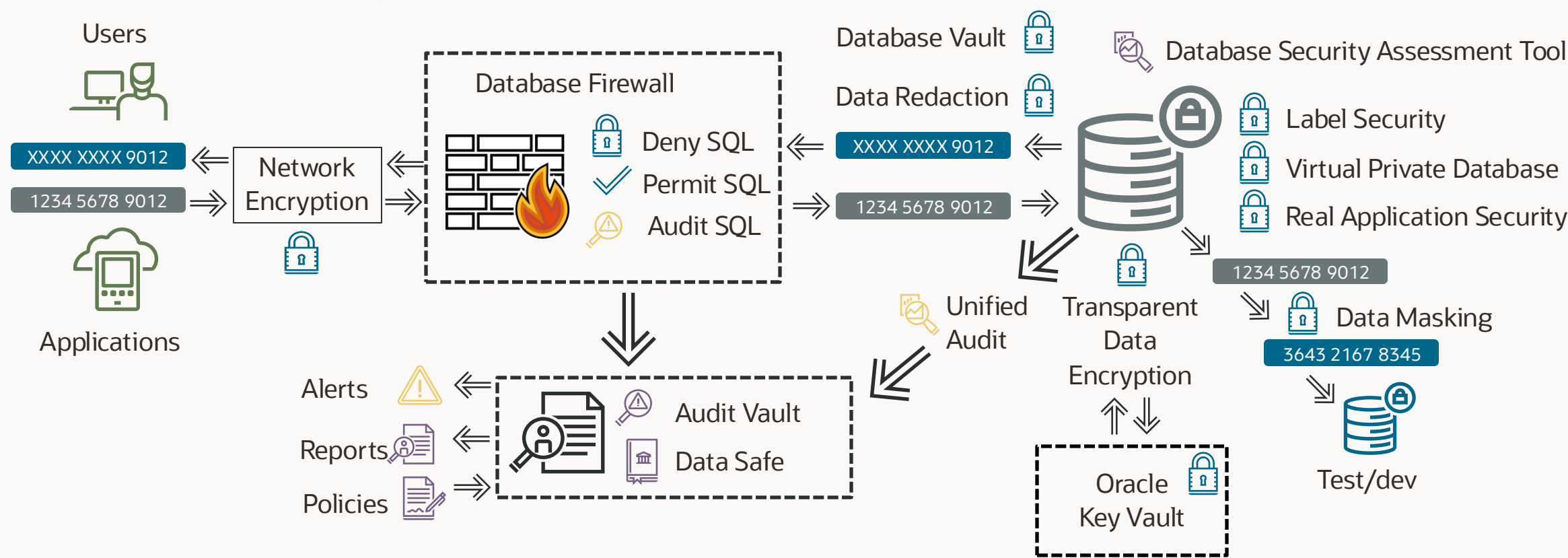
- Administrator access to manage DB related resources assign user to groups
 - odbg-db-family-administrators
 - odbg-network-administrators
 - odbg-dbmgmt-administrators
 - odbg-metrics-readers
- Administrator access to manage DB related resources assign user to groups but read only network:
 - odbg-db-family-administrators
 - odbg-network-readers
 - odbg-dbmgmt-administrators
 - odbg-metrics-readers
- Readers access
 - odbg-db-family-readers
 - odbg-network-readers
 - odbg-metrics-readers



Security

Oracle Maximum Security Architecture

Defense in Depth to protect your data



Securing your Databases

Control User and Network Access: Use passwords, private subnets, and network security groups to control user and network access.

Restrict Permissions for Deleting Database Resources: To prevent inadvertent or malicious deletion of databases, grant the delete permissions (DATABASE_DELETE and DB_SYSTEM_DELETE) to a minimum set of users and groups.

Data Encryption: All databases created in OCI are encrypted using transparent data encryption (TDE). Ensure that any migrated databases are also encrypted. Rotate the TDE master key every 90 days or less.

Patching: Apply Oracle Database security patches (Oracle Critical Patch Updates) to mitigate known security issues, and keep the patch levels up-to-date.

Use DB Security Tools: Established Data Masking and Auditing tools to secure your data are available for Oracle Database@Google Cloud.

Enable Data Safe: Data Safe is a unified control center for Oracle cloud and on-premises databases. Use Data Safe to assess database and data security configuration, detect associated risk for user accounts, identify existing sensitive data, implement controls to protect data, and audit user activity.

Securing database traffic – in transit

Configuring Oracle Database Native Network Encryption and Data Integrity

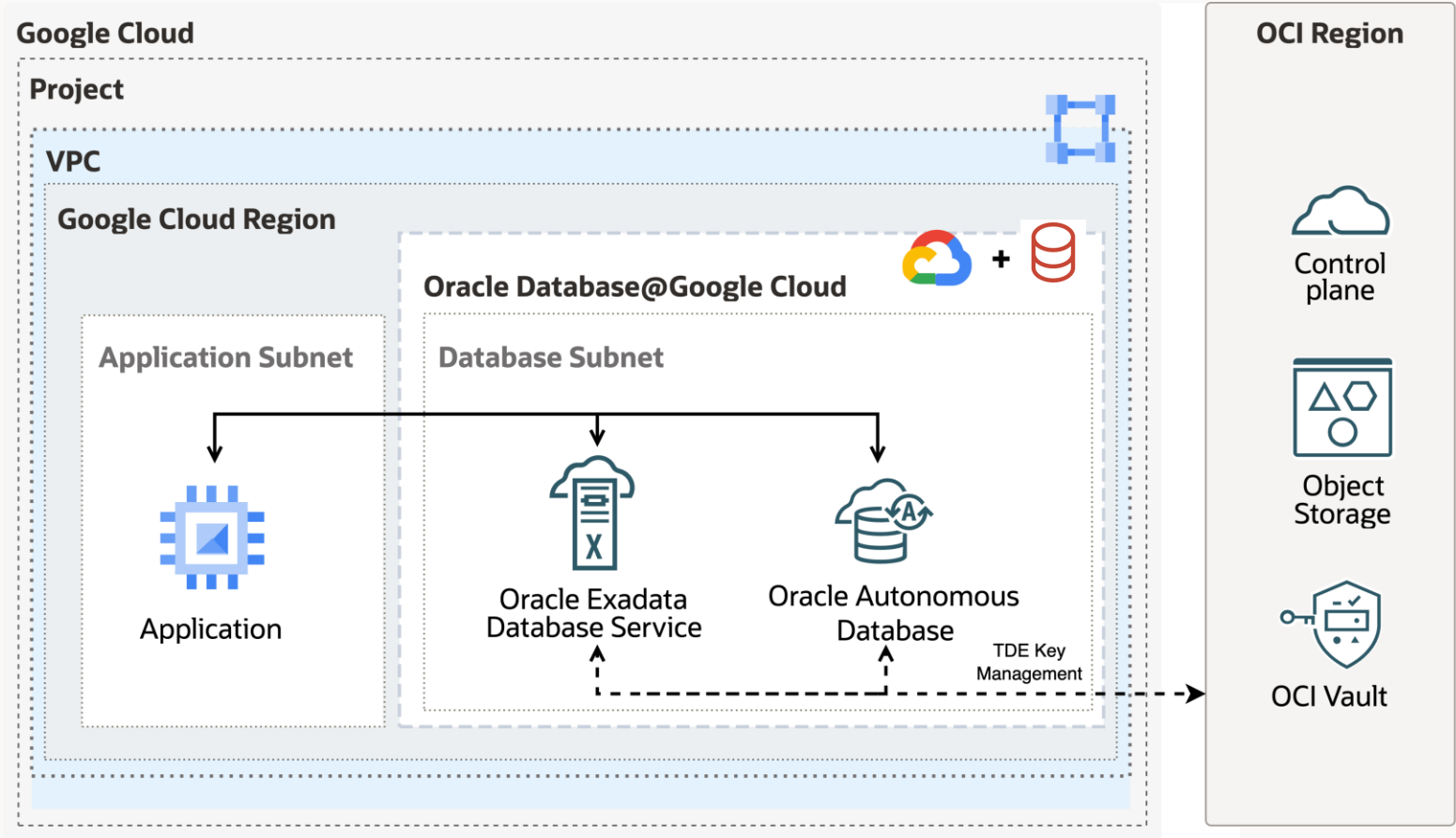
Configure native Oracle Net Services data encryption and data integrity for both servers and clients.

Oracle Database enables you to encrypt data that is sent over a network

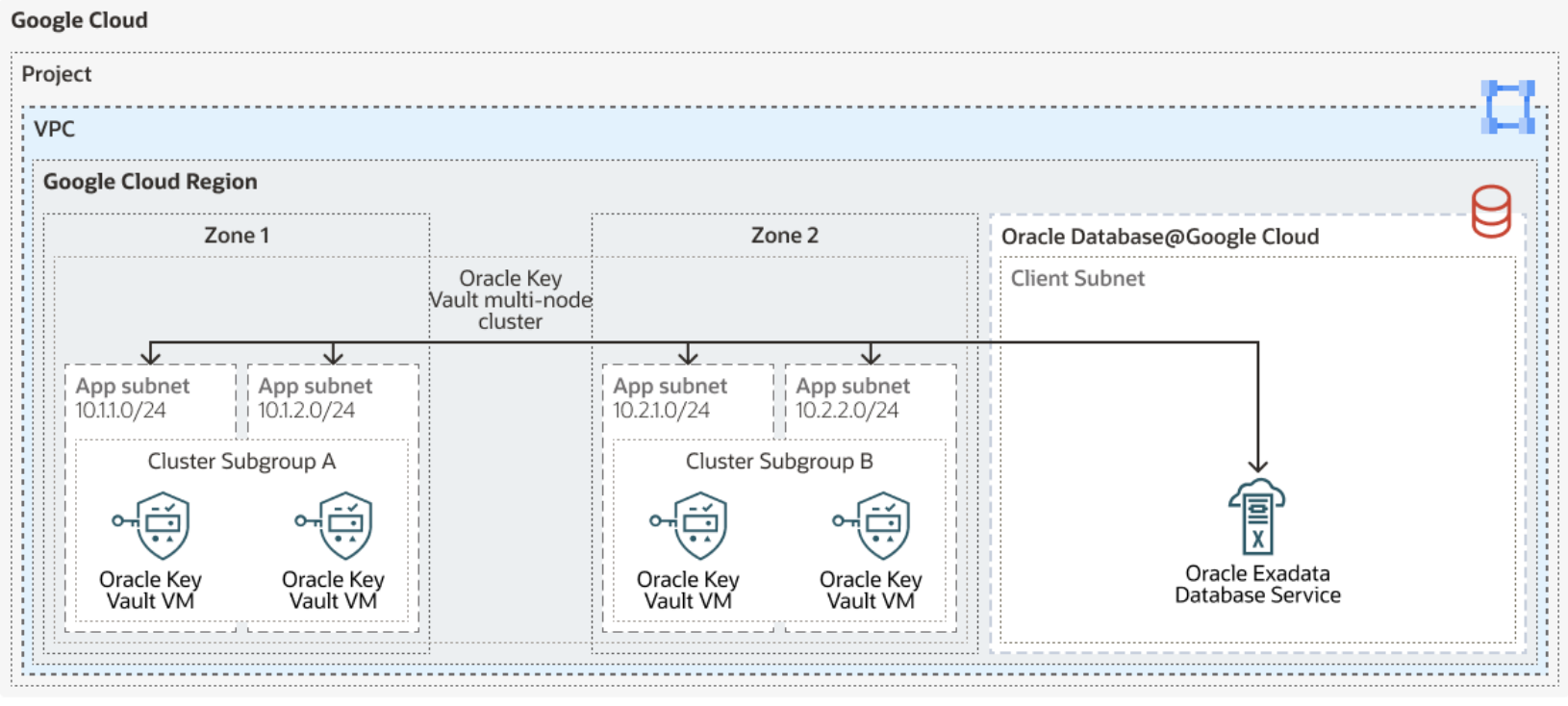
- Oracle Database provides **native data network encryption and integrity** to ensure that data is secure as it travels across the network
- Oracle Database supports the **Federal Information Processing Standard (FIPS)** encryption algorithm, **Advanced Encryption Standard (AES)**
- **Triple-DES encryption (3DES)** encrypts message data with three passes of the DES algorithm
- Oracle offers two ways to encrypt data over the network, native network encryption and **Transport Layer Security (TLS)**
- The Physical layer between the Oracle Database@Google Cloud and OCI are **encrypted**.

<https://docs.oracle.com/en/database/oracle/oracle-database/19/dbseg/configuring-network-data-encryption-and-integrity.html>

Encryption using OCI Vault



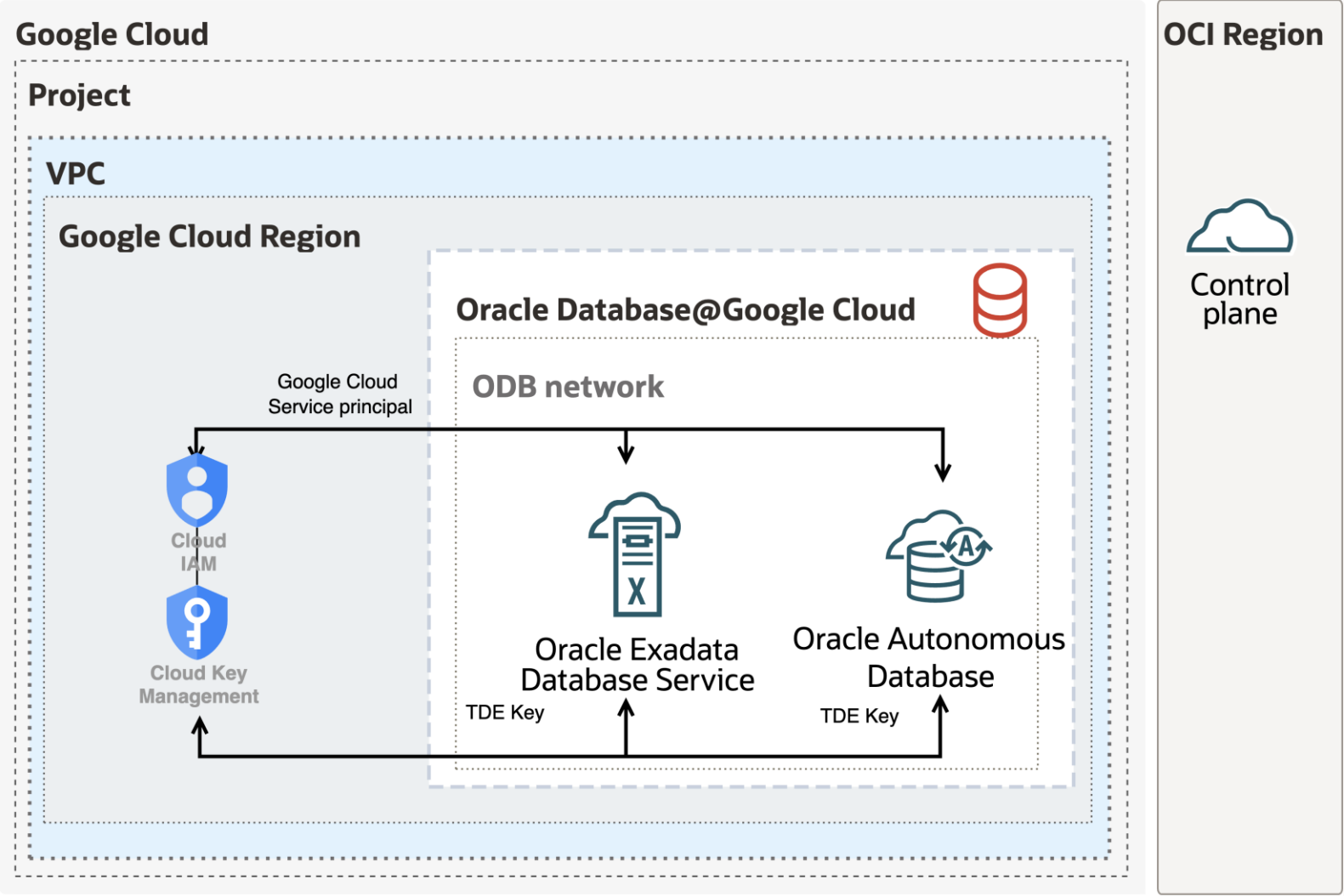
Deploy OKV with Oracle Exadata Database Service



Deploy Oracle Key Vault on GCE



Encryption using Google Cloud KMS



Support

Oracle Database@Google Cloud - Support

Collaboration support model - Customers raise tickets to Oracle

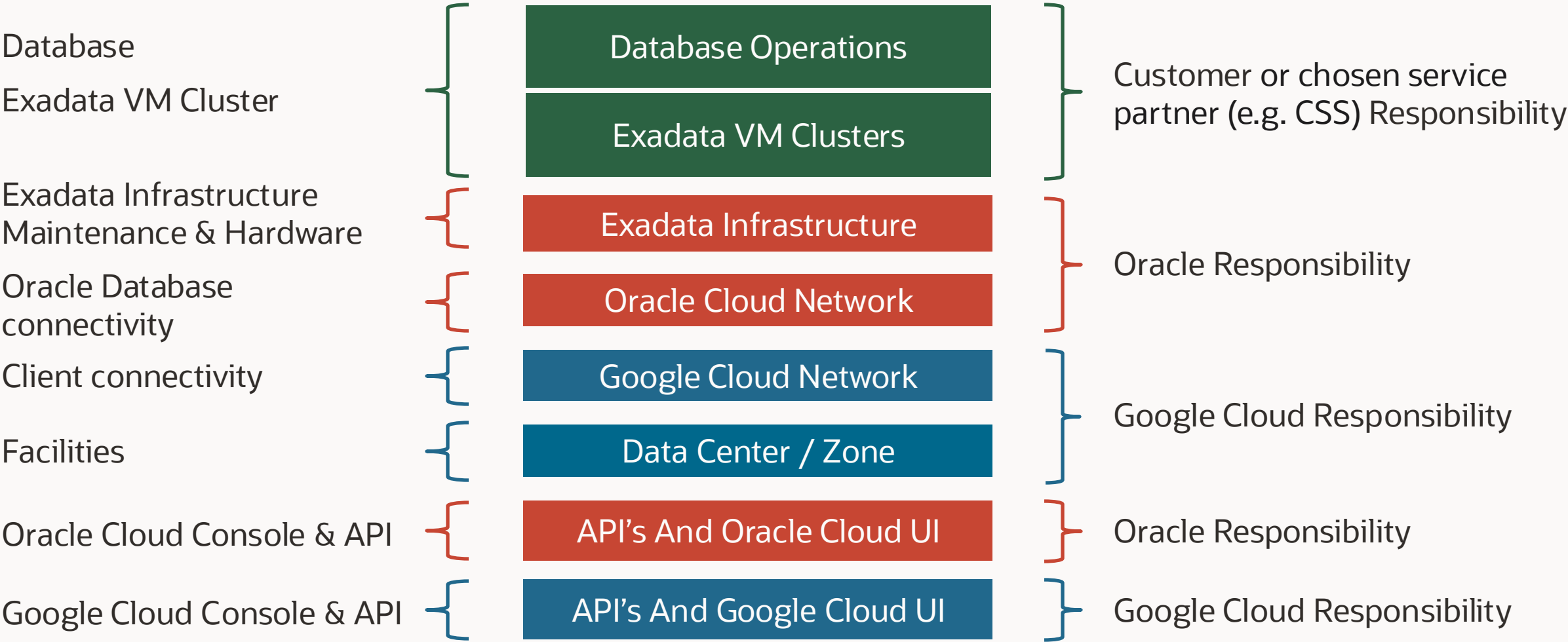
I. Oracle Support Scope

- Oracle Cloud Support is the first line of support for all Oracle Database@Google Cloud questions.
 - Database connection issues (Oracle TNS)
 - Oracle Database performance issues
 - Oracle Database error resolution
 - Networking issues related to communications with the OCI tenancy associated with the service
 - Quota (limits) increases to receive more capacity
 - Scaling to add more compute and storage capacity to your Oracle Database infrastructure
 - New generation hardware upgrades
 - Billing issues

II. Google Cloud Support Scope

- Google Cloud Support is your first line of support for all Google Cloud related issues and questions.
 - Virtual networking issues including those involving network address translation (NAT), firewalls, DNS and traffic management, and Google Cloud subnets.
 - Bastion and virtual machine (VM) issues including database host connection, software installation, latency, and host performance.
 - Exadata VM cluster metrics reporting within the Google Cloud Monitoring service.

Oracle Database@Google Cloud Responsibility Matrix



Resources available

Product web pages

- [Oracle Database@Google Cloud](#)

Documentation

- [Oracle Multicloud](#)
- [Google Cloud](#)

Oracle Architecture Center

- 20 [reference architectures](#)

Oracle Learn

- [Step by steps tutorials](#)

[Oracle Multicloud Capabilities](#)

▼ Oracle Multicloud

Learning Resources and
Reference Information

Cloud Terminology Mapping

Requesting a Limit Increase for
Database Resources

▶ Oracle Database@AWS

▼ Oracle Database@Google Cloud

What's New in Oracle
Database@Google Cloud

▶ Getting Started

Oracle Database@Google
Cloud Support Information

Deleting Google Cloud Projects

▶ Oracle Database@Azure

Latest Oracle Multicloud Updates

<https://bit.ly/multicloud-regions>

Oracle Multicloud Capabilities

Updated 27 May 2025FeedbackSign In

ResourcesRegionsServicesComplianceHigh AvailabilityMigrationCompare DB service capabilities, pricing & TCO at DBExpert >

AzureGoogle CloudAWS

Azure

GA Region: a region with 2 AZs or a single-AZ region with another single-AZ region in the same country

Live Region: Single AZ where a customer can run a workload

Q

Go

Download

Azure Region	City	Country	Geo	Status	Availability Zones	Oracle Exadata Database Service on Dedicated Infrastructure	Oracle Exadata Database Service on Exascale Infrastructure	Oracle Autonomous Database Serverless	Oracle Base Database Service	Oracle Database Zero Data Loss Autonomous Recovery Service
Canada Central	Toronto	Canada	North America	GA	2	✓	✓	✓	○	✓
Central US	Des Moines	USA	North America	GA	2	✓	○	✓	○	✓
East US	Ashburn	USA	North America	GA	2	✓	✓	✓	○	✓
East US 2	Ashburn	USA	North America	GA	2	✓	○	○	○	✓
South Central US	Dallas	USA	North America	To access this region, contact your Oracle customer sales representative.	1	○	○	○	○	○
West US	Santa Clara	USA	North America	GA	1	✓	○	✓	○	○



API Management

Cloud Automation for Lifecycle Management

Google Cloud Console UI REST APIs, SDK, CLI, Terraform

- Create Oracle Exadata Infrastructure and Exadata VM Clusters
- Schedule Infrastructure Maintenance
- Scale Servers and OCPUs

Oracle Cloud UI REST APIs, SDK, CLI, Terraform

- Create Database Homes and Databases
- Update Operating System, Grid Infrastructure, and Databases
- Backup and Recover Databases
- Enable Data Guard

Google Cloud CLI

gcloud oracle-database group examples

- *gcloud oracle-database autonomous-databases*
- *gcloud oracle-database cloud-exadata-infrastructures*
- *gcloud oracle-database cloud-vm-clusters commands group*
- *gcloud oracle-database gi-versions list command*

Cloud Automation with Terraform

Google Cloud

- Create Oracle Exadata Infrastructure and Exadata VM Clusters
- ADB-S deployment

```
resource "google_oracle_database_autonomous_database" "myADB"{
  autonomous_database_id = "my-instance"
  location = "us-east4"
  project = "my-project"
  database = "testdb"
  admin_password = "123Abpassword"
  network = data.google_compute_network.default.id
  cidr = "10.5.0.0/24"
  properties {
    compute_count = "2"
    data_storage_size_tb="1"
    db_version = "19c"
    db_workload = "OLTP"
    license_type = "LICENSE_INCLUDED"
  }
}

data "google_compute_network" "default" {
  name = "new"
  project = "my-project"
}
```

Oracle Cloud

- Create Database Homes and Databases
- Update Operating System, Grid Infrastructure, and Databases
- Scale Servers and OCPUs
- Backup and Recover Databases
- Enable Data Guard

```
resource "oci_database_autonomous_database_backup"
"test_autonomous_database_backup"{

  #Required

  autonomous_database_id =
oci_database_autonomous_database.test_autonomous_database.id


  #Optional

  display_name = var.autonomous_database_backup_display_name
  is_long_term_backup = var.autonomous_database_backup_is_long_term_backup
  retention_period_in_days =
var.autonomous_database_backup_retention_period_in_days
}
```

Provisioning and Deployment

Steps to provision Exadata Infrastructure

1. Enter required information
 1. Infrastructure display name
 2. Infrastructure ID
 3. Region
 4. Exadata Infrastructure model
 5. Number of Database servers
 6. Number of Storage services
2. Select the managed patching for the infrastructure as well as the schedule
3. Once these details are entered, you can create and provision

Google Cloud omcpmout2 Search (/) for resources, docs, products, and more

Oracle Database@Google Cloud / Exadata Database Service: Dedicated Infrastructure / Create Exadata Infrastructure

← Create Exadata Infrastructure

Instance details

Infrastructure display name *

Infrastructure ID *

Place parent/child resources and your ODB network in the same zone. This ensures optimal performance and seamless communication.

Region * us-east4 Region choice is permanent

GCP Oracle Zone * us-east4-b-r1 Zone choice is permanent

Oracle Cloud account omcpmout2

Machine configuration

Exadata Infrastructure model * Exadata.X11M

Database server type X11M 760 ECUs, 1.5 TB memory per server

Storage server type X11M-HC 80TB storage per server

Database servers * 2 2 - 32

Storage servers * 3 3 - 64

ECUs 1,520

Storage size 189 TB

Create Database (CDB and PDB) in OCI

- Must be part of the Oracle Database@Google Cloud CDB or PDB admin groups, depending on the operation
- In the Google Cloud console VM cluster resource page, click **Manage in OCI** using the identity federation
- On the OCI console, create CDB and/or PDB

ORACLE Cloud Search resources, services, documentation, and Marketplace Netherlands Northwest (Amsterdam)

Created: Wed, Dec 14, 2022, 15:48:08 UTC
Cluster name: cl-u6q5dfja
Time zone: UTC
License type: Bring Your Own License (BYOL)
Lifecycle state: Available
IORM status: Disabled
Diagnostics collection: Disabled [Edit](#)

Resource allocation
VM Count: 2
Enabled OCPUs: 4
Memory: 2780 GB
Local Storage: 1800 GB
Exadata storage: 192 TB
Storage for local backups: Not enabled
Storage for Exadata sparse snapshots: Not enabled

Infrastructure
Exadata cloud infrastructure:

Resources
Databases (3)
Database homes (3)
Virtual IP address (0)
Virtual Machines (2)
Metrics
Work requests (0)

Databases
[Create database](#)

Name	State
	Available
	Available
	Available

Create database

Basic information for the database

Provide the database name
DB1109

Provide a unique name for the database *Optional* ⓘ

Select a Database version
19c

Provide a PDB name *Optional* ⓘ

Specify a database home

Database Home source
☒ Select an existing Database Home ☐ Create a new Database Home

Database Home display name
Select a Database Home for the new database

Create administrator credentials

Username *Read-only*
sys

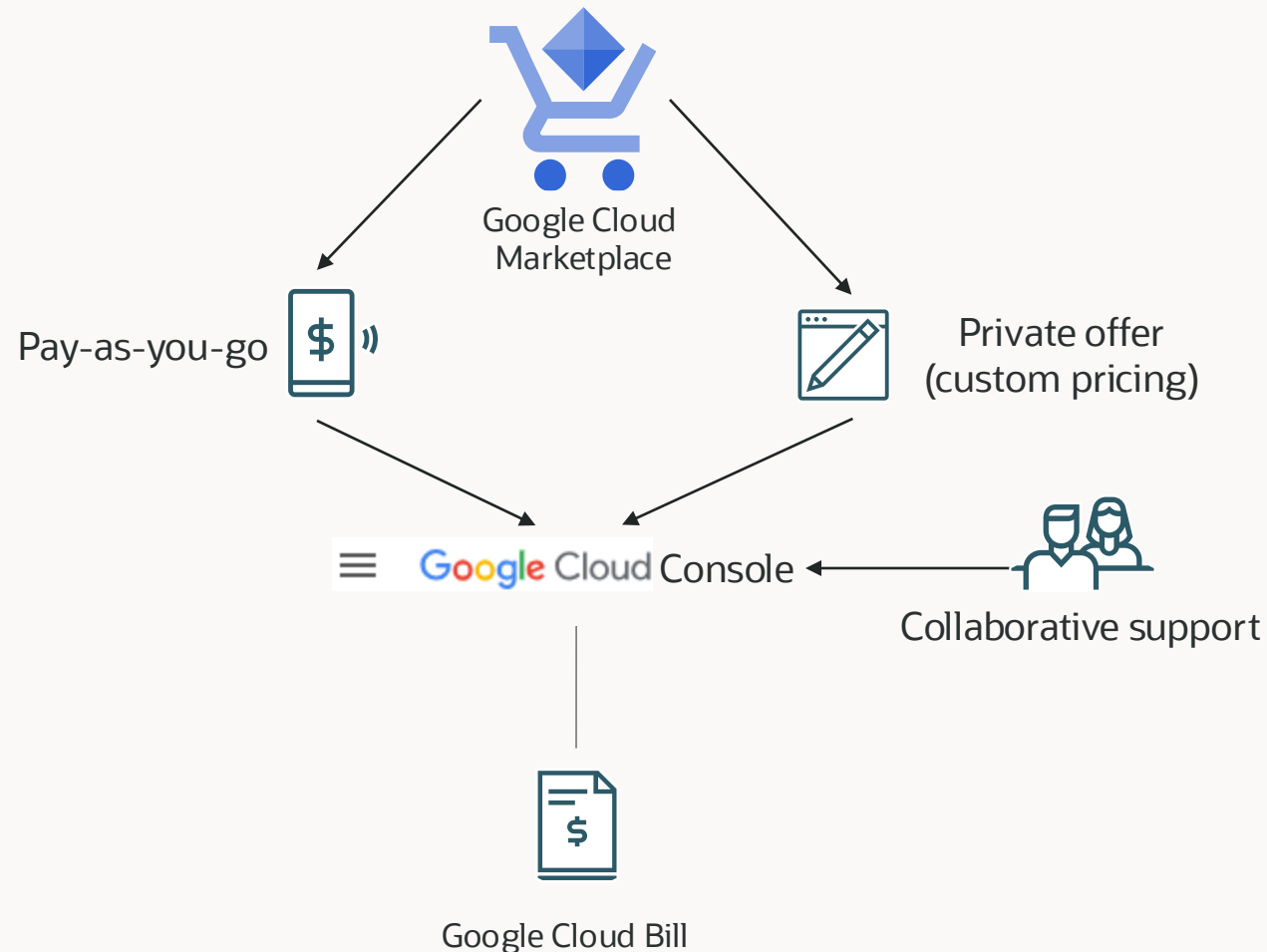
Password ⓘ

Confirm password

[Create database](#) [Cancel](#)

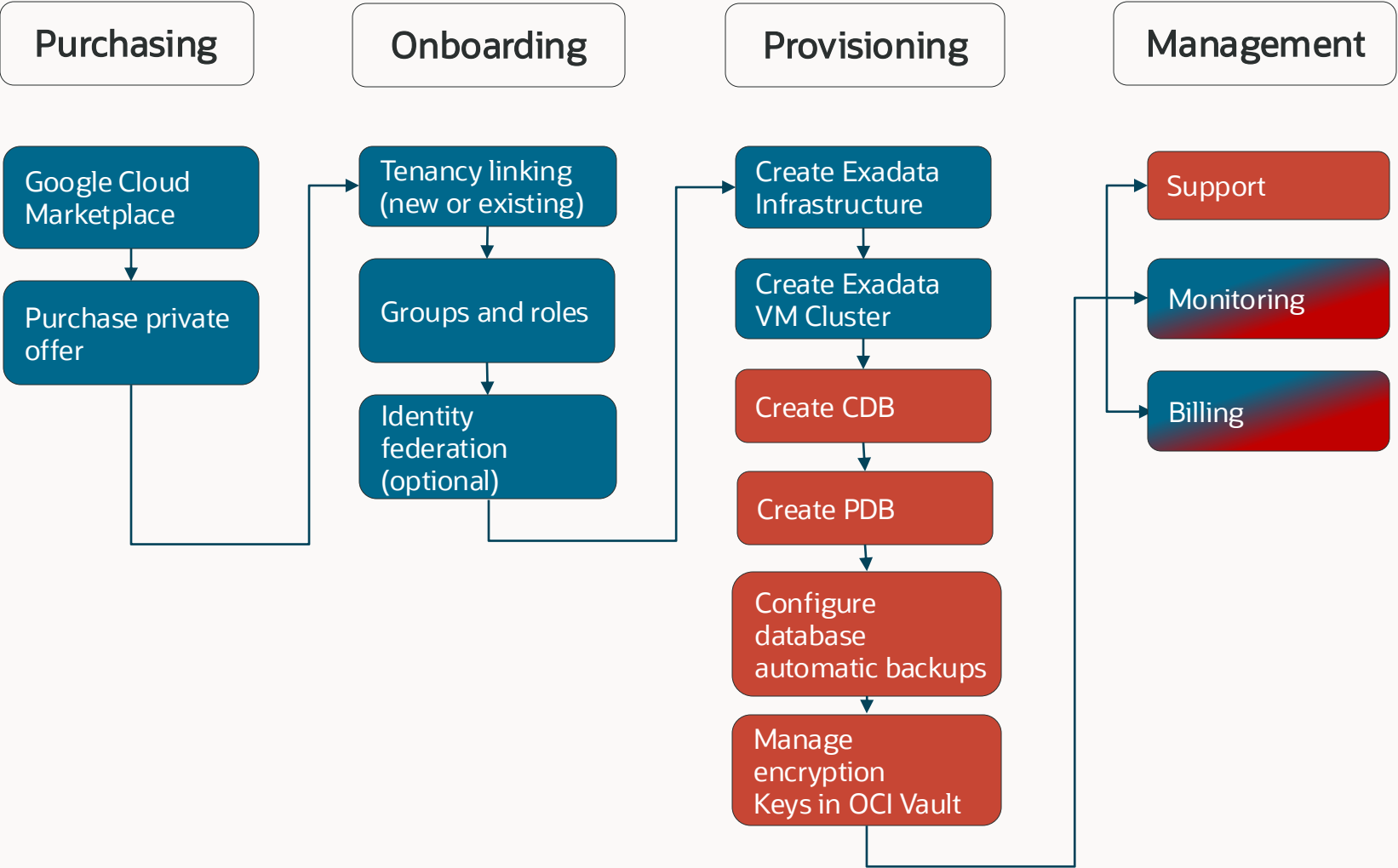
Purchasing and Enablement

Seamless multicloud experience freeing you to focus on your business

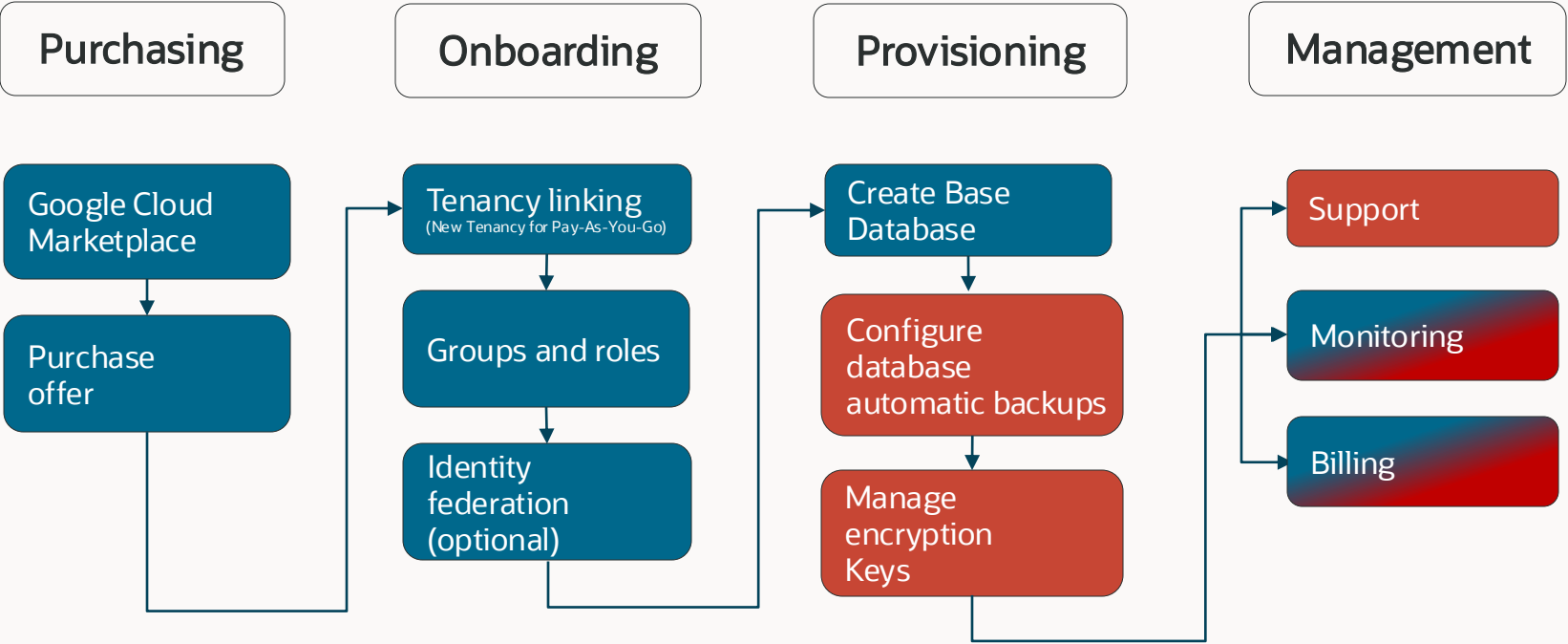


- Frictionless end-to-end experience across purchasing, billing and collaborative support
- Purchase Oracle database services through Google Cloud Marketplace using your Google Cloud Commitments and get a single bill for Oracle Database and Google Cloud services
- Option to use existing Oracle Database licenses and accrue the same Oracle Support Rewards as when using OCI directly

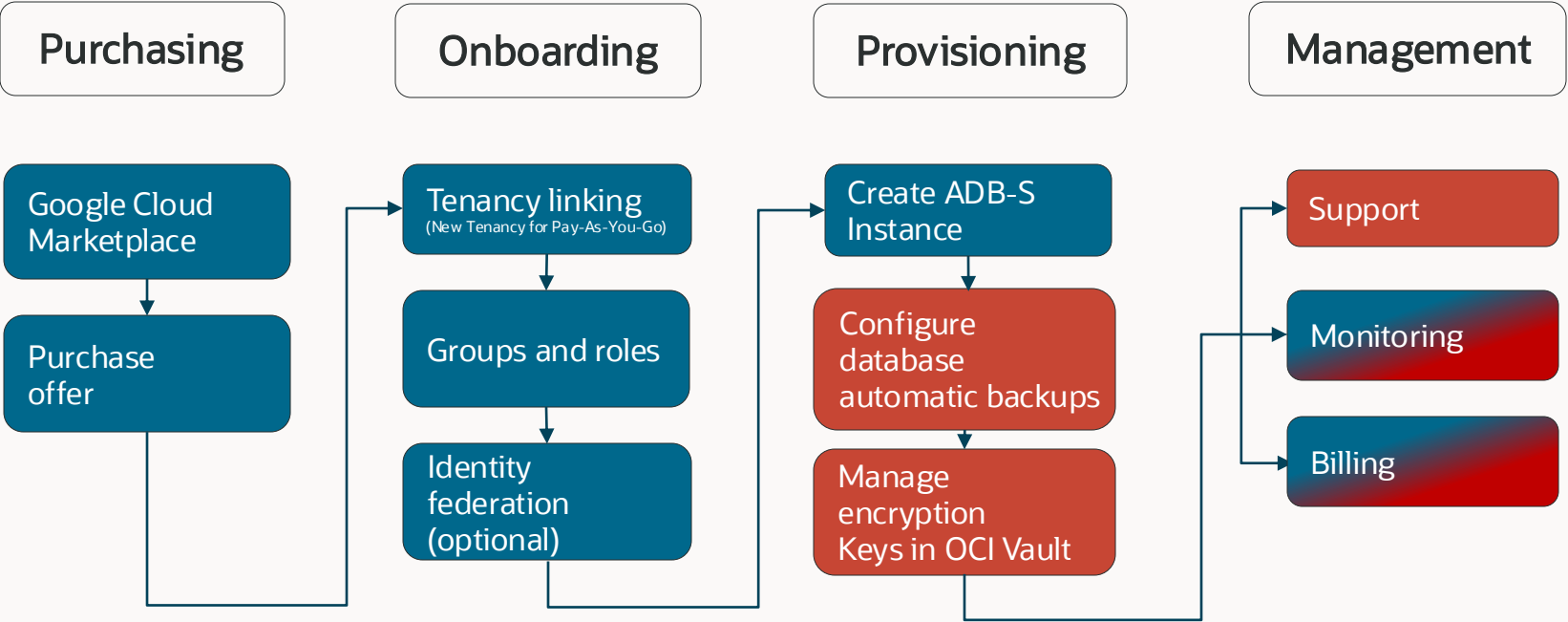
User Journey – Private Offer (Exadata Database Service)



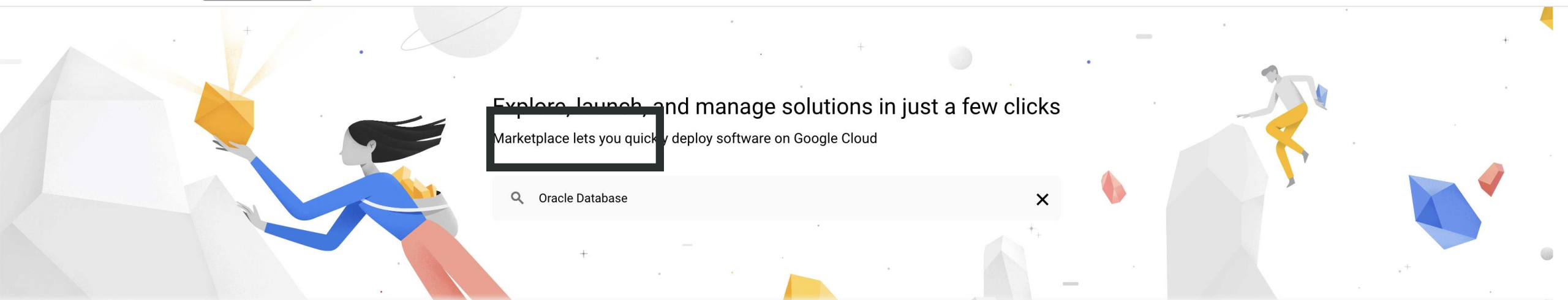
User Journey – Private & Pay-As-You-Go (BaseDB)



User Journey – Private & Pay-As-You-Go Offer (ADB-S)



Public Offer



Explore, launch, and manage solutions in just a few clicks

Marketplace lets you quickly deploy software on Google Cloud

 Oracle Database 

Your products

Your orders

Marketplace governance

 Filter Type to filter

Category ^

Analytics (966) 

Big data (575)

Databases (444)

Machine learning (526)

Retail

[VIEW ALL \(202\)](#)



commercetools
commercetools

Reimagining digital commerce for
the world's leading companies

Type SaaS & APIs



FullStory
FullStory, INC

Digital Experience Intelligence and
Product Analytics

Type SaaS & APIs



LiveRamp
LiveRamp

Data connectivity platform for
exceptional customer experiences

Type SaaS & APIs



Quantum Metric
Quantum Metric

The Platform For Continuous
Product Design

Type SaaS & APIs



Shopify
Shopify Inc

The leading global commerce
company that provides essential
internet infrastructure for...

Type SaaS & APIs

Analytics

[VIEW ALL \(966\)](#)

Marketplace

oracle database

Marketplace > "oracle database"

- Marketplace home
- Your products
- Your orders

Filter Type to filter

- Category
- Analytics (15)
 - Big data (14)
 - Databases (37)
 - Machine learning (4)
 - Data Sources (3)

- Type
- Google Cloud Platform (2)

50 results



Oracle Database@Google Cloud

Oracle - SaaS & APIs

Take control of your mission-critical database workloads with Oracle Database@Google Cloud, a groundbreaking service that brings the power of Oracle Autonomous Database and Oracle Exadata Database Service to Google Cloud. With this innovative offering, you can deploy, manage, and scale your Oracle Databases directly within Google Cloud. Oracle Autonomous Database on Oracle Database@Google Cloud A fully automated...



Oracle Database@Google Cloud API

Google - SaaS & APIs

The Oracle Database@Google Cloud API provides a set of APIs to manage Oracle database services, such as Exadata and Autonomous Databases.



FlashGrid Cluster for Oracle RAC / SI Database (RH7)

FlashGrid Inc. - Virtual machines

FlashGrid Cluster is an engineered cloud system designed for database high availability. FlashGrid Cluster comes as a fully integrated Infrastructure-as-Code template that you can customize and deploy to your IaaS cloud account with a few mouse clicks. Run your most critical databases on GCP Highly available clustered database with 2, 3, or more nodes Proven Oracle RAC database engine Multi-AZ deployment for...

31 Day Free Trial





Oracle Database@Google Cloud

Oracle

Accelerate innovation and supercharge database applications with Oracle Database@Google Cloud

SUBSCRIBE

CONTACT SALES

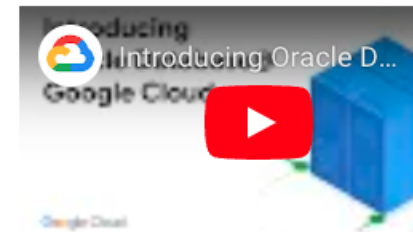
OVERVIEW

PRICING

DOCUMENTATION

SUPPORT

RELATED PRODUCTS



Overview

Take control of your mission-critical database workloads with Oracle Database@Google Cloud, a groundbreaking service that brings the power of Oracle Autonomous Database and Oracle Exadata Database Service to Google Cloud. With this innovative offering, you can deploy, manage, and scale your Oracle Databases directly within Google Cloud.

Oracle Autonomous Database on Oracle Database@Google Cloud

A fully automated database service that makes it easy for all organizations to develop and deploy application workloads, regardless of complexity, scale, or criticality. Oracle Autonomous Database is the only database cloud service that meets the needs ranging from self-service analytics to sophisticated, large-scale data management applications. Autonomous Database supports multiple workloads, including Data Warehousing, Transaction Processing, JSON-centric Applications, and Oracle APEX Low Code Development.

Additional details

Type: [SaaS & APIs](#)

Last product update: 9/16/24

Category: [Databases](#)



Order Summary

1. Select Plan

Plan
Oracle Database at Google Cloud pay-as-you-go

Features

- Oracle Database@Google Cloud: Yes

Pricing

Usage fee

Oracle Database at Google Cloud pay-as-you-go	USD 1.00
	/Oracle Database@Google Cloud credit

2. Purchase details

Oracle Database@Google Cloud By Oracle

Free

Estimated total cost

Adjust estimated timeframe



Monthly usage fee

Oracle Database at Google Cloud pay-as-you-go usage

Estimated Oracle Database@Google Cloud credit
0 Oracle Database@Google Cloud credit/mo

USD 0.00/mo

Pricing

Usage fee

Oracle Database at Google Cloud pay-as-you-go usage	USD 1.00 /Oracle Database@Google Cloud credit

2. Purchase details

Select a billing account *
My Billing account

3. Terms

Cancellation and change policy

- Your subscription fee is billed every month.

Additional terms

☐ By purchasing, deploying, accessing, or using this product, you acknowledge that Google is the merchant of record and Vendor's reseller with respect to this transaction and you agree to comply with the [Google Cloud Marketplace Terms of Service](#), [Oracle Terms of Service](#), [Terms of Use](#), [Privacy](#) and the terms of applicable open source software licenses bundled with the product.

SUBSCRIBE



Oracle Database@Google Cloud
By Oracle

Free

Estimated total cost

Adjust estimated timeframe



Monthly usage fee

Oracle Database at Google Cloud pay-as-you-go usage

Estimated Oracle Database@Google Cloud credit	USD 0.00/mo
0 Oracle Database@Google Cloud credit/mo	



[←](#) New Oracle Database@Google Cloud subscription[X](#) Pricing Calculator

Oracle Database at Google Cloud pay-as-you-go

USD 1.00

/Oracle Database@Google Cloud credit

2. Purchase details

Select a billing account *

Mon compte de facturation

3. Terms

Cancellation and change policy

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Additional terms

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Your order request has been sent to Oracle



Your new subscription to the Oracle Database at Google Cloud pay-as-you-go plan for Oracle Database@Google Cloud will begin after Oracle approves your request. Your subscription will then be active and you will begin getting charged. This processing time will depend on the vendor, and you should reach out to Oracle directly with any questions. You can manage your subscriptions on the Marketplace Orders Page

[MANAGE ORDERS](#)[GO TO PRODUCT PAGE](#)

Oracle Database@Google Cloud

By Oracle

Free

Estimated total cost

Adjust estimated timeframe

day

1 month

1 year

Monthly usage fee

Oracle Database at Google Cloud pay-as-you-go usage

Estimated Oracle Database@Google Cloud credit

0 Oracle Database@Google Cloud credit/mo

USD 0.00/mo

Your orders for this product

VIEW BILLING REPORTS

VIEW BILLING COMMITMENTS

SEND FEEDBACK

Select a billing account *
Research

This page contains all your orders for the Oracle Database@Google Cloud Dev product.
To manage other product orders visit [Your orders](#).

Filter

Filter by column name or chart value

Status	Order number	Order title	Provider	Product	Plan	Next plan	Auto-renew	Purchase date	Start date	End date	Payment schedule	
Pending	85c7a0...		Oracle	Oracle Database@Google Cloud Dev	Oracle Database at Google Cloud Pay as you go	N/A	N/A	08/21/2024	Pending	Pending	Postpay	



Your orders for this product

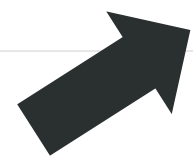
[VIEW BILLING REPORTS](#) [VIEW BILLING COMMITMENTS](#) [SEND FEEDBACK](#)

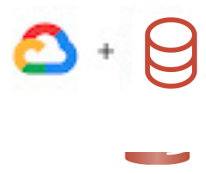
Select a billing account *
Research

This page contains all your orders for the Oracle Database@Google Cloud Dev product.
To manage other product orders visit [Your orders](#).

Filter Filter by column name or chart value

Status	Order number	Order title	Provider	Product	Plan	Next plan	Auto-renew	Purchase date	Start date	End date	Payment schedule	
Active ⓘ	85c7a0... 📋		Oracle	Oracle Database@Google Cloud Dev	Oracle Database at Google Cloud Pay as you go	N/A	N/A	08/21/2024	08/21/2024	-	Postpay	⋮





Oracle Database@Google Cloud

[Oracle](#)

ODBG brings the highest level of Oracle Database performance, scalability, and availability.

MANAGE ON CONSOLE [↗](#)

CONTACT SALES

✓ [Last purchased on 8/21/24](#)

- OVERVIEW
- PRICING

Overview

Run your workloads on Google Cloud with the fully managed Oracle database services. Experience the highest level of Oracle Database performance, scale, availability, and feature and pricing parity. With familiar application development tools and frameworks Google Cloud Platform supports, you can now migrate, modernize, and innovate using Oracle database services and Google cloud resources such as Google Compute Engine and Google Kubernetes Engine.

About managed services

Managed Services are fully hosted, managed, and supported by the service providers. Although you register with the service provider to use the service, Google handles all billing.

Additional details

Type: [SaaS & APIs](#)
Last product update: 8/13/24
Category: [Databases](#)

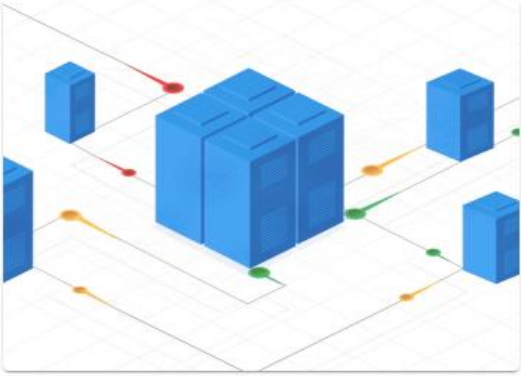


Connect your Marketplace offer with an Oracle account to start provisioning resources.

CREATE ACCOUNT

Get started with Oracle Databases

Choose the right Oracle database solution for your needs. Exadata for unmatched performance and scalability, Autonomous Database for effortless automation.



Oracle Exadata Database

The ultimate database platform for performance, scale, and simplicity. Consolidate all your workloads, from OLTP to analytics, on a single, powerful system.

EXPLORE EXADATA DATABASE



Oracle Autonomous Database

Autonomous Databases delivers fast performance and requires no database administration. It performs all routine database maintenance tasks without human intervention while the system is running.

EXPLORE AUTONOMOUS DATABASE



Private Offer

Back

Compose

Reply

Reply all

Forward

Delete

Move

Print

Mark

More

Inbox1

Drafts

Junk

Trash

New Private Offer from Oracle

Message 1 of 3

 From Google Cloud Platform
To john.wargo@acmedynamite.com
Date Tue 14:35



MY CONSOLE

 Google Cloud Marketplace

New private offer from Oracle - Oracle Database@Google Cloud Dev

Hello,

Your private offer for Oracle Database@Google Cloud Dev is ready. Please review and accept the offer by Aug 31, 2024, 11:45 PM PDT

REVIEW OFFER

- Access to the offer link requires appropriate permissions [Learn more](#)



New offer

Order title

Offer for Demo

Partner details

Customer details

Oracle

Oracle Sales

sales@oracle.com

Oracle

John Doe

john.doe@acmedynamite.com

Billing account: Production

Billing account number: 010631-0C1FA8-B9BDDC

Deadline to accept: Aug 31, 2024, 11:45 PM PDT

Key Events

Created

by Oracle, Aug 21, 2024, 12:16 PM PDT

PDF

Offer overview

Total contract value

USD 1.00

Offer Dates	
Start date	On offer approval
End date	Aug 31, 2024, 11:45 PM PDT



Offer Dates	
Start date	On offer approval
End date	Aug 31, 2025, 11:59 PM PDT
Offer duration	12 months
Billing frequency	Annually, 1 year(s)
Auto-renew availability	Not currently available

Product and plan

Product	Oracle Database@Google Cloud
Plan	Oracle Database at Google Cloud Private Offer

Pricing

Pricing Model: Flat fee with usage discount

Prices are in US Dollars (USD). If your billing account is configured to a different currency then the charge will be converted based on the current exchange rate.

Installment schedule ?

Upcoming installments

	Installment date	Flat fee	Discount on usage
	September 1, 2024	USD 1.00	100.00%

Next bill will have 100.00% off monthly usage charges

USD 1.00 + 100.00% off additional usage on Sep 1, 2024

Key Events

Created

by Oracle, Aug 21, 2024, 12:16 PM PDT

PDF



USD 1.00
/per credit

Oracle Database at Google Cloud Private Offer

Oracle Database at Google Cloud Private Offer

Oracle Database at Google Cloud Private Offer

Key Events

✓ Created [📄 PDF](#)
by Oracle, Aug 21, 2024, 12:16 PM PDT

Review and accept

Purchase policy details

- You can't cancel your subscription after the start date
- You'll pay a flat fee regardless of actual usage. You may also be charged additional fees as described in the usage metric charges table.
- This offer only applies to metrics listed in the usage metric charges table. New metrics that the partner launches are out of scope for this offer and will be charged at list price.
- Auto-renew is not available. Therefore, at the end of contract duration you will lose access to the service if a replacement offer hasn't been signed.

Terms

- By purchasing, deploying, accessing, or using this product, you acknowledge that Google or an affiliate is the Vendor's agent with respect to this transaction and you agree to comply with the [Oracle Terms of Service](#), [Google Cloud Marketplace Terms of Service](#) (including the GPC terms set forth in Appendix A of the Marketplace ToS), and the terms of applicable open source software licenses bundled with the product.

☐ I agree to the terms above

ACCEPT

Download PDF

What's included:

- ☒ Offer details
Overview, pricing and payments, and Oracle contact information

What else you can include:

- ☐ Notes from Oracle
Notes weren't added when creating the offer
- ☒ EULA and Terms of Service

[CANCEL](#) [DOWNLOAD](#)

	USD 1.00
Oracle Database at Google Cloud Private Offer	/per credit
Oracle Database at Google Cloud Private Offer	
Oracle Database at Google Cloud Private Offer	

Key Events

- ✓

Created

by Oracle, Aug 21, 2024, 12:16 PM PDT

📄 PDF

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☐ I agree to the terms above

ACCEPT



Oracle Database at Google Cloud Private Offer	USD 1.00
Oracle Database at Google Cloud Private Offer	/per credit
Oracle Database at Google Cloud Private Offer	
Oracle Database at Google Cloud Private Offer	

Key Events

- ☒ Created [PDF](#)
- by Oracle, Aug 21, 2024, 12:16 PM PDT

Accepted! Now set up your account with partner



Make this offer active by setting up your account with Oracle. You will receive an email from Oracle for the detailed next step. If you don't sign up before the order start date, you won't be able to access Oracle's services and they'll have to create (and you'll need to accept) a new offer.

MANAGE ORDERS

GO TO PRODUCT PAGE

Review and accept

Purchase policy details

- You can't cancel your subscription after the start date
- You'll pay a flat fee regardless of actual usage. You may also be charged additional fees as described in the usage metric charges table.
- This offer only applies to metrics listed in the usage metric charges table. New metrics that the partner launches are out of scope for this offer and will be charged at list price.
- Auto-renew is not available. Therefore, at the end of contract duration you will lose access to the service if a replacement offer hasn't been signed.

Terms

- By purchasing, deploying, accessing, or using this product, you acknowledge that Google or an affiliate is the Vendor's agent with respect to this transaction and you agree to comply with the [Oracle Terms of Service](#), [Google Cloud Marketplace Terms of Service](#) (including the GPC terms set forth in Appendix A of the Marketplace ToS), and the terms of applicable open source software licenses bundled with the product.

☒ I agree to the terms above





Oracle Database@Google Cloud

[Oracle](#)



Accelerate innovation and supercharge database applications with Oracle Database@Google Cloud

🕒 [Purchase pending provider approval](#)

OVERVIEW

PRICING

Overview

Accelerate innovation and supercharge database applications with Oracle database services on Oracle Cloud Infrastructure in Google Cloud.

- Oracle Exadata Database Service
- Oracle Autonomous Database

Gain deeper and faster insights by combining capabilities like Vertex AI and Oracle Database 23ai. Run your Oracle Database workloads with the highest performance, security, and availability. Leverage a frictionless end-to-end experience across purchasing, operations, and collaborative support.

[Learn more](#)

About managed services

Managed Services are fully hosted, managed, and supported by the service providers. Although you register with the service provider to use the service, Google handles all billing.

Additional details

Type: [SaaS & APIs](#)

Last product update: 8/26/24

Category: [Databases](#)



Your orders for this product

VIEW BILLING REPORTS

VIEW BILLING COMMITMENTS

SEND FEEDBACK

Select a billing account *
Production

This page contains all your orders for the Oracle Database@Google Cloud Dev product.
To manage other product orders visit [Your orders](#).

Filter

Filter by column name or chart value

Status	Order number	Order title	Provider	Product	Plan	Next plan	Auto-renew	Purchase date	Start date	End date	Payment schedule	
Active	5bec82...	Offer for Demo	Oracle	Oracle Database@Google Cloud Dev	Oracle Database at Google Cloud Private Offer Monthly Subscription	N/A	N/A	08/27/2024	08/27/2024	08/27/2025	Annual	

Google Cloud

omcpmout2

Search (/) for resources, docs, products and more

Search

J

Oracle database @Google Cloud / Overview

Overview

Autonomous database se...

Autonomous database

Exadata database service

Dedicated infrastructure

Exascale infrastructure

Core database service

Base database service

Connectivity

ODB network

Overview

Get started with Oracle databases

Choose the right Oracle database solution for your needs.

Autonomous database

Delivers fast performance and requires no database administration. It performs all routine database maintenance tasks without human intervention while the system is running. [Learn more](#)

Explore service

Exadata Database Service on Dedicated Infrastructure

The ultimate database platform for performance, scale and simplicity. Consolidate all your workloads, from OLTP to analytics, on a single, powerful system. [Learn more](#)

Explore service

Exadata Database Service on Exascale Infrastructure

Powers next-gen Exadata Database Service, delivering extreme performance, automatically scaling resources, and simplified database management. [Learn more](#)

Explore service

Base Database Service

Delivers a versatile, cost-effective and highly available relational database service, offering granular resource control and simplified operations for a wide range of workloads. [Learn more](#)

Explore service

Support

<1

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Thank you

ORACLE

Oracle Cloud Infrastructure for Google Cloud professionals

Concept	Google Cloud	Oracle Cloud Infrastructure
Cluster of data centers and services	Region	Region
Abstracted data center	Zone	Availability Domain
Account	Organization	Tenancy
Organizing resources	Project	Compartments
Virtual Network	Virtual Private Cloud (VPC)	Virtual Cloud Network (VCN)
Dedicated Private Connectivity	Partner Interconnect	FastConnect
Identity and Access Management	Cloud IAM	OCI IAM Domains

